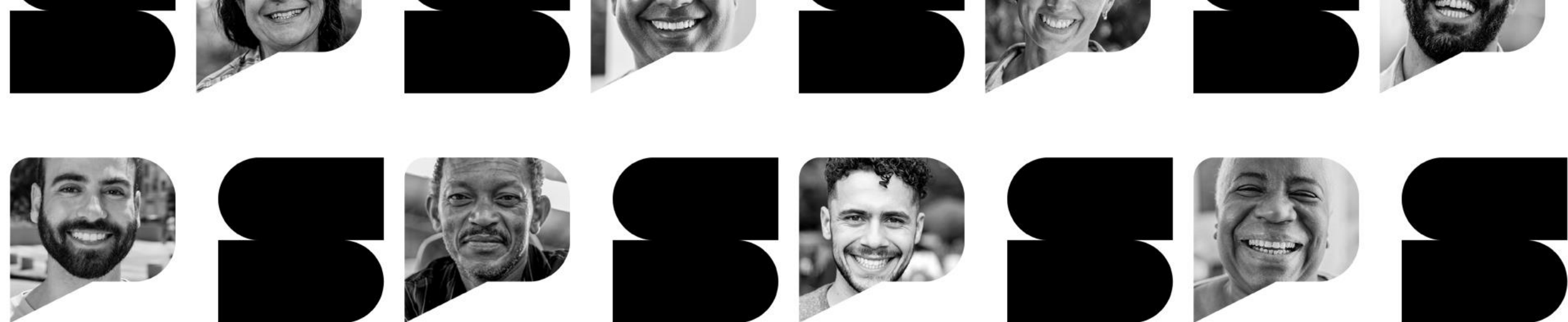
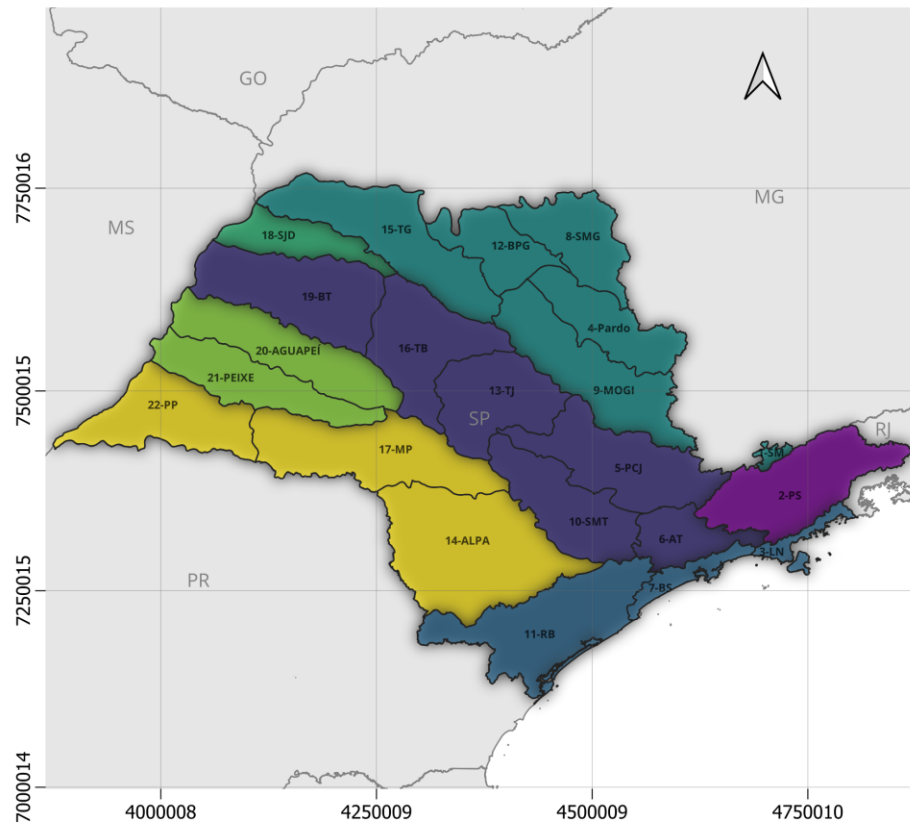




Chuvas no Estado de São Paulo
Índice de Precipitação Padronizado (SPI)
Magnitude da Seca (DM)

DAEE, Agosto de 2024

UNIDADES DE GERENCIAMENTO DE RECURSOS HÍDRICOS AGRUPADAS POR REGIÃO HIDROGRÁFICA



Legenda

Regiões Hidrográficas

- Bacia do Rio Paraíba do Sul
- Bacia do Rio Tietê
- Vertente Litorânea
- Vertente Paulista do Rio Grande
- Vertente Paulista do Rio Grande/São José dos Dourados
- Aguapeí/Peixe
- Vertente Paulista do Rio Paranapanema

Unidade de Gerenciamento de Recursos Hídricos

- Limite UGRHI

Contexto espacial

Estado inserido em 3 Regiões
Hidrográficas Brasileiras: Rio Paraná;
Atlântico Sudeste e Atlântico Sul

- 7 Regiões Hidrográficas Estaduais (vertentes);
- 22 Unidades de Gerenciamento de Recursos Hídricos (UGRHI);
- 9 Regiões Metropolitanas;
- 645 municípios;
- 45,1 milhões de habitantes

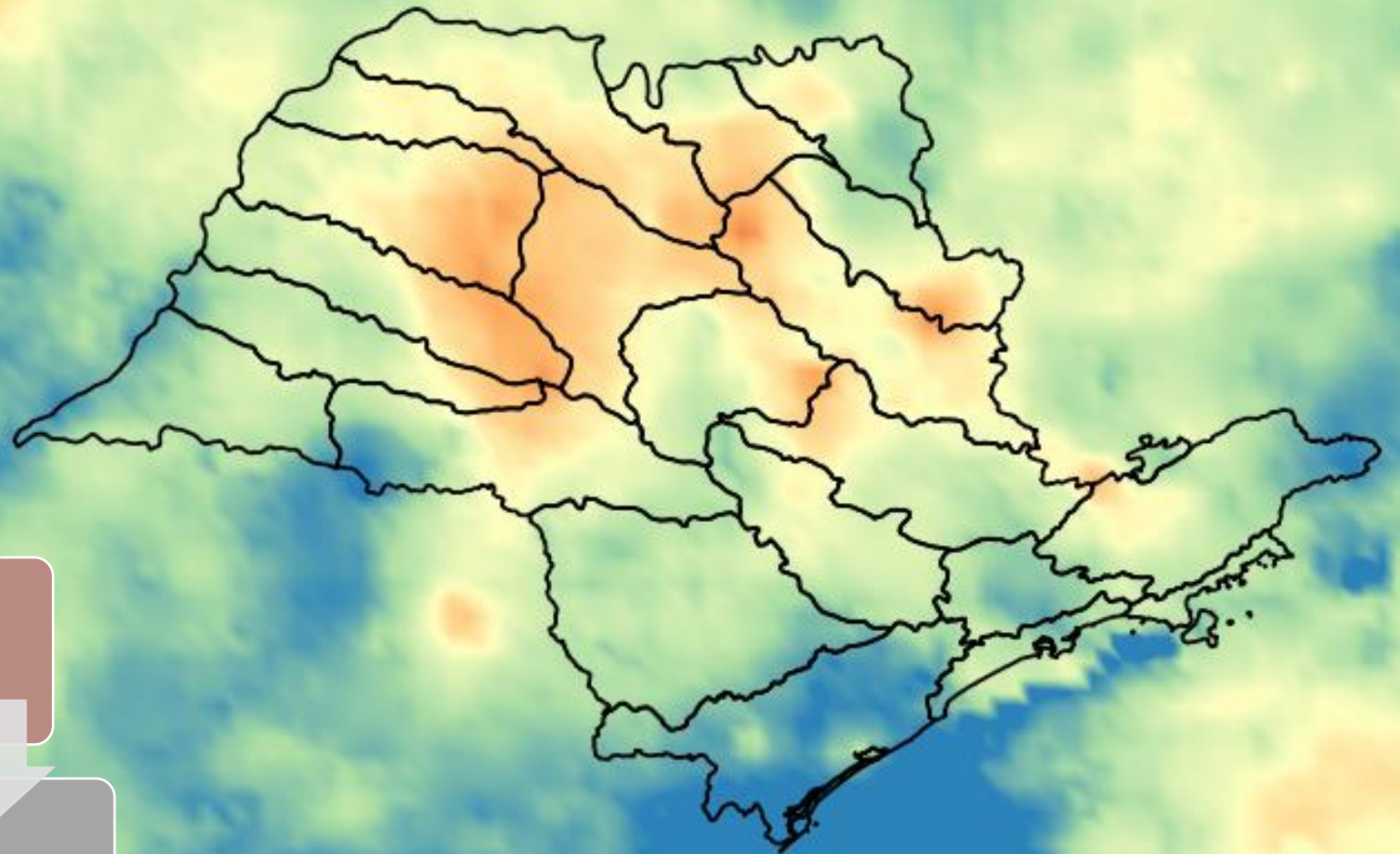
Metodologia

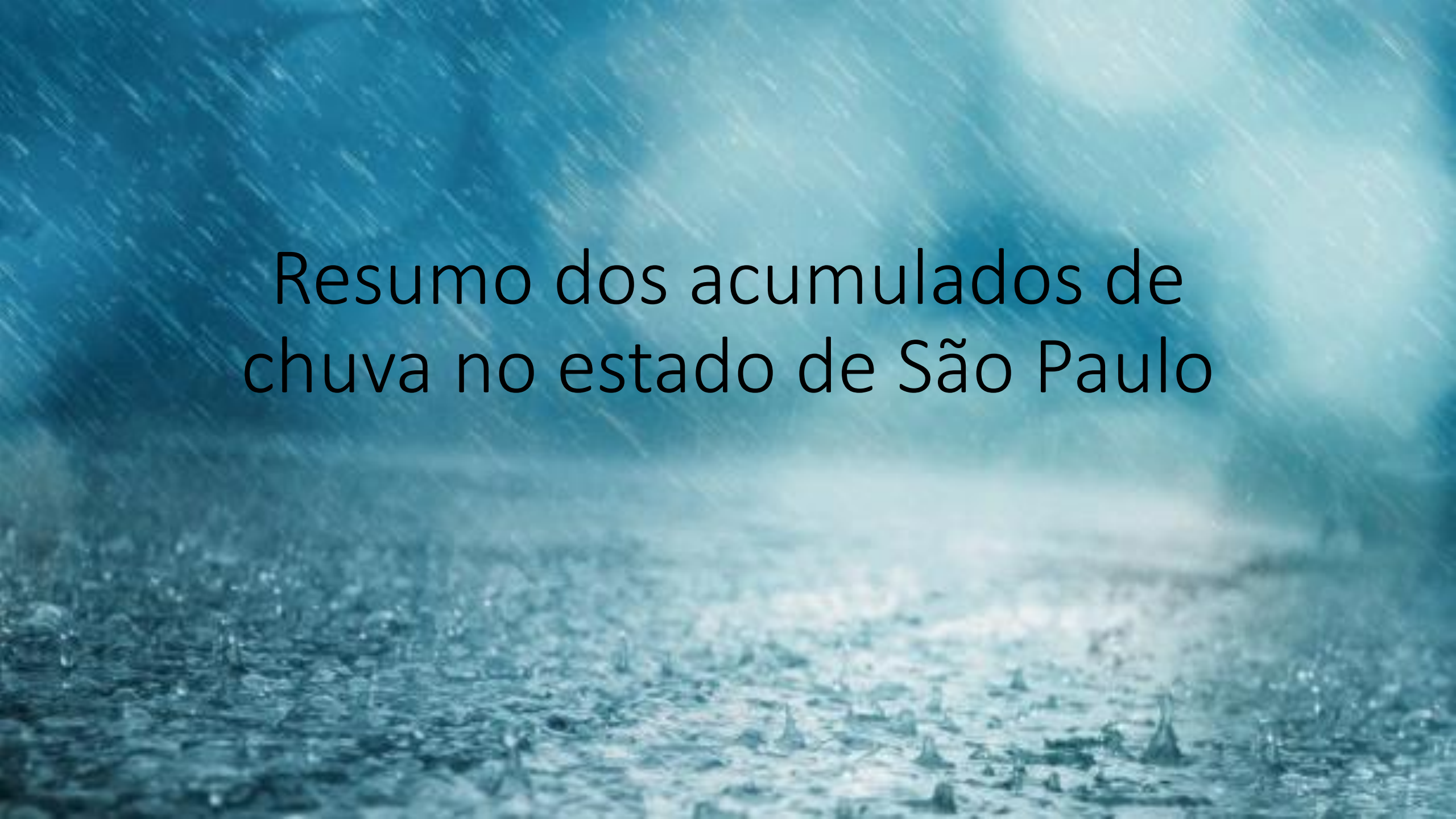
Coleta dados da chuva diária raster (Merge/INPE): combinação da precipitação observada (pluviômetros) com estimativa de precipitação por satélite.

Processamento com uma biblioteca para dados geoespaciais (GDAL)

Obtenção da chuva diária e mensal na área da UGRHI

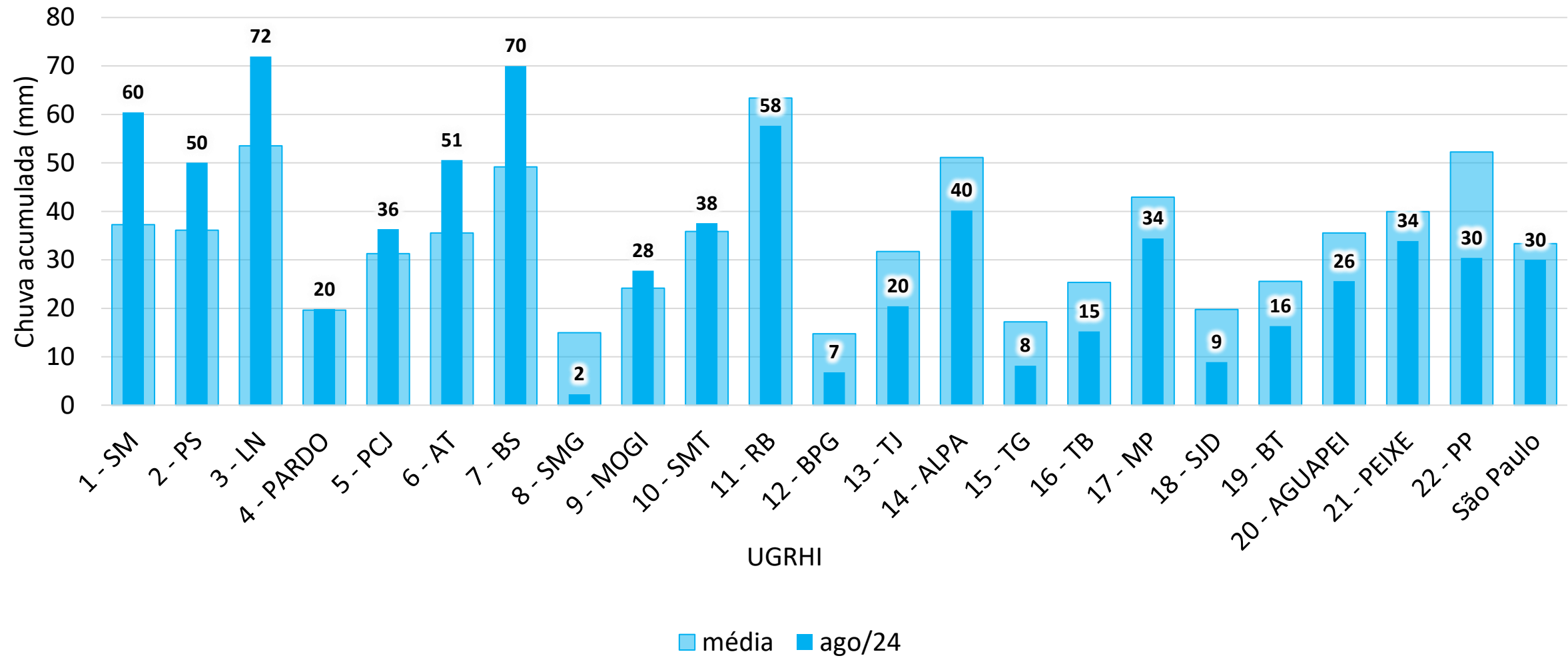
Cálculo de indicadores (SPI e DM)



A photograph of a city street during a heavy rain. The rain is captured as numerous white streaks against a dark, blurred background of buildings and streetlights. The overall color palette is cool, dominated by blues and greys.

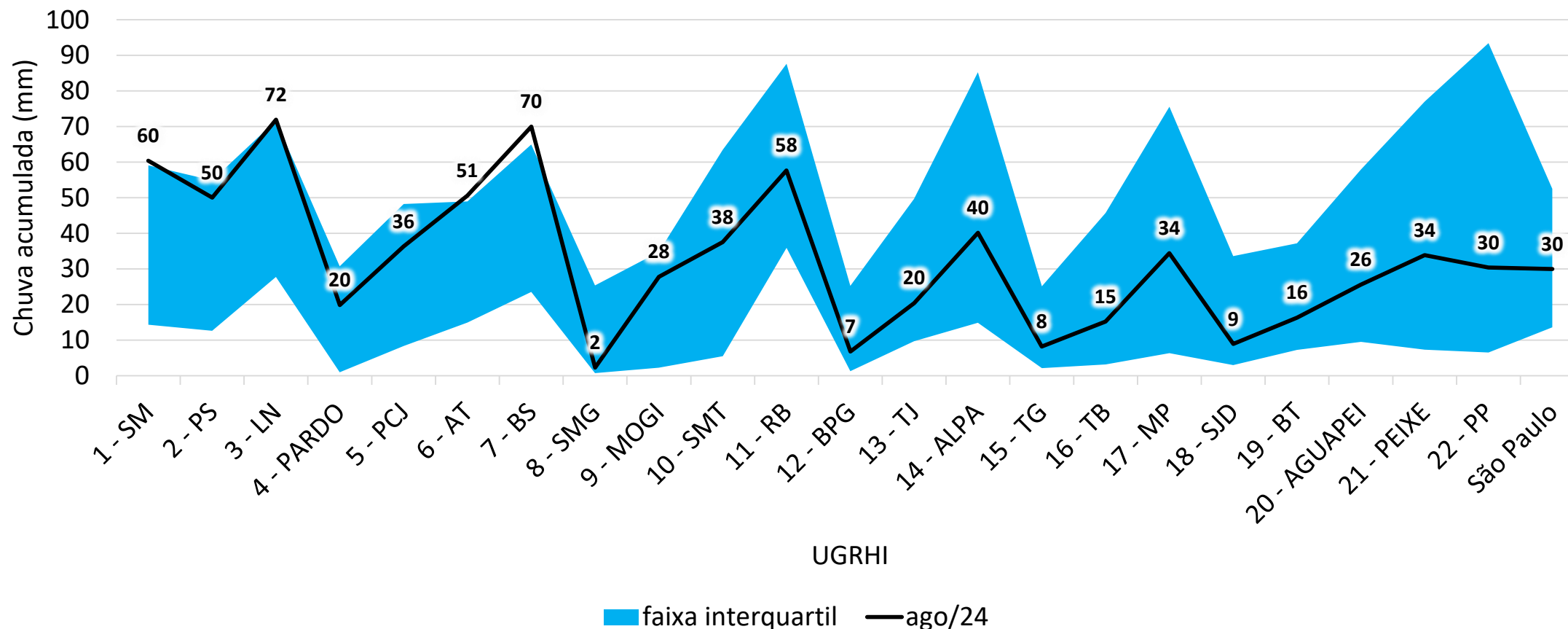
Resumo dos acumulados de chuva no estado de São Paulo

Chuva por UGRHI em Agosto de 2024

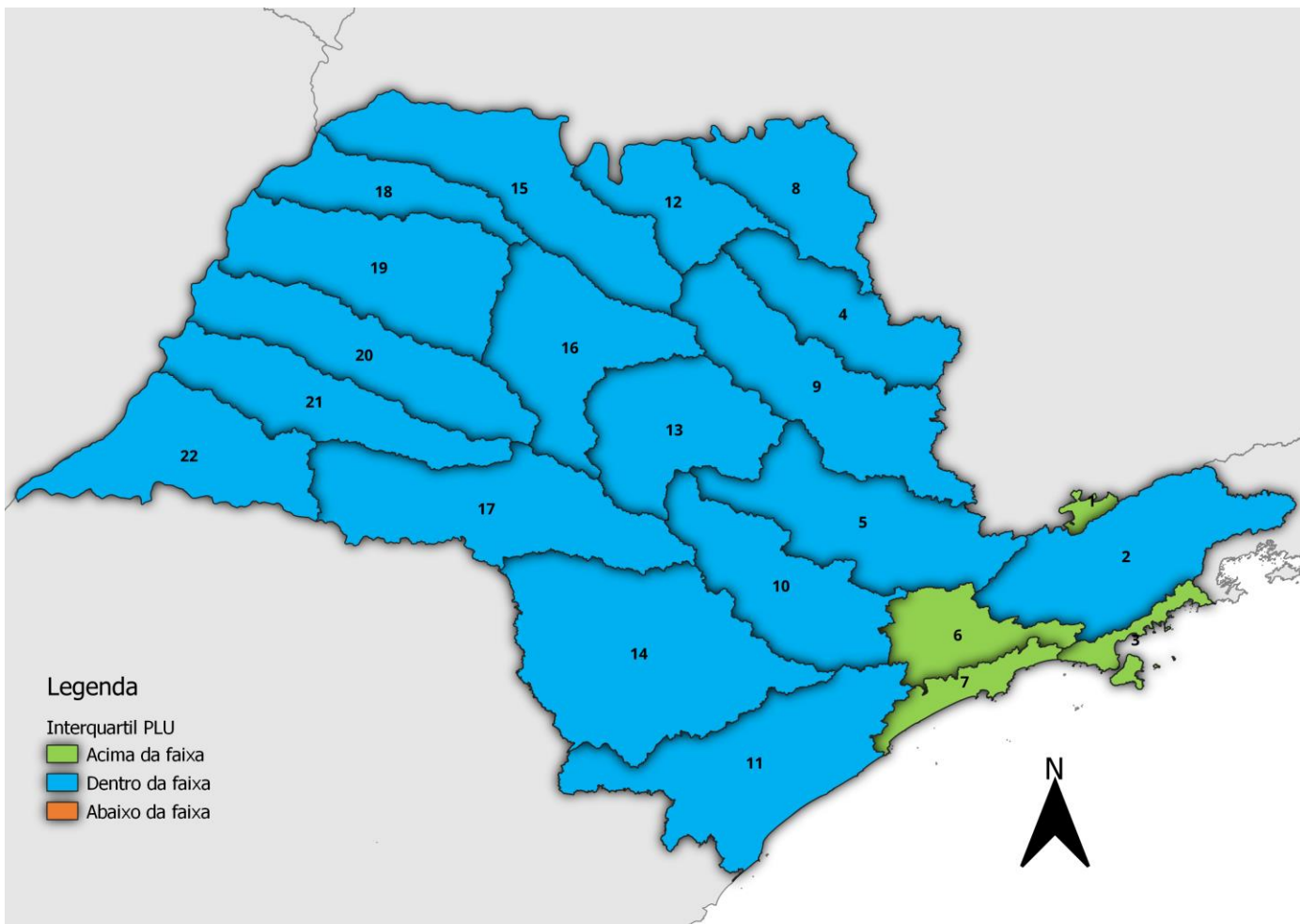


Análise de dispersão – intervalo interquartil

- Precipitação pluviométrica acumulada do mês de análise comparada com o intervalo interquartil, calculado a partir da série histórica de dados do produto MERGE/Inpe;
- Faixa interquartil: valores entre o primeiro quartil (25%) e o terceiro quartil (75%).



Análise de dispersão – intervalo interquartil



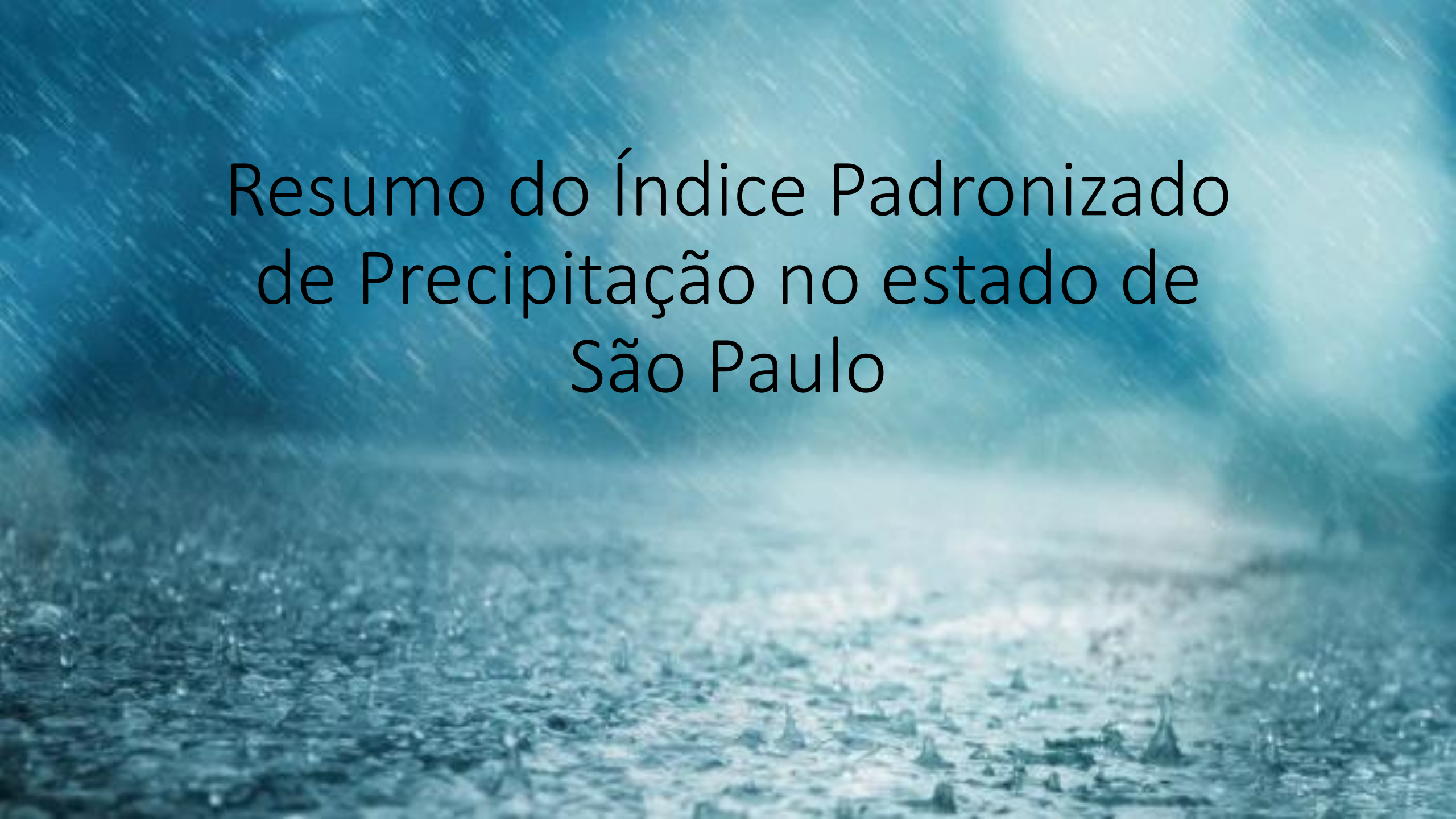
Mapa construído a partir da análise de quartis aplicada na série de dados de chuva de cada UGRHI.

Critérios:

Acima da faixa: valor de chuva acumulada acima do terceiro quartil (75%)

Dentro da faixa: valor de chuva dentro da faixa interquartil, ou seja, entre o primeiro quartil (25%) e o terceiro quartil (75%)

Abaixo da faixa: valor de chuva acumulada abaixo do primeiro quartil (25%)

A background image showing a cityscape with many buildings, viewed from a high angle, with rain falling diagonally across the entire scene. The rain is represented by many thin, white, slanted lines. The city below is somewhat blurred and has a blueish tint.

Resumo do Índice Padronizado de Precipitação no estado de São Paulo

Índice Padronizado de Precipitação

O Índice Padronizado de Precipitação (Standardized Precipitation Index - SPI) é uma ferramenta consistente de detecção e monitoramento probabilístico/padronizado de déficit ou excesso de precipitação para diferentes escalas temporais, possibilitando a comparação entre regiões com diferentes características climáticas. O índice é negativo para condições de déficit hídrico, nas quais a precipitação é inferior à média histórica (climatológica), e positivo para condições de excesso hídrico, que indicam precipitação superior à média histórica.

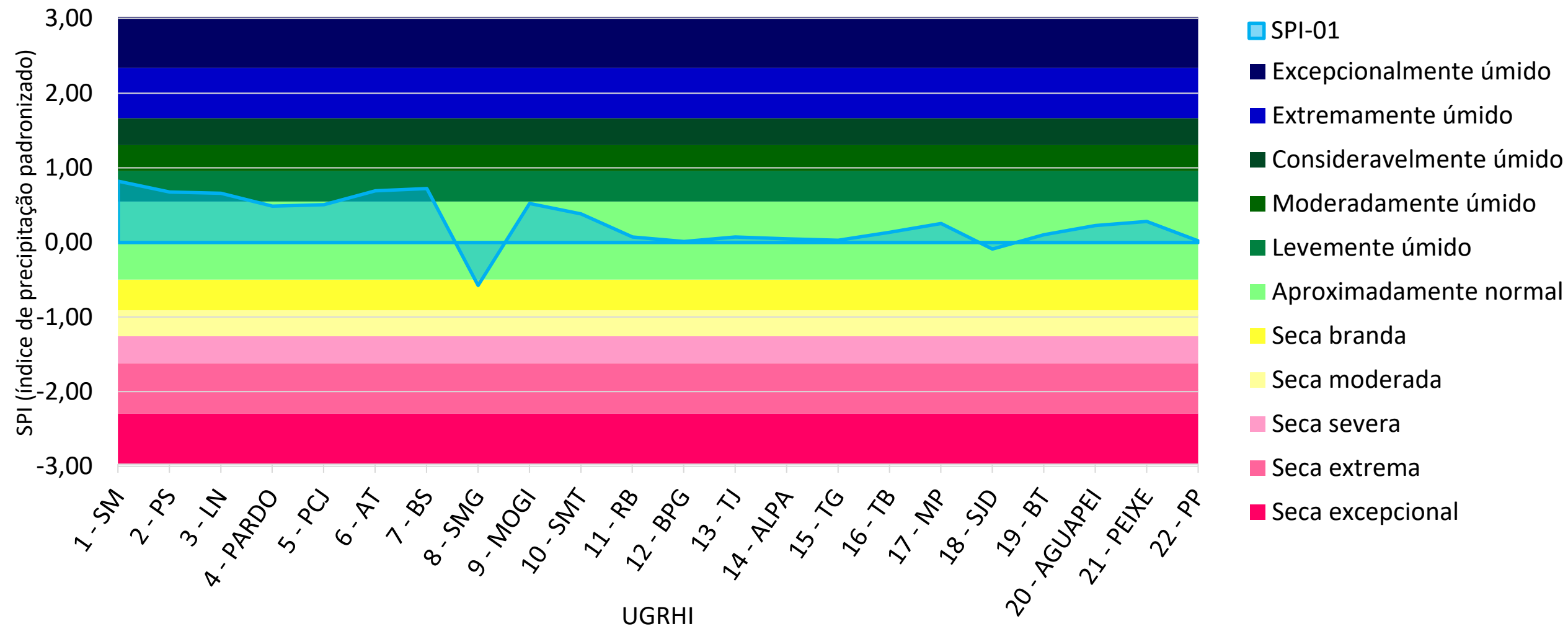
Sendo assim, o conhecimento do SPI pode fornecer informações importantes para a tomada de decisões quanto às medidas necessárias ao combate e a mitigação dos efeitos associados às secas e as enchentes.

- DM (Drought Magnitude): a magnitude da seca é a soma positiva do SPI para todos os meses dentro de um evento de seca, ou seja, meses consecutivos com $SPI < -0,524$.
- Semestre úmido: outubro a março
- Semestre seco: abril a setembro

Classificação SPI

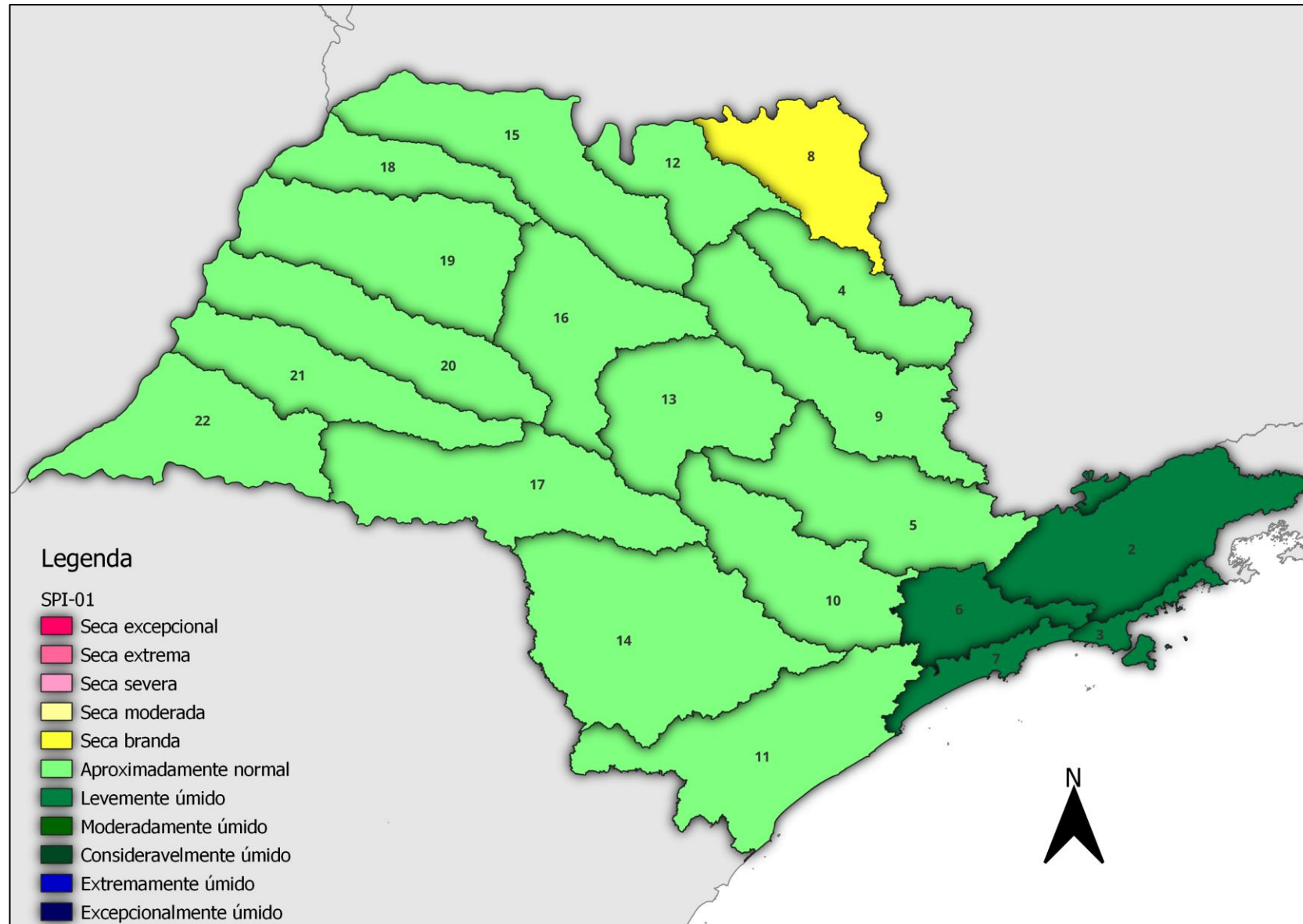
Valores	Condição	Probabilidade
$SPI \leq -2.326$	Seca excepcional	1,0%
$-2.326 < SPI \leq -1.645$	Seca extrema	4,0%
$-1.645 < SPI \leq -1.282$	Seca severa	5,0%
$-1.282 < SPI \leq -0.935$	Seca moderada	7,5%
$-0.935 < SPI \leq -0.524$	Seca branda	12,5%
$-0.524 < SPI < +0.524$	Aproximadamente normal	40,0%
$+0.524 \leq SPI < +0.935$	Levemente úmido	12,5%
$+0.935 \leq SPI < +1.282$	Moderadamente úmido	7,5%
$+1.282 \leq SPI < +1.645$	Consideravelmente úmido	5,0%
$+1.645 \leq SPI < +2.326$	Extremamente úmido	4,0%
$SPI \geq +2.326$	Excepcionalmente úmido	1,0%

SPI-01 em Agosto de 2024

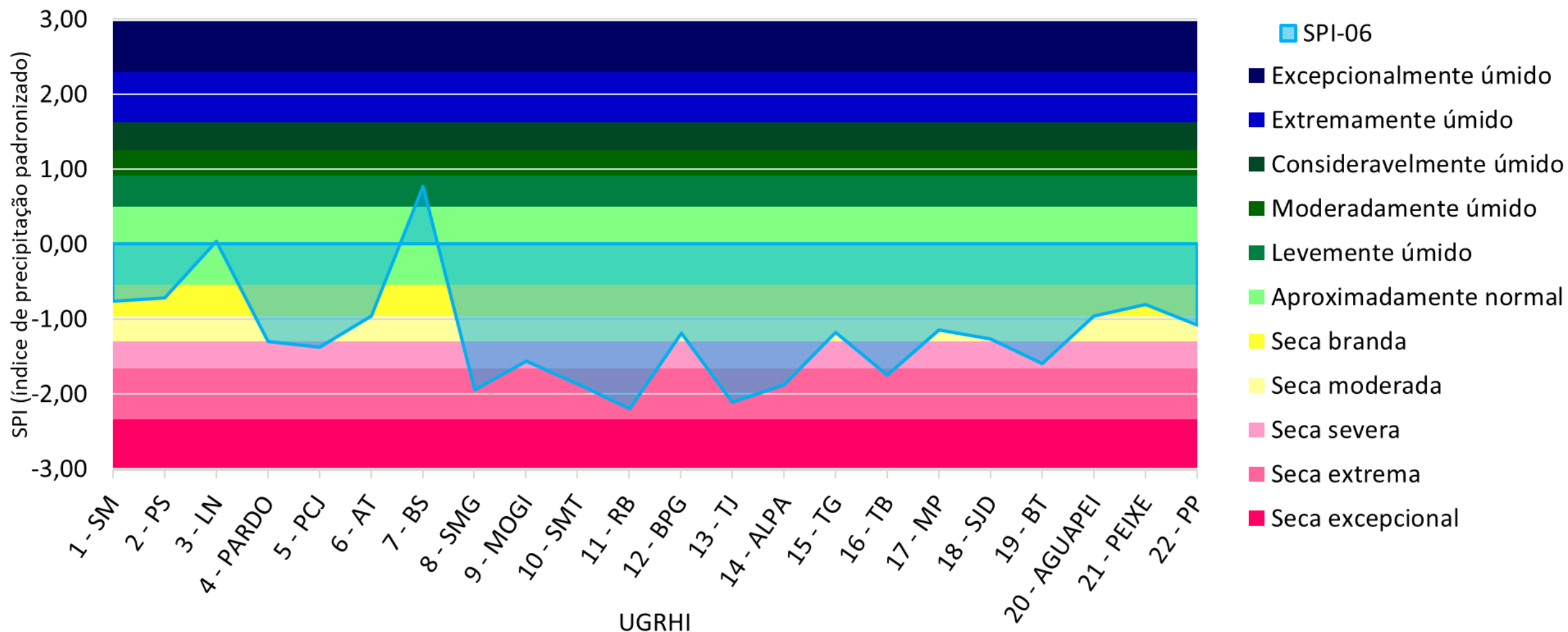


SPI-01 se baseia na análise dos dados de precipitação do mês de Agosto de 2024

Mapa SPI-01 em Agosto de 2024

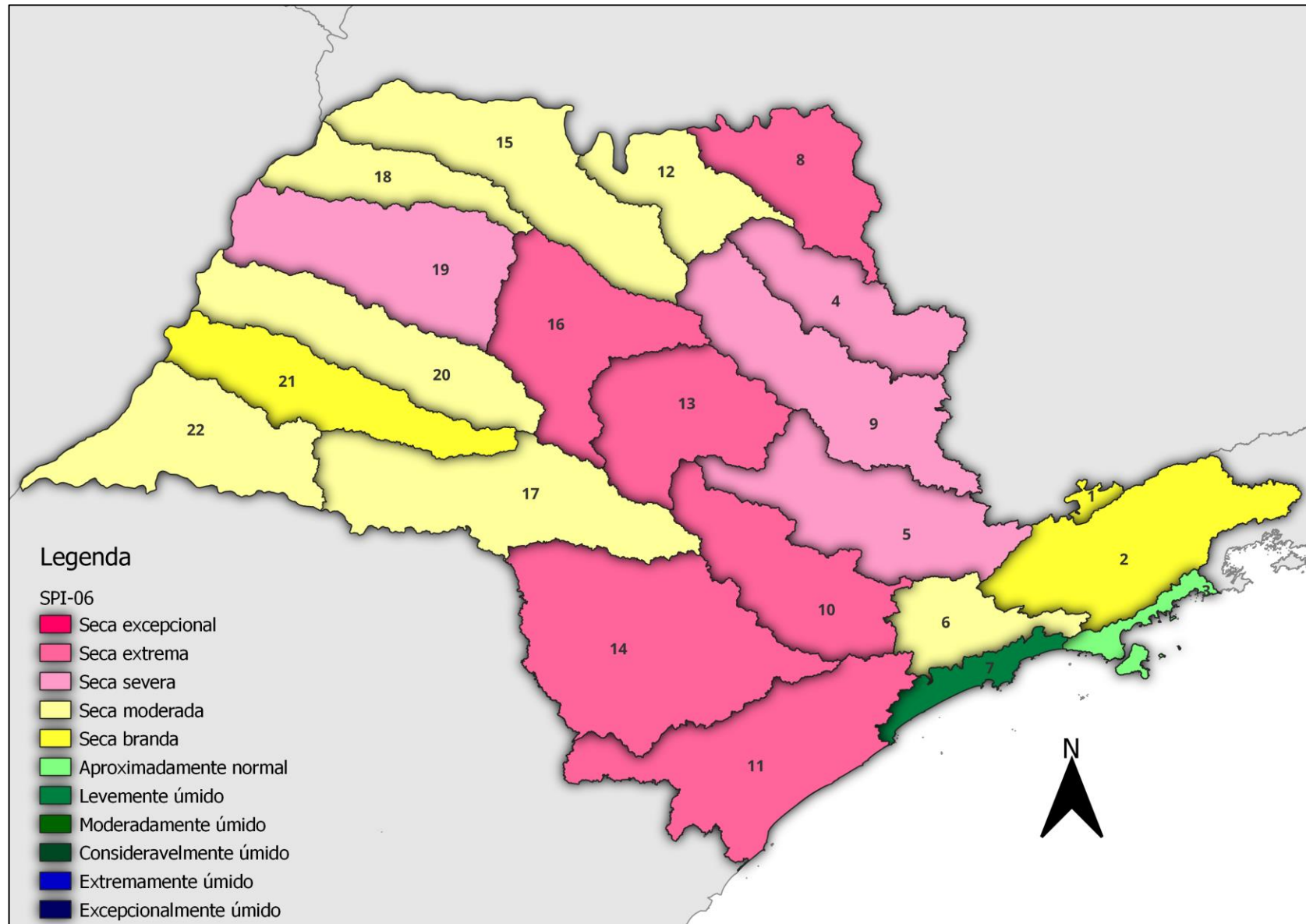


SPI-06 em Agosto de 2024

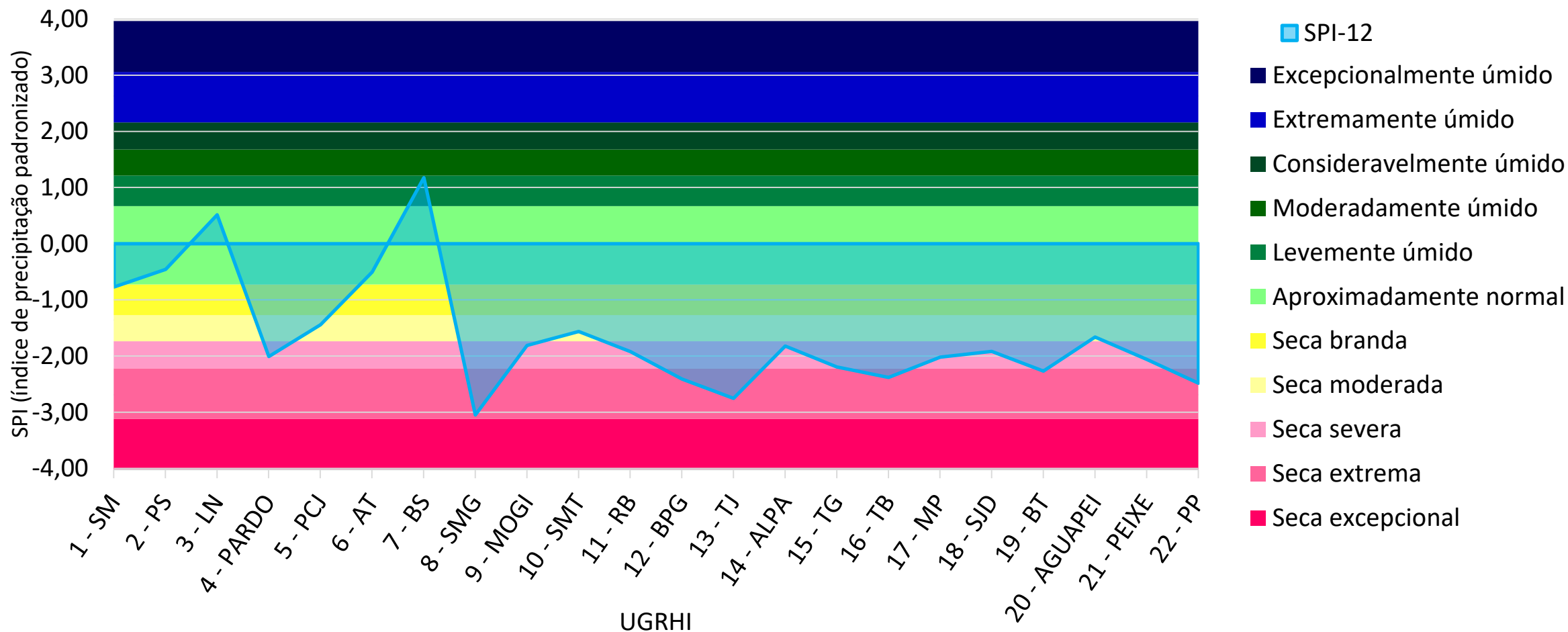


SPI-06 se baseia na análise dos dados de precipitação dos últimos 06 meses.

Mapa SPI-06 em Agosto de 2024

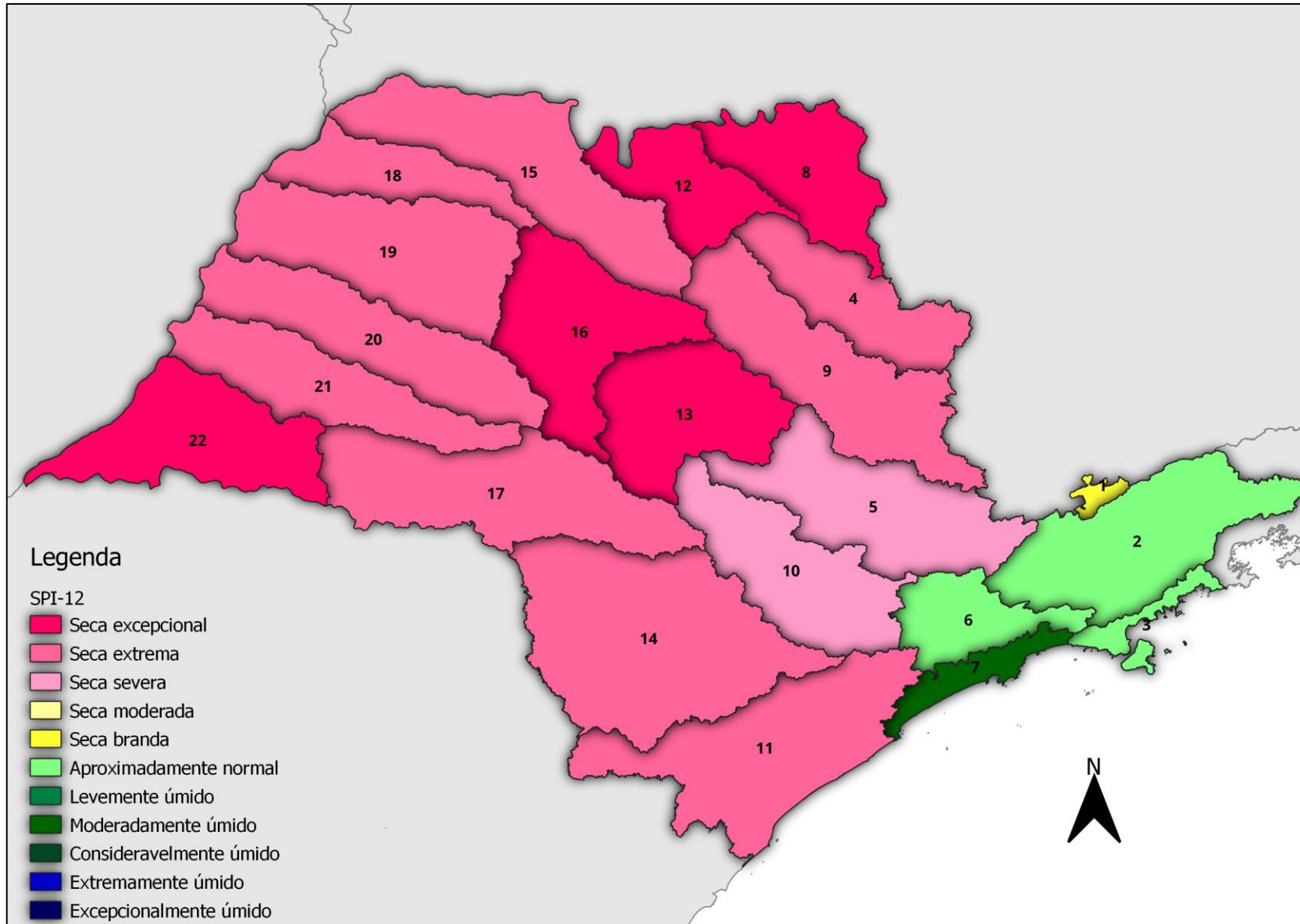


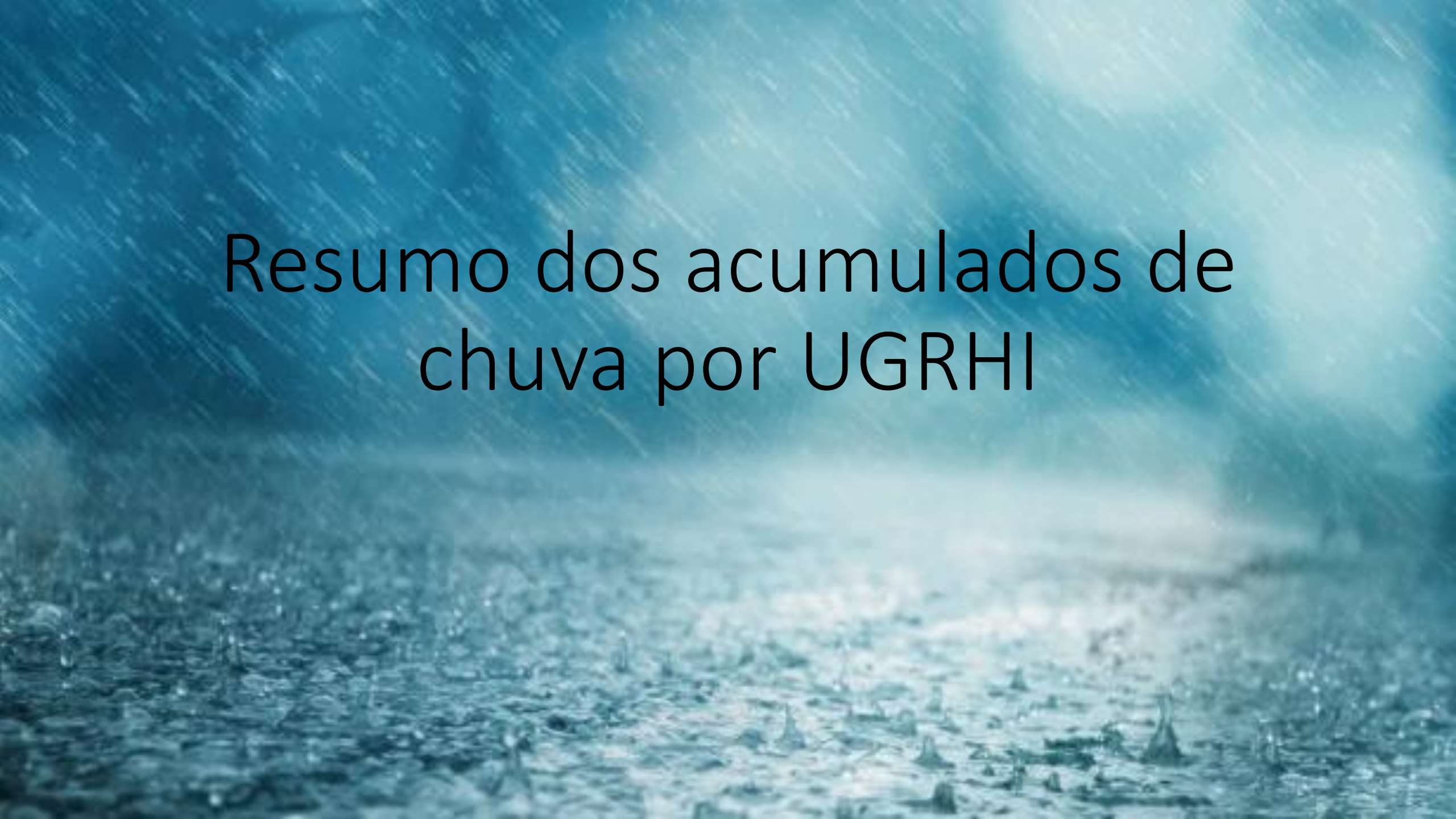
SPI-12 em Agosto de 2024



SPI-12 se baseia na análise dos dados de precipitação dos últimos 12 meses (escala anual).

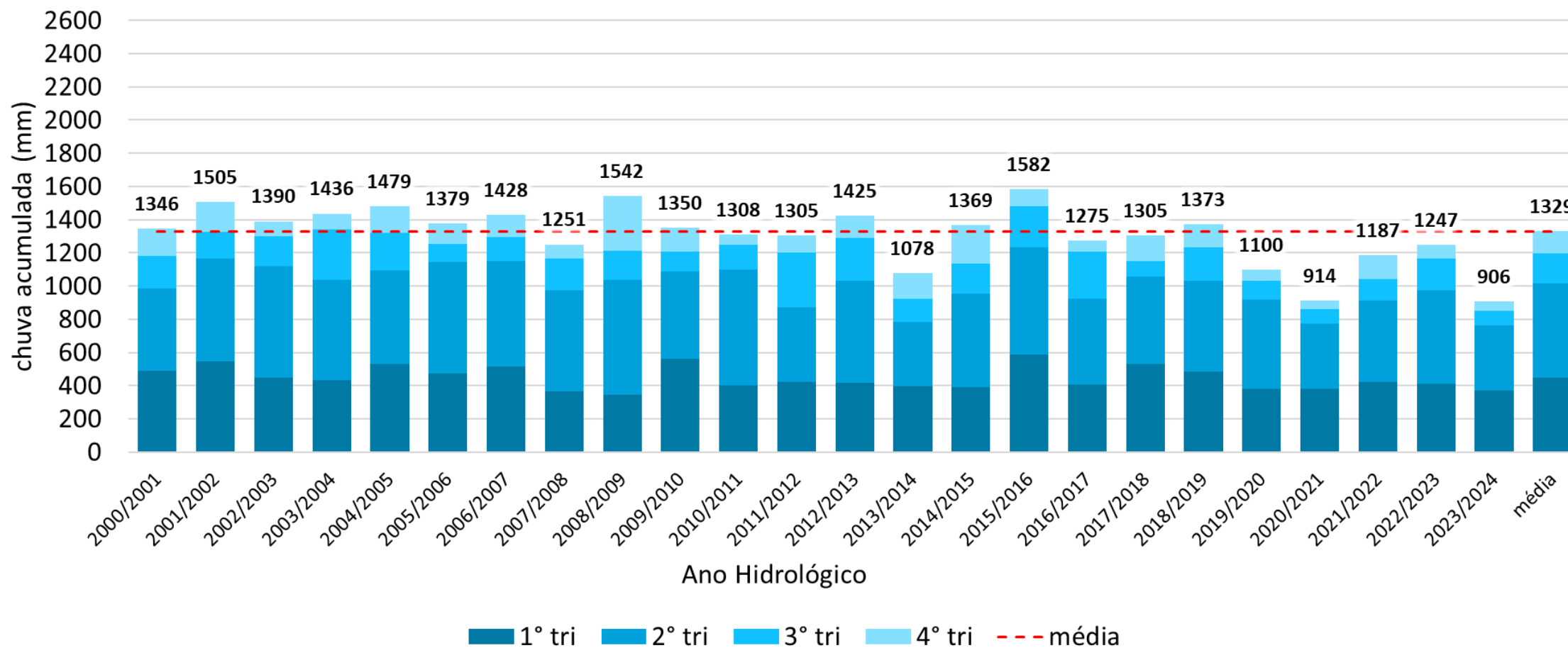
Mapa SPI-12 em Agosto de 2024



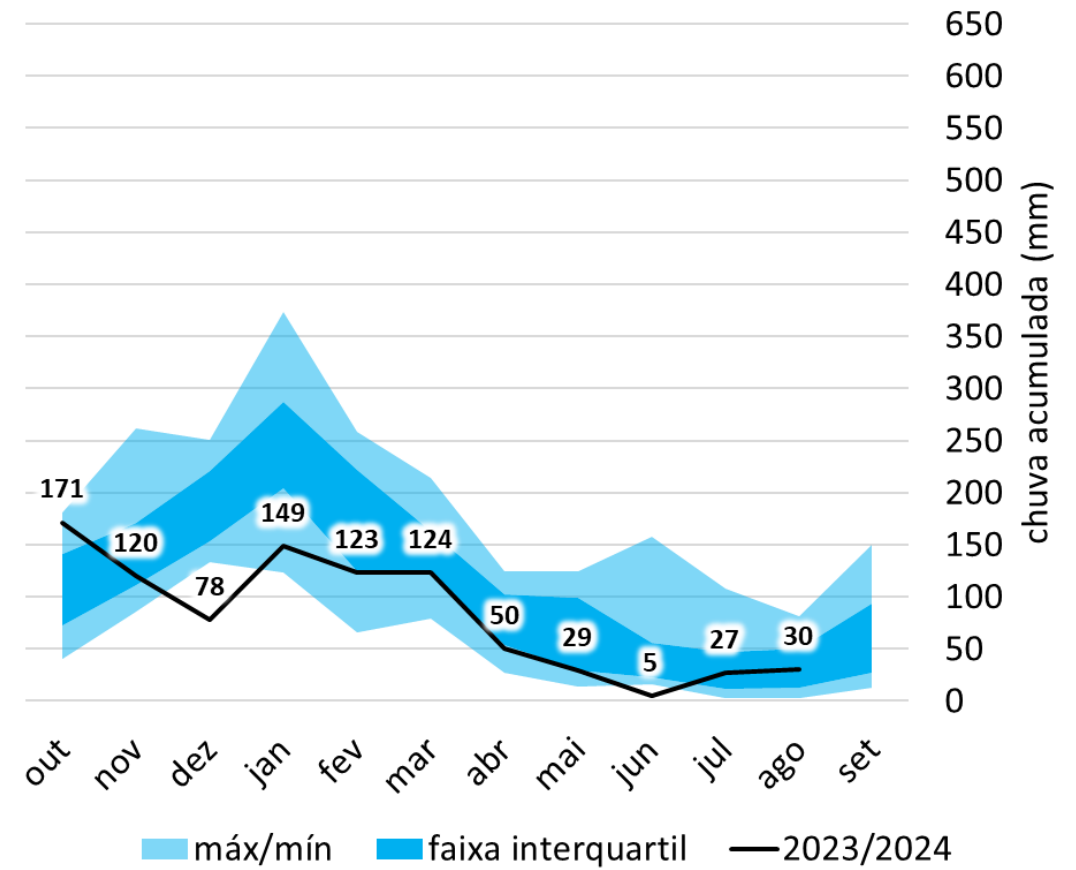
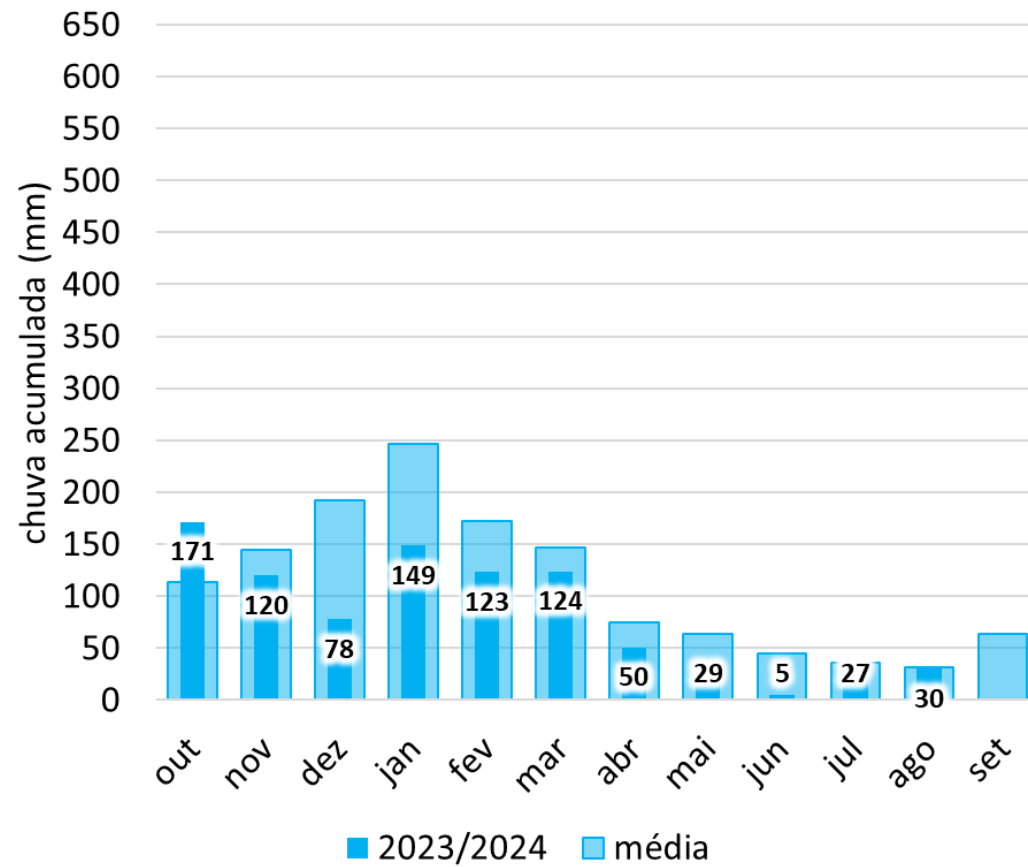
A background image showing rain falling on a city street. The rain is captured as many diagonal white streaks against a blurred background of city buildings and a street. The overall color palette is cool, with various shades of blue and teal.

Resumo dos acumulados de chuva por UGRHI

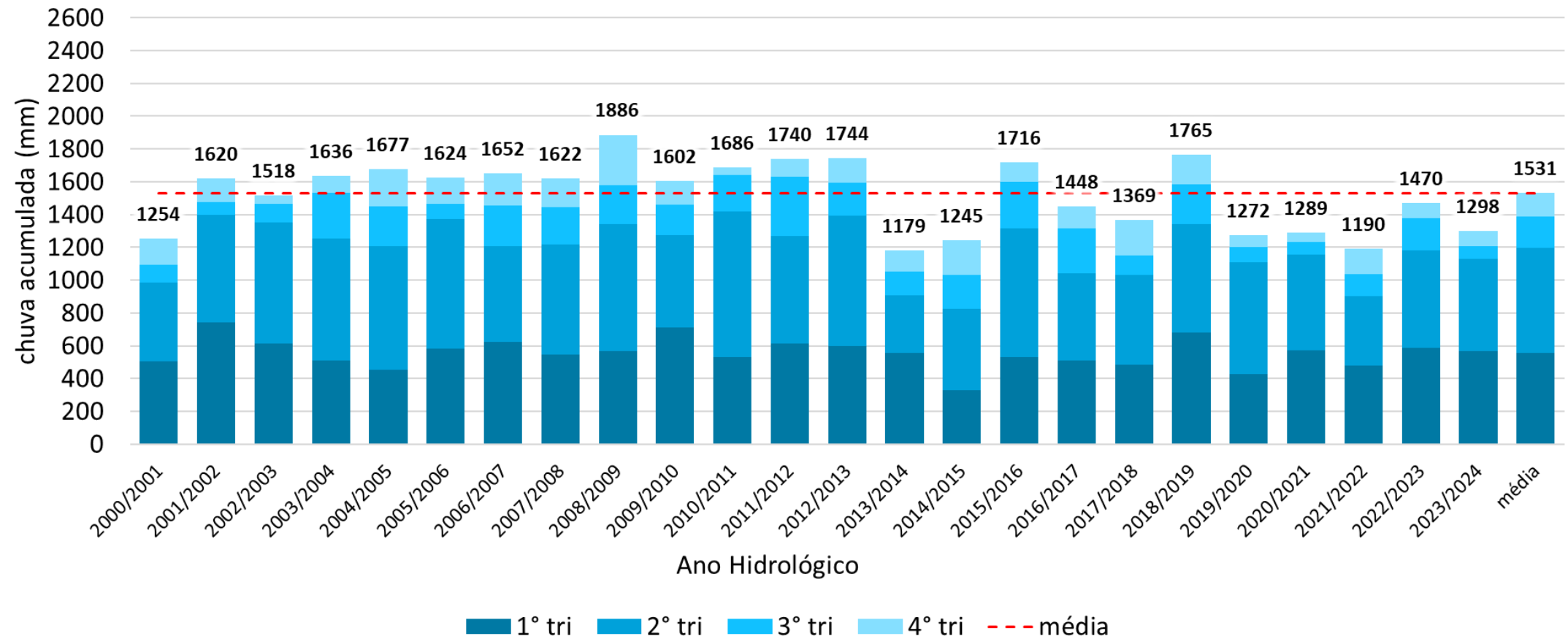
Estado de São Paulo



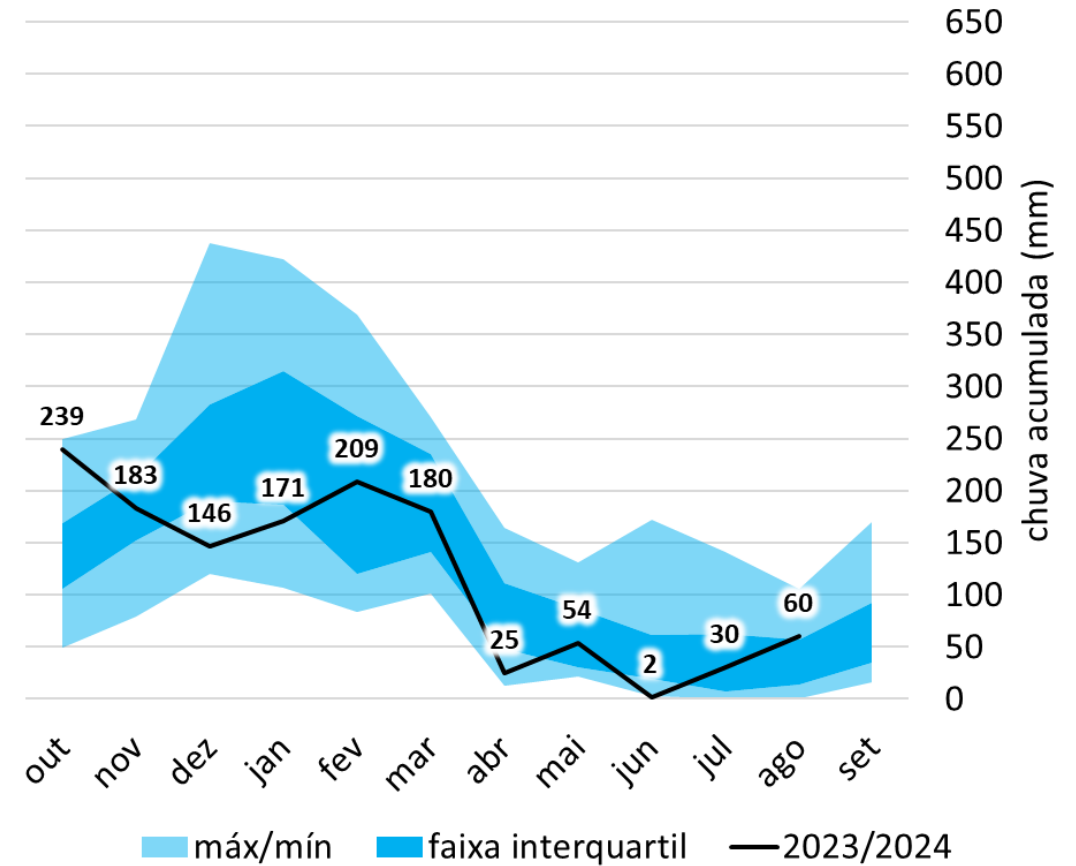
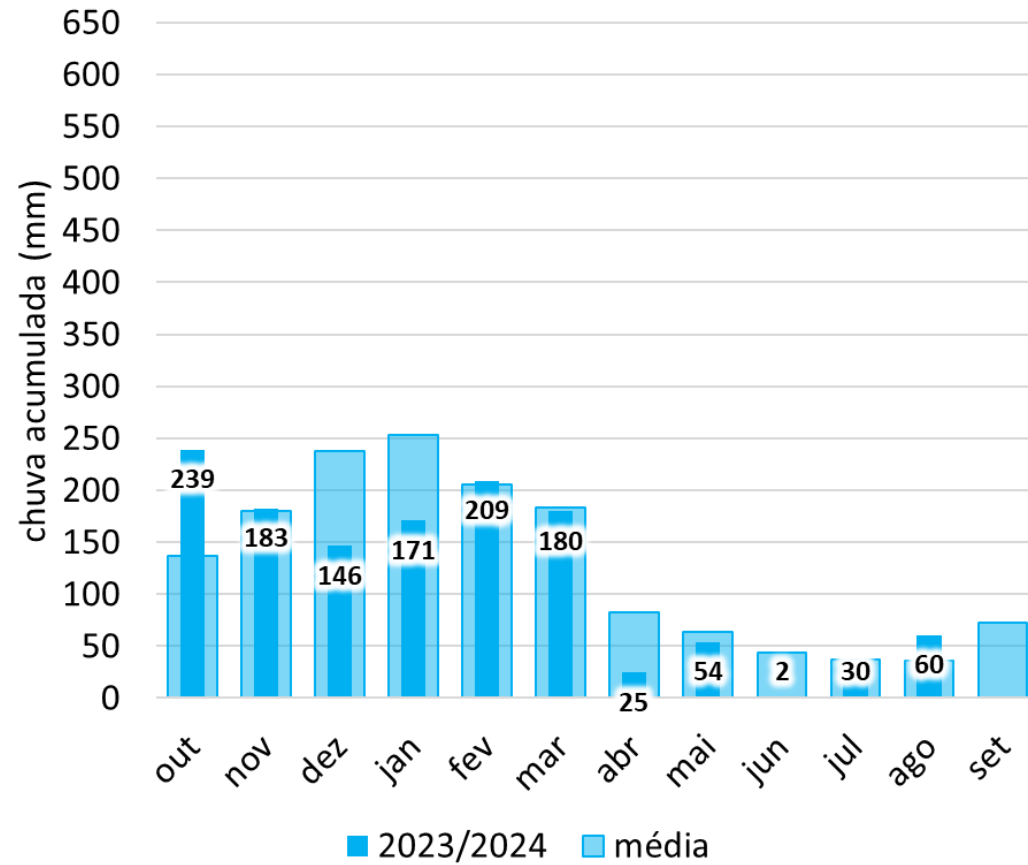
Estado de São Paulo



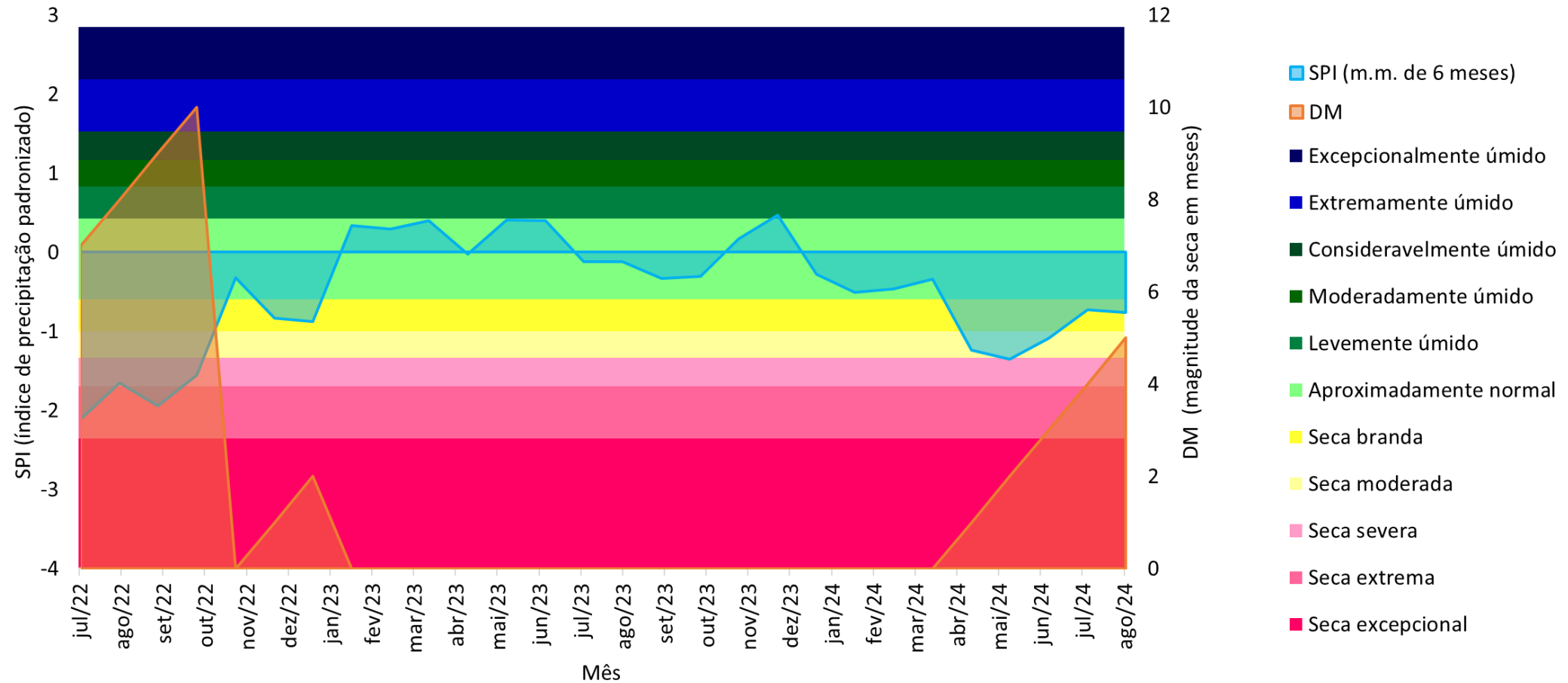
UGRHI 1 - Mantiqueira



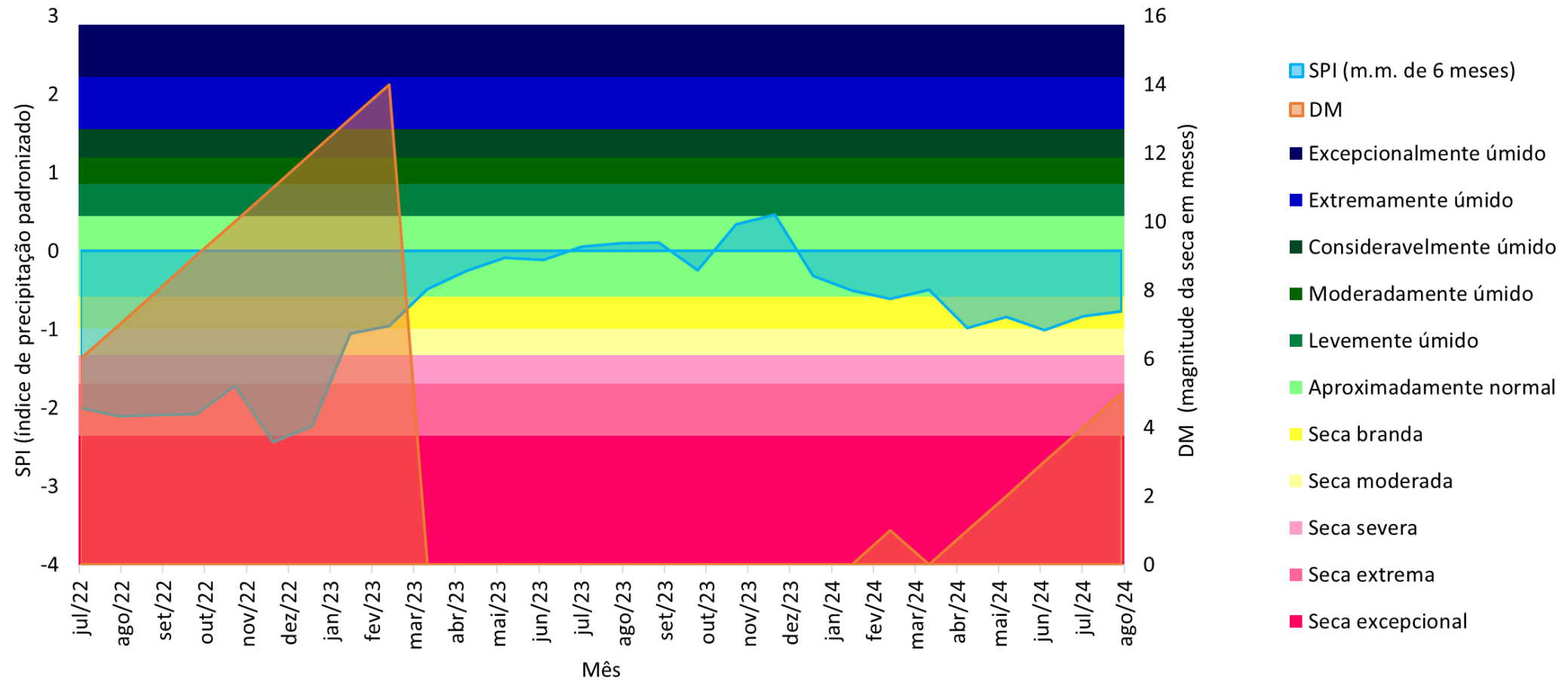
UGRHI 1 - Mantiqueira



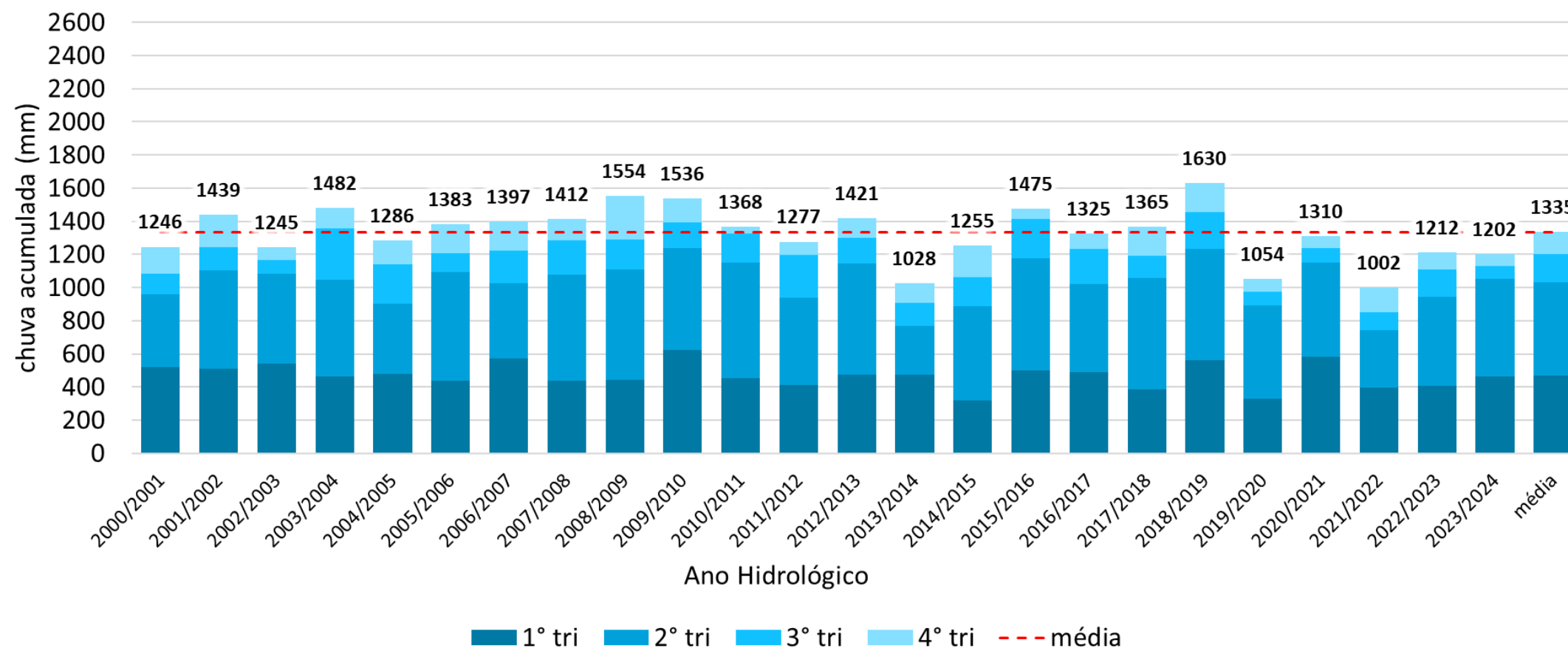
UGRHI 1 – Mantiqueira | SPI-06



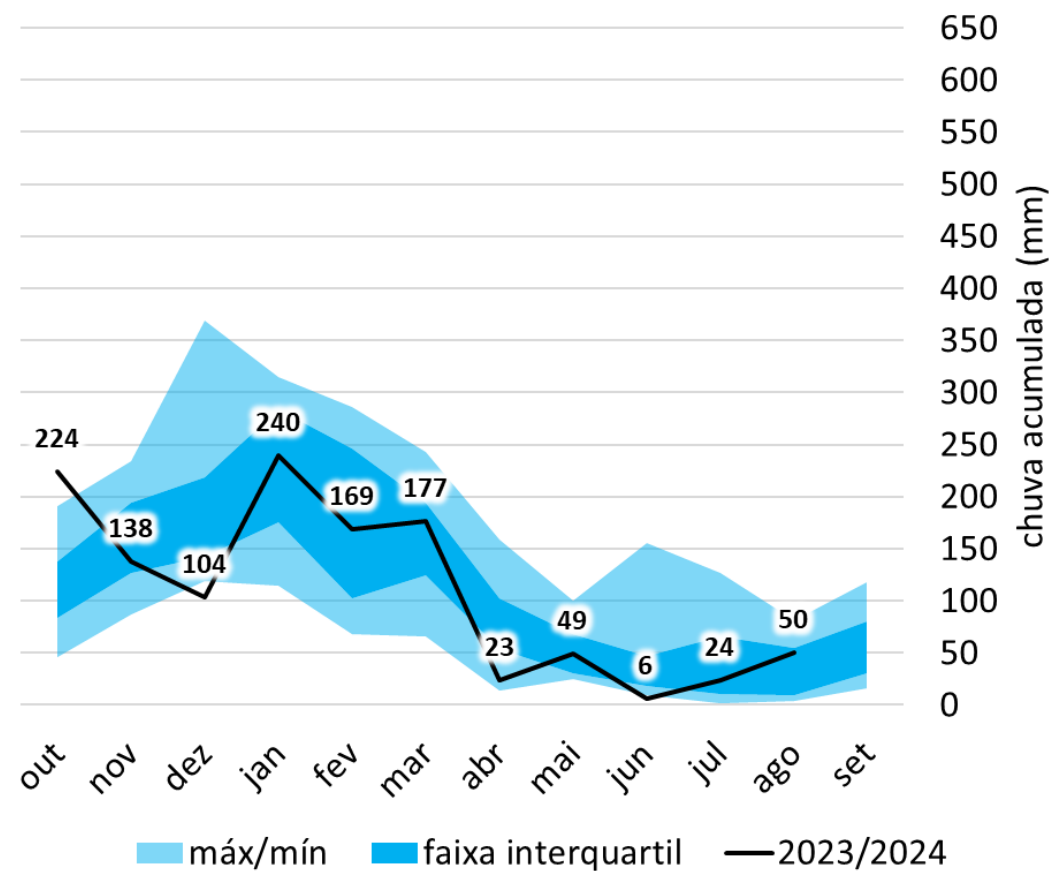
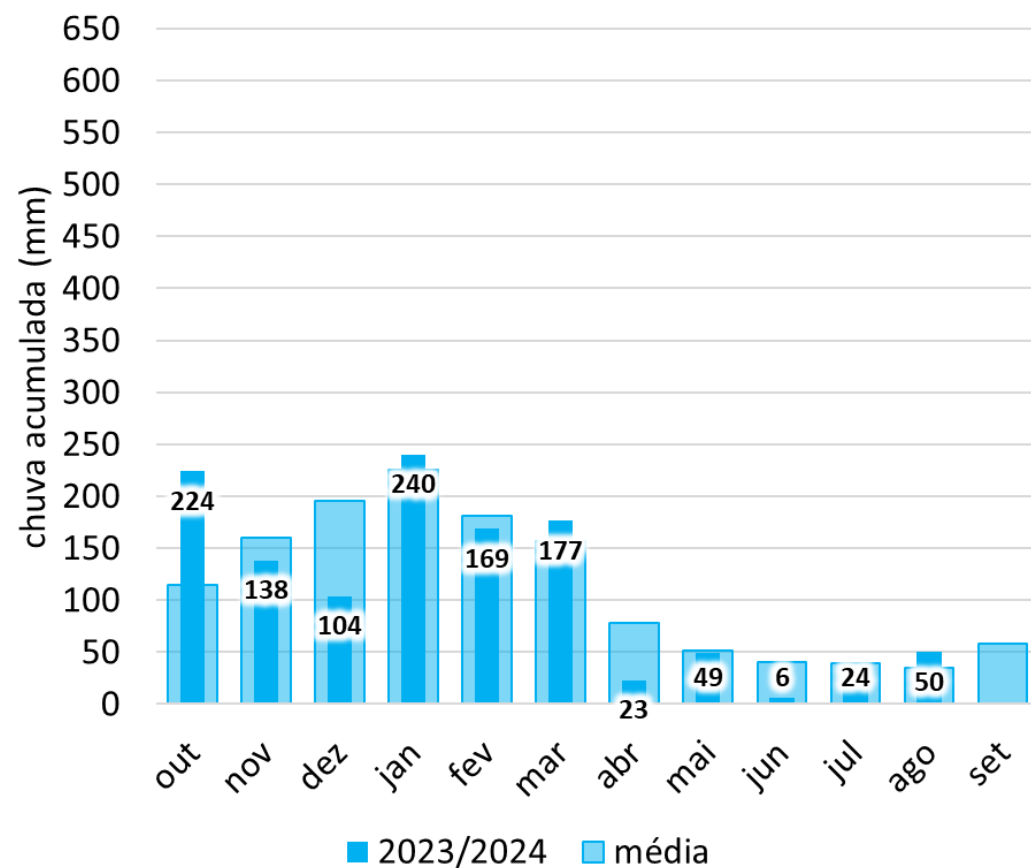
UGRHI 1 – Mantiqueira | SPI-12



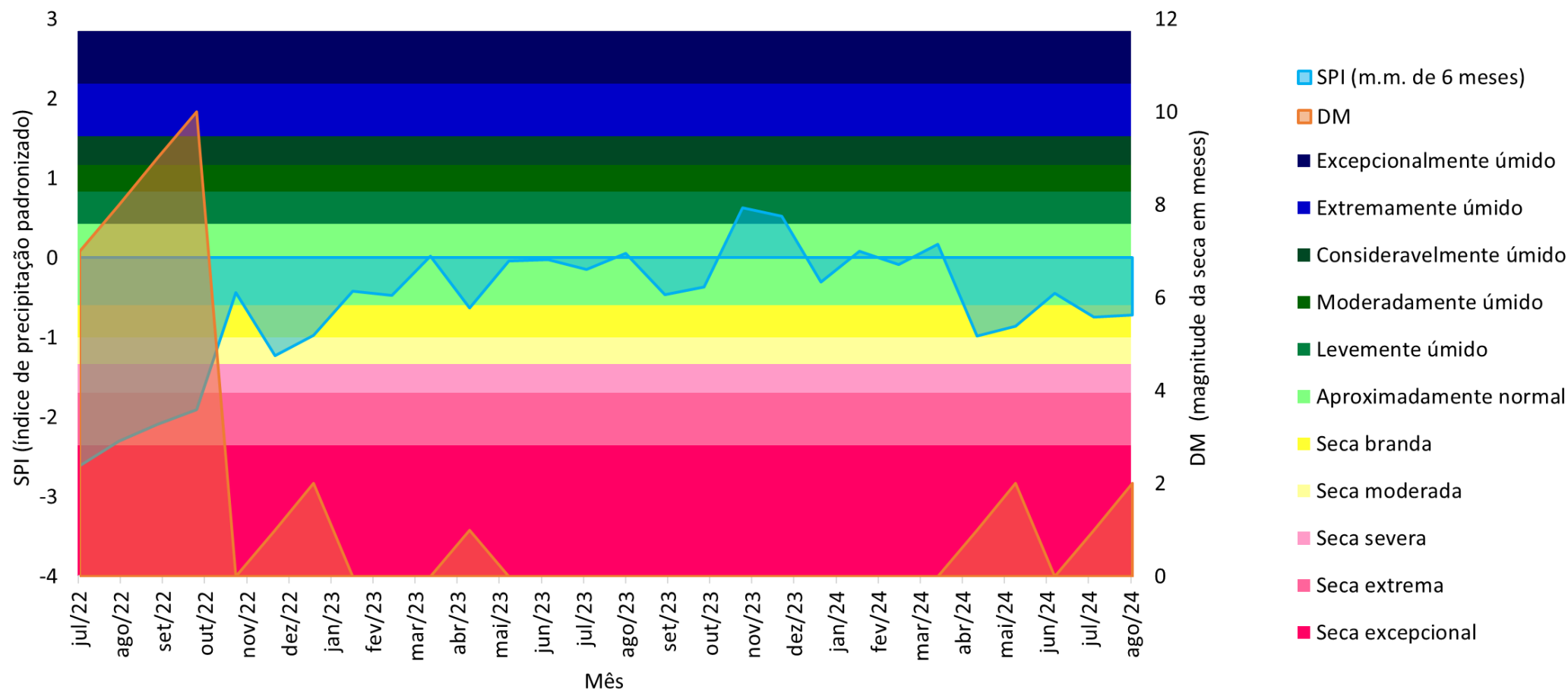
UGRHI 2 - Paraíba do Sul



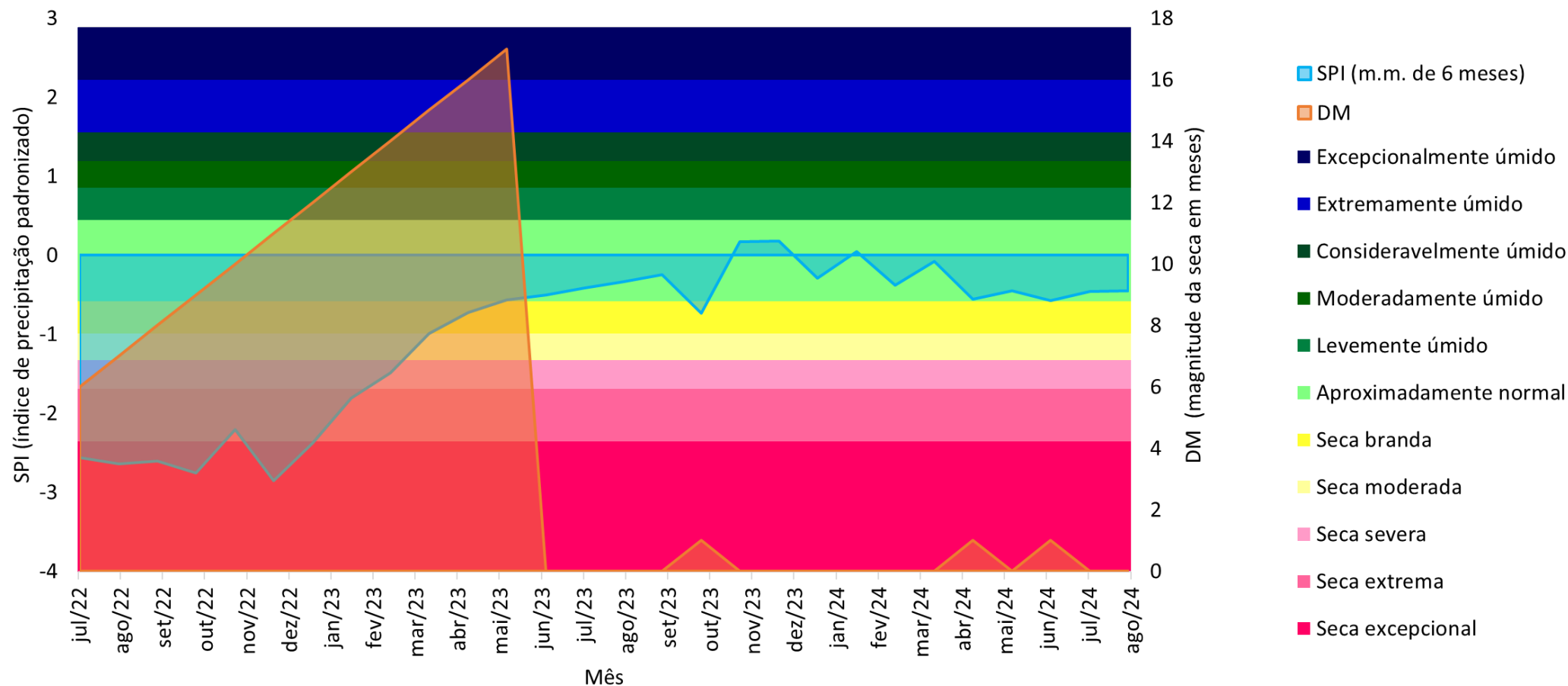
UGRHI 2 - Paraíba do Sul



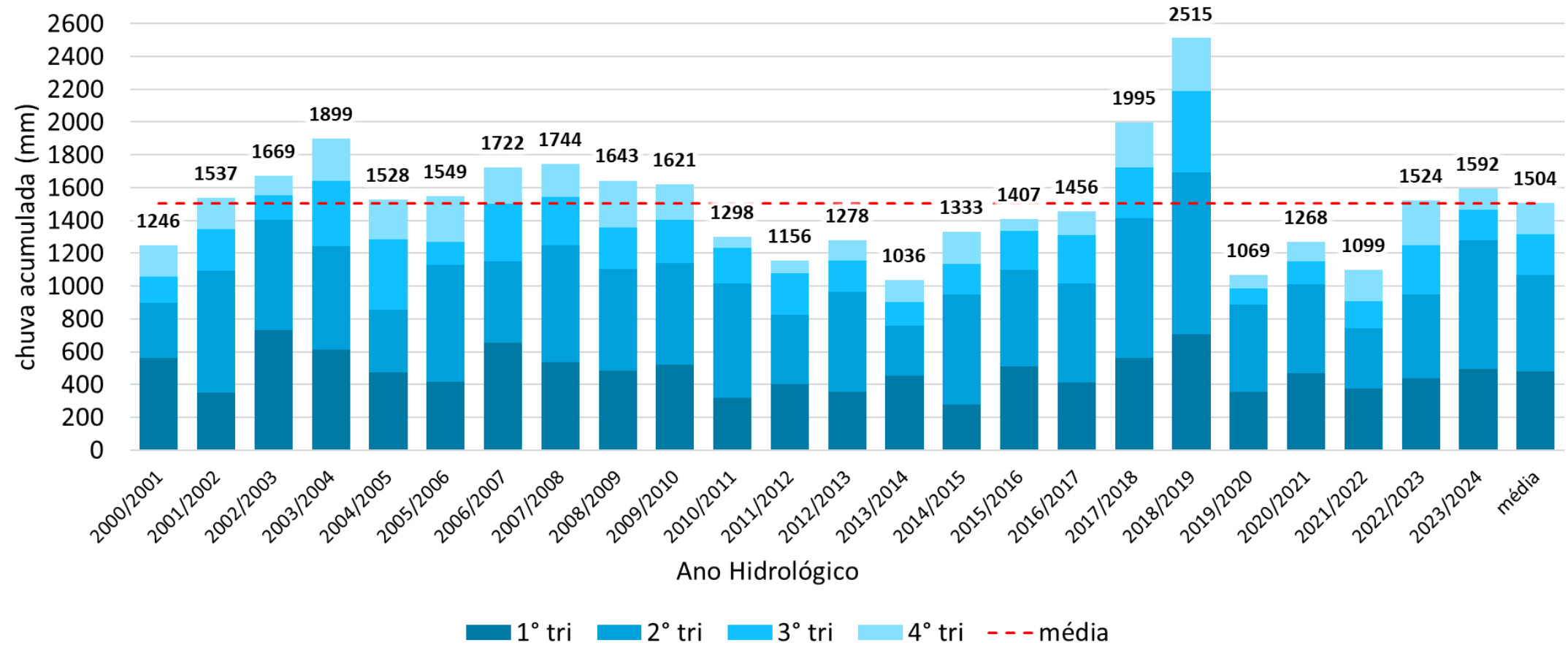
UGRHI 2 - Paraíba do Sul | SPI-06



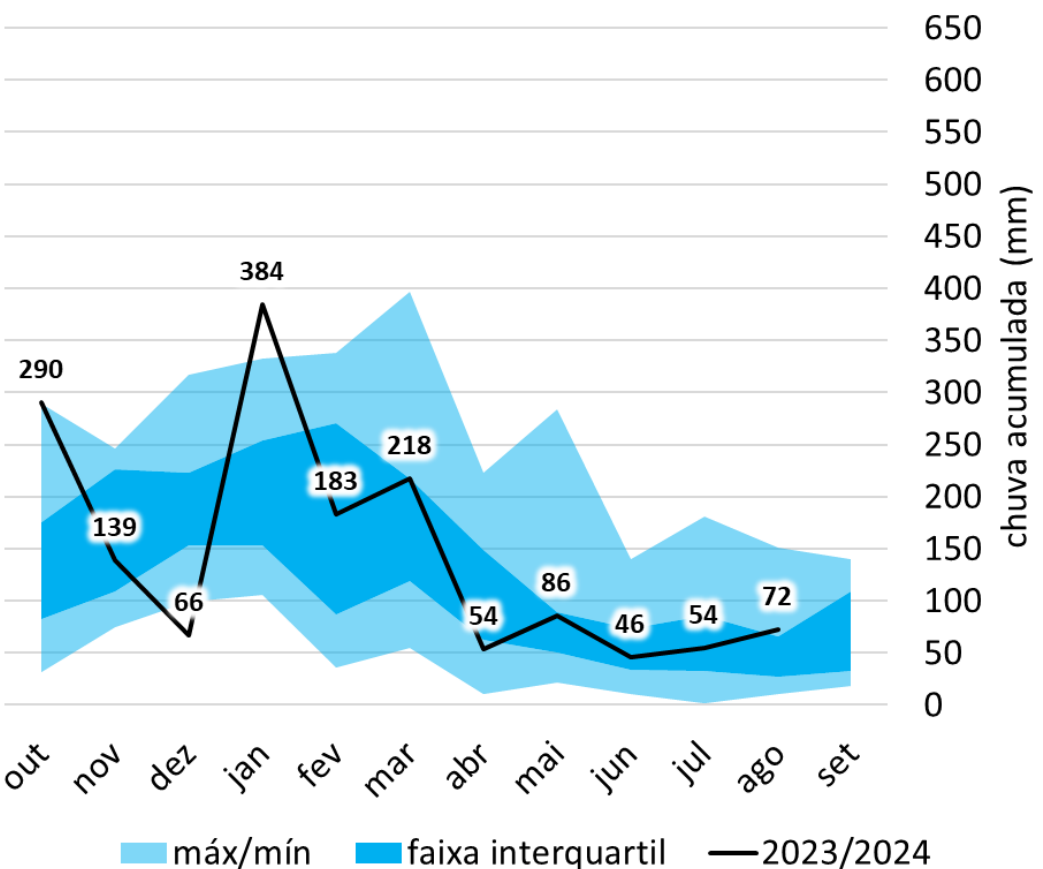
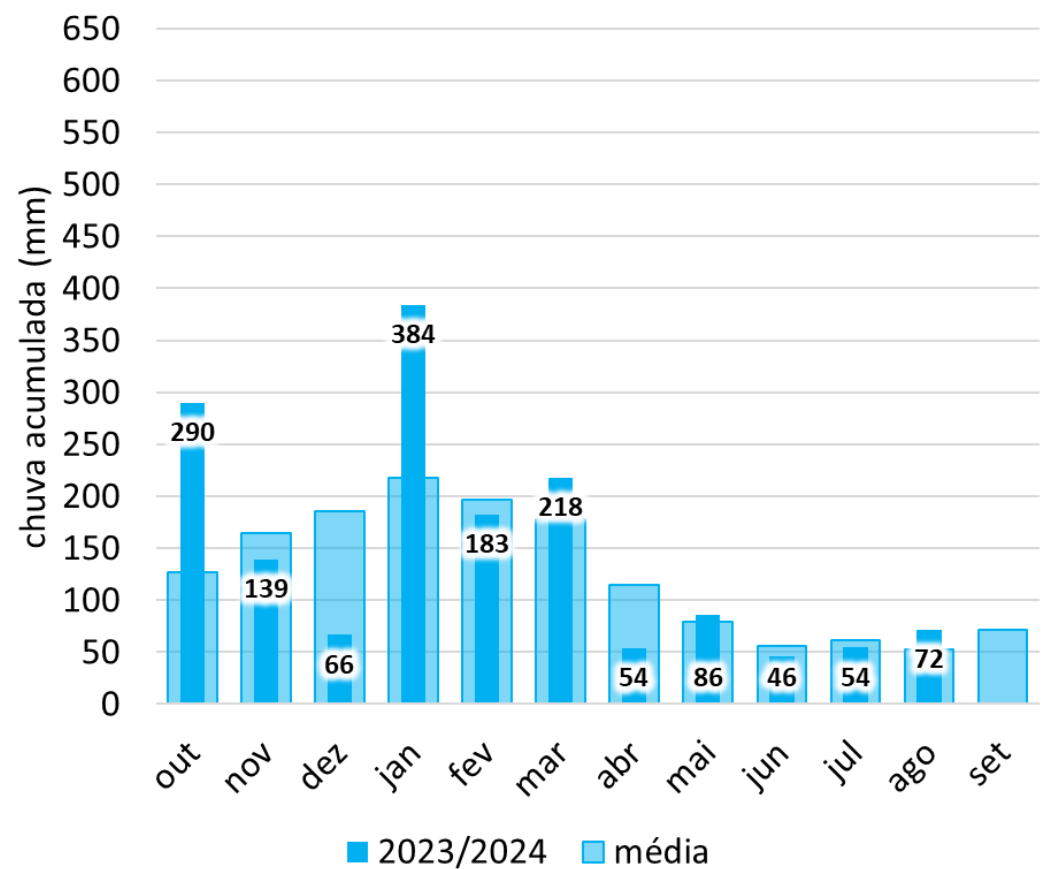
UGRHI 2 - Paraíba do Sul | SPI-12



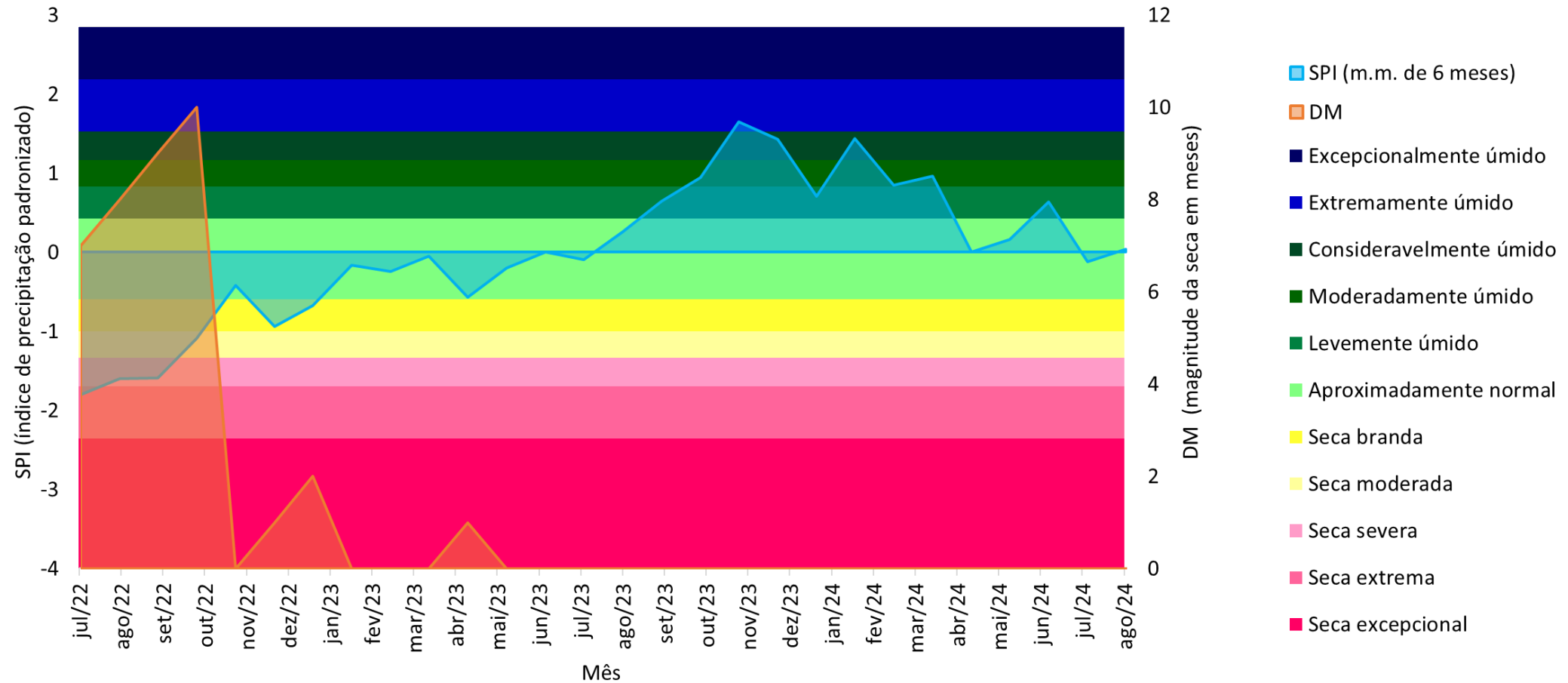
UGRHI 3 - Litoral Norte



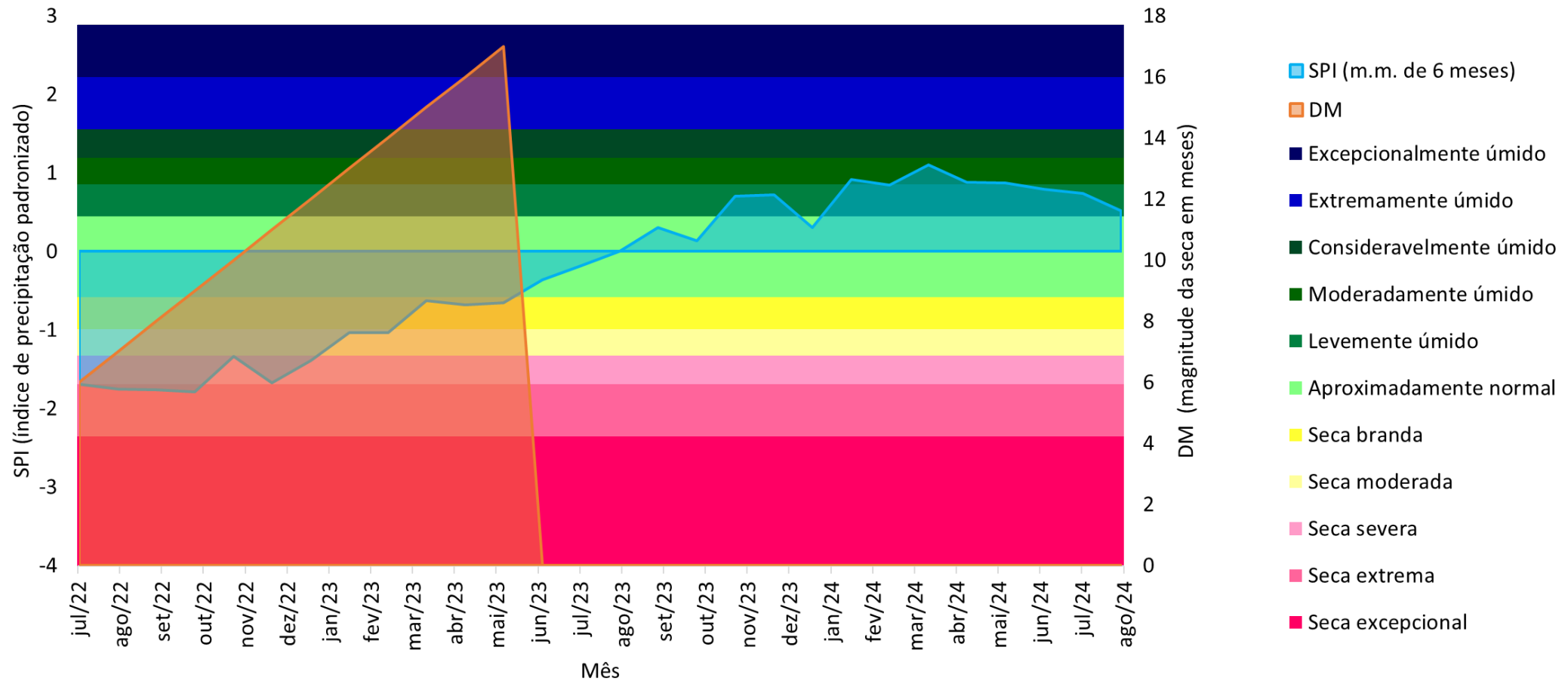
UGRHI 3 - Litoral Norte



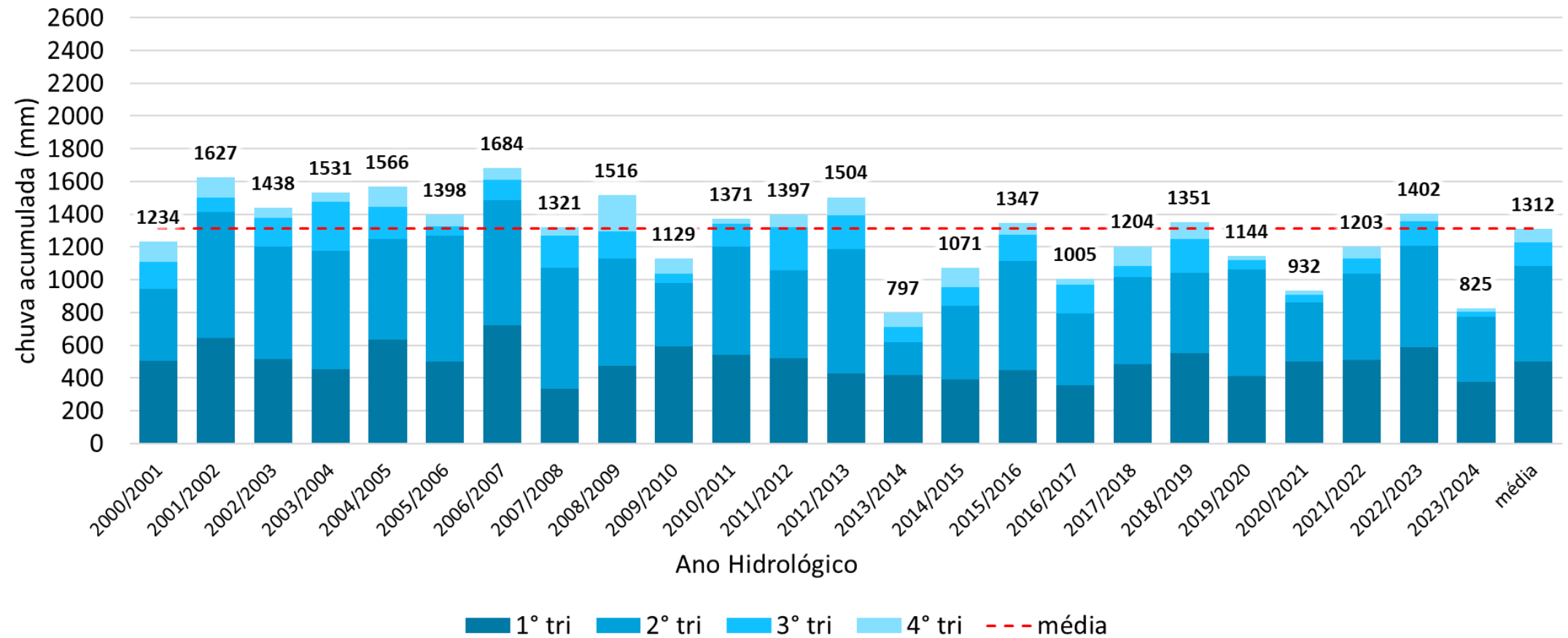
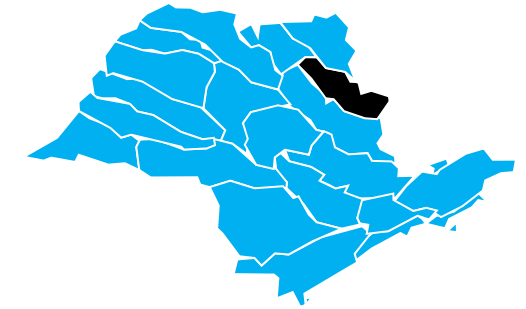
UGRHI 3 - Litoral Norte | SPI-06



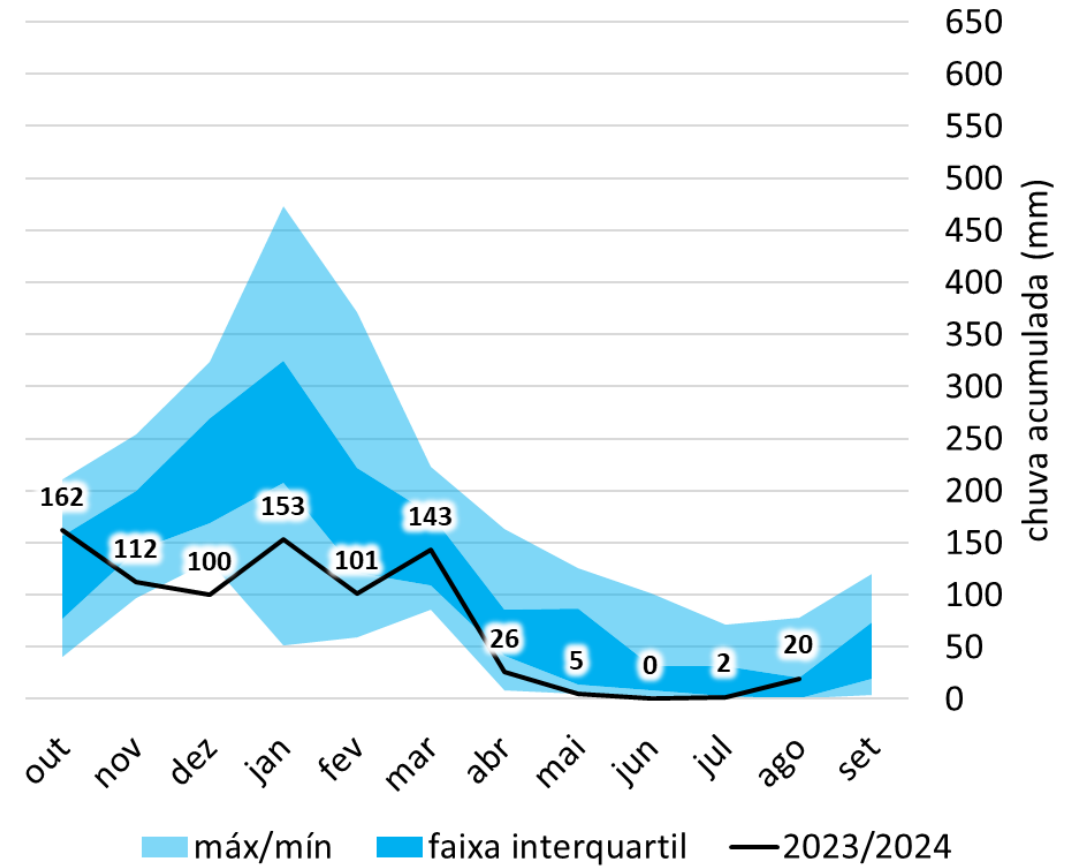
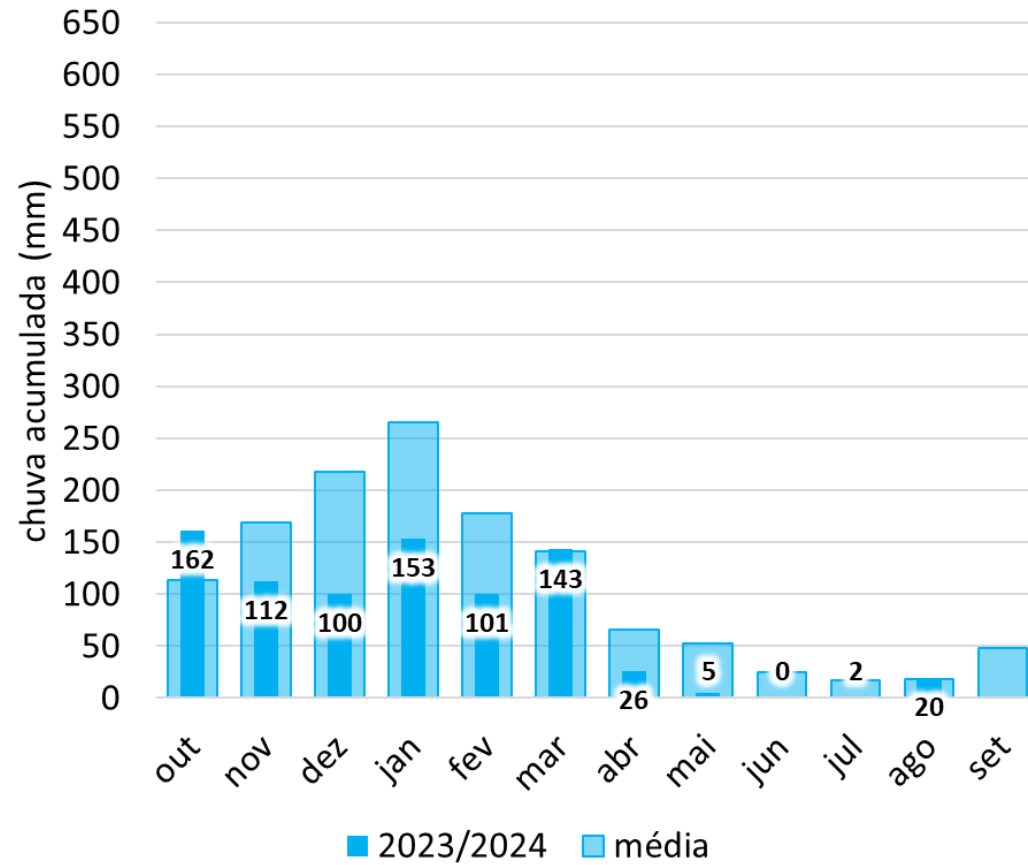
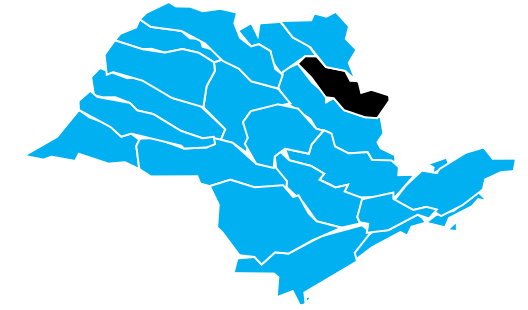
UGRHI 3 - Litoral Norte | SPI-12



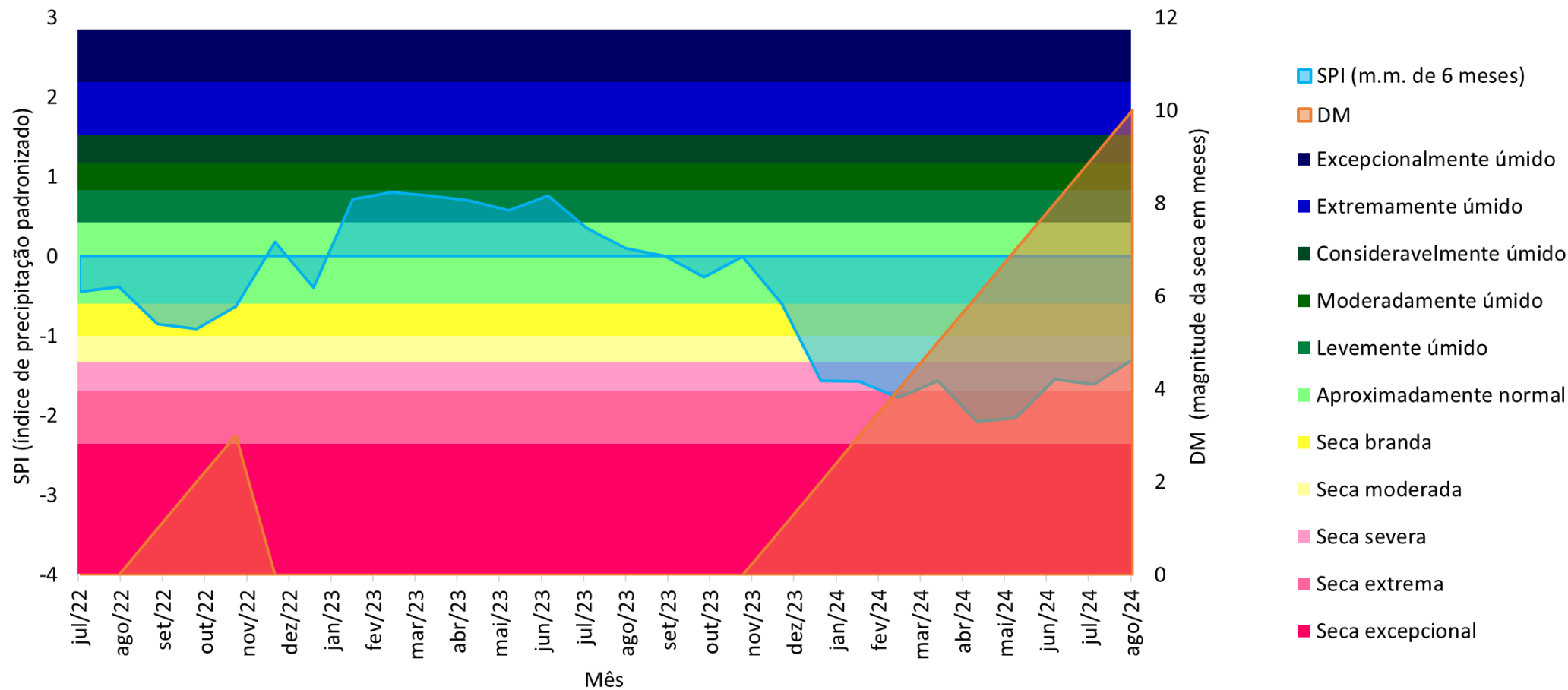
UGRHI 4 - Pardo



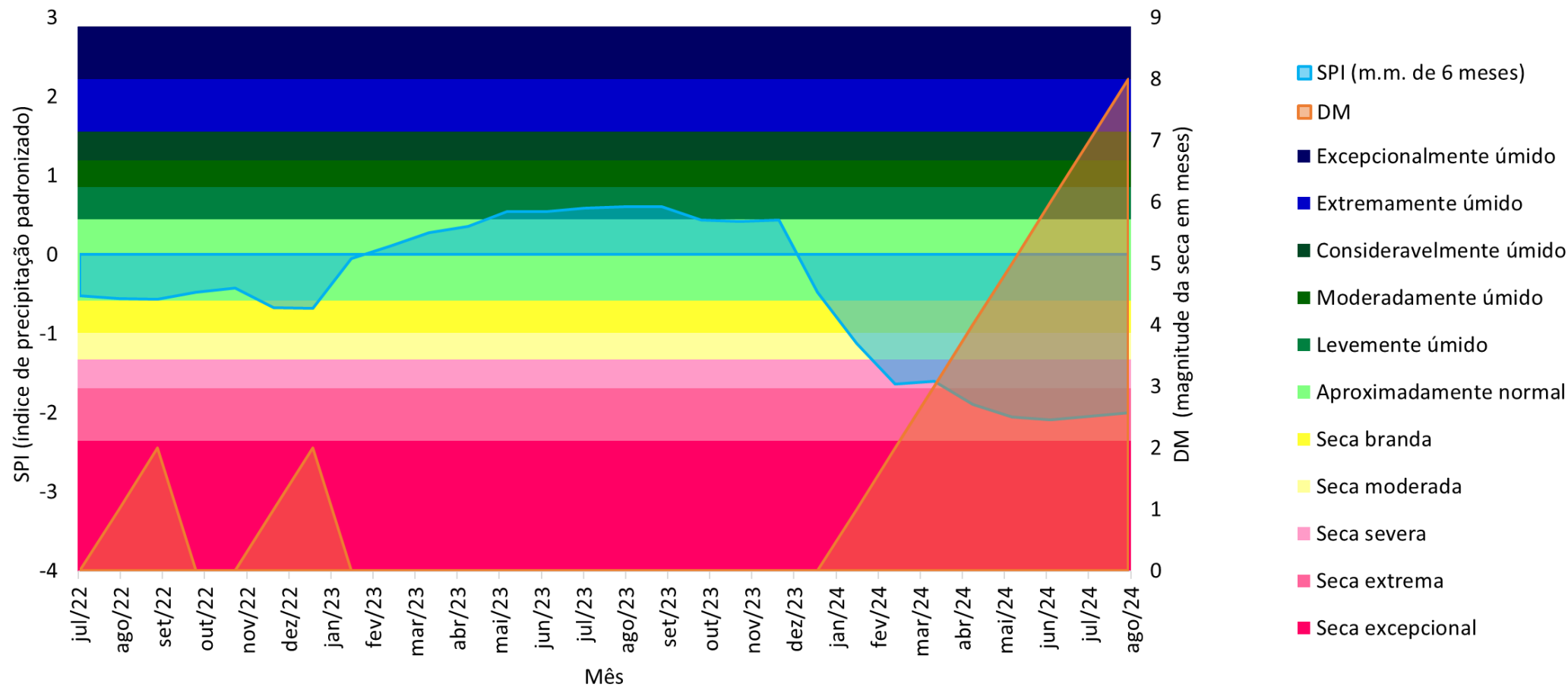
UGRHI 4 - Pardo



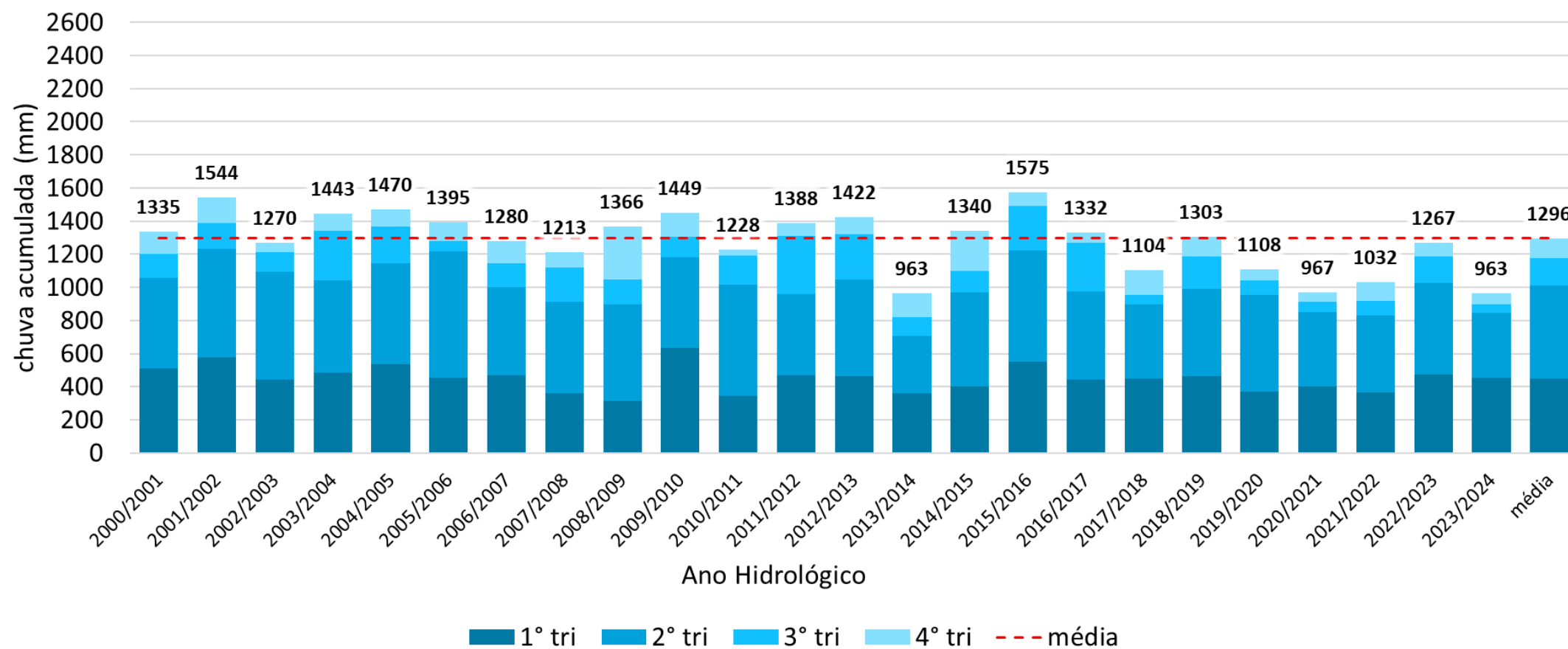
UGRHI 4 - Pardo | SPI-06



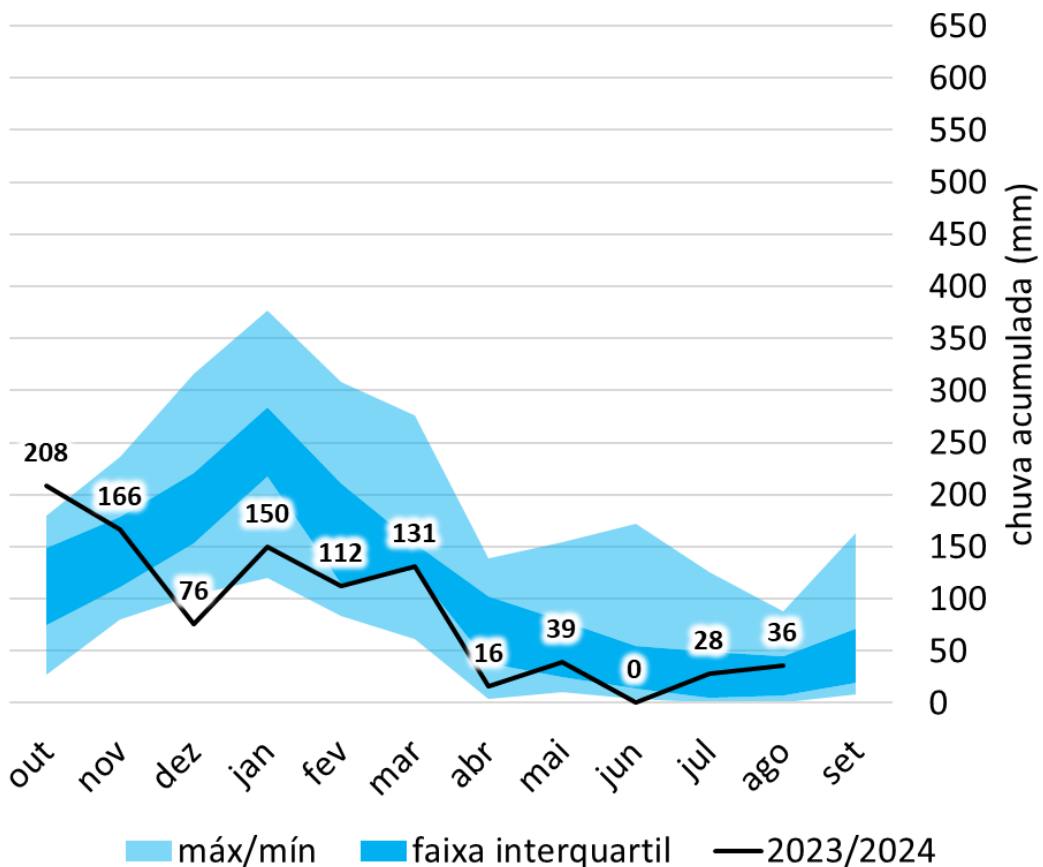
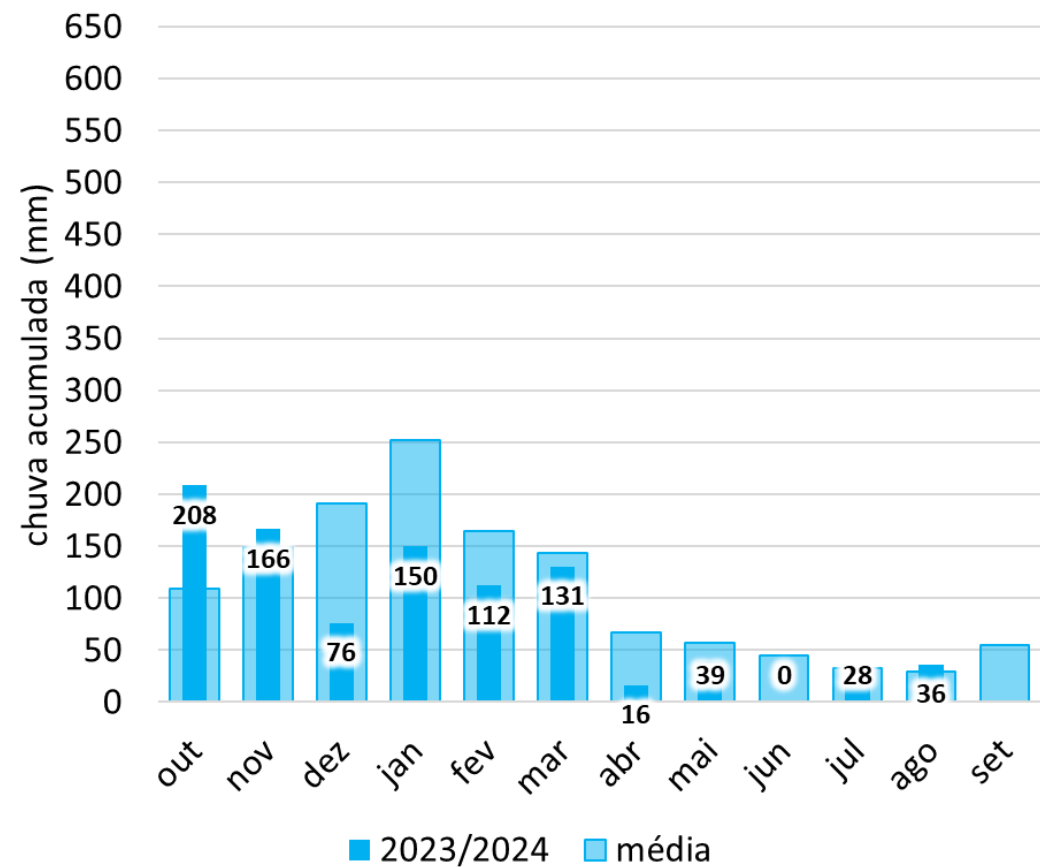
UGRHI 4 - Pardo | SPI-12



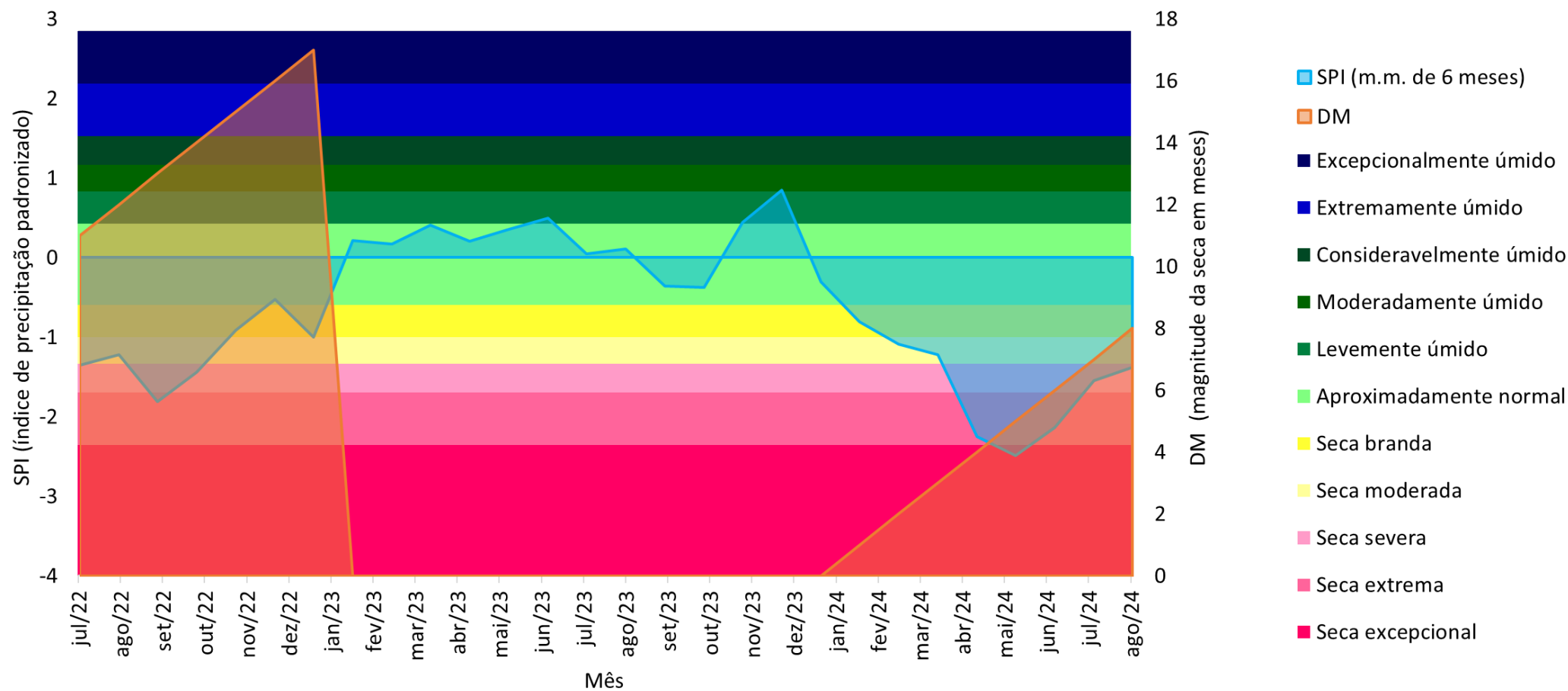
UGRHI 5 - Piracicaba/Capivari/Jundiaí



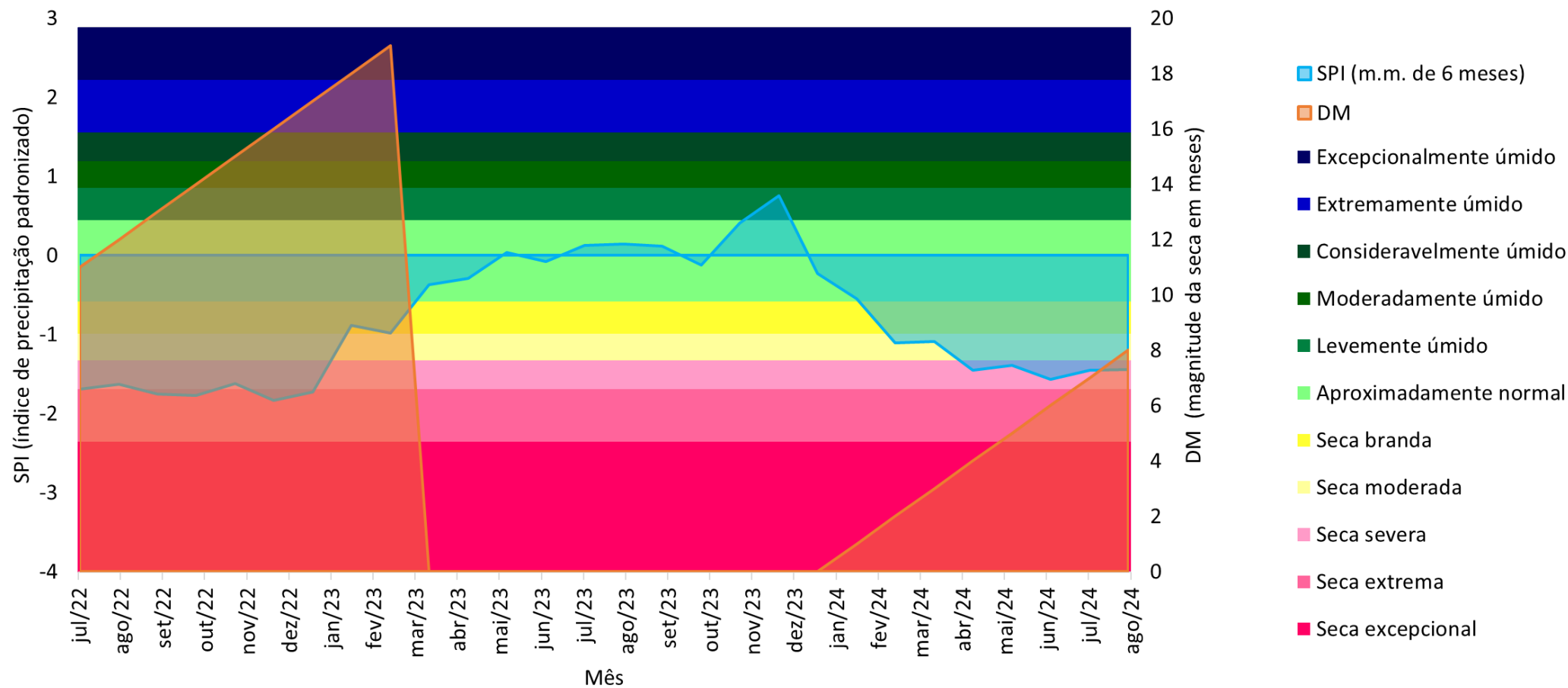
UGRHI 5 - Piracicaba/Capivari/Jundiaí



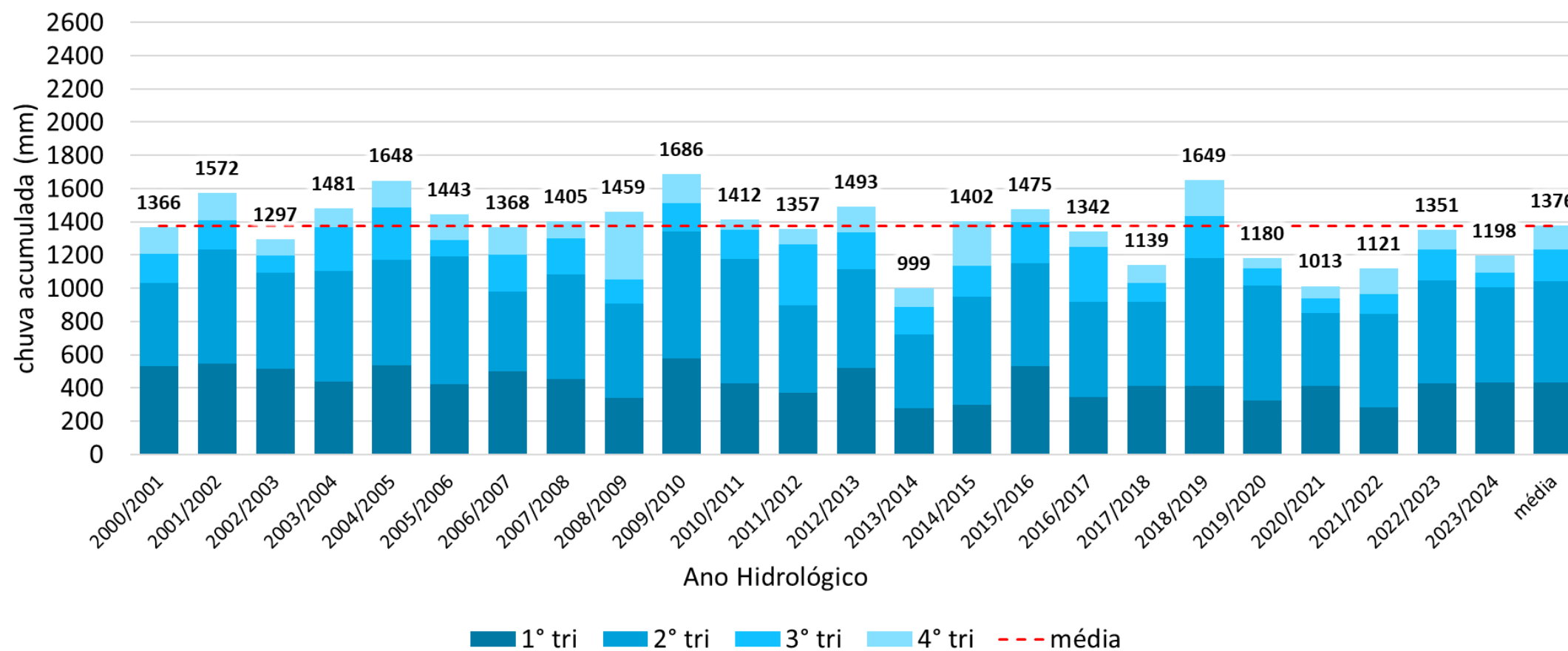
UGRHI 5 - Piracicaba/Capivari/Jundiaí | SPI-06



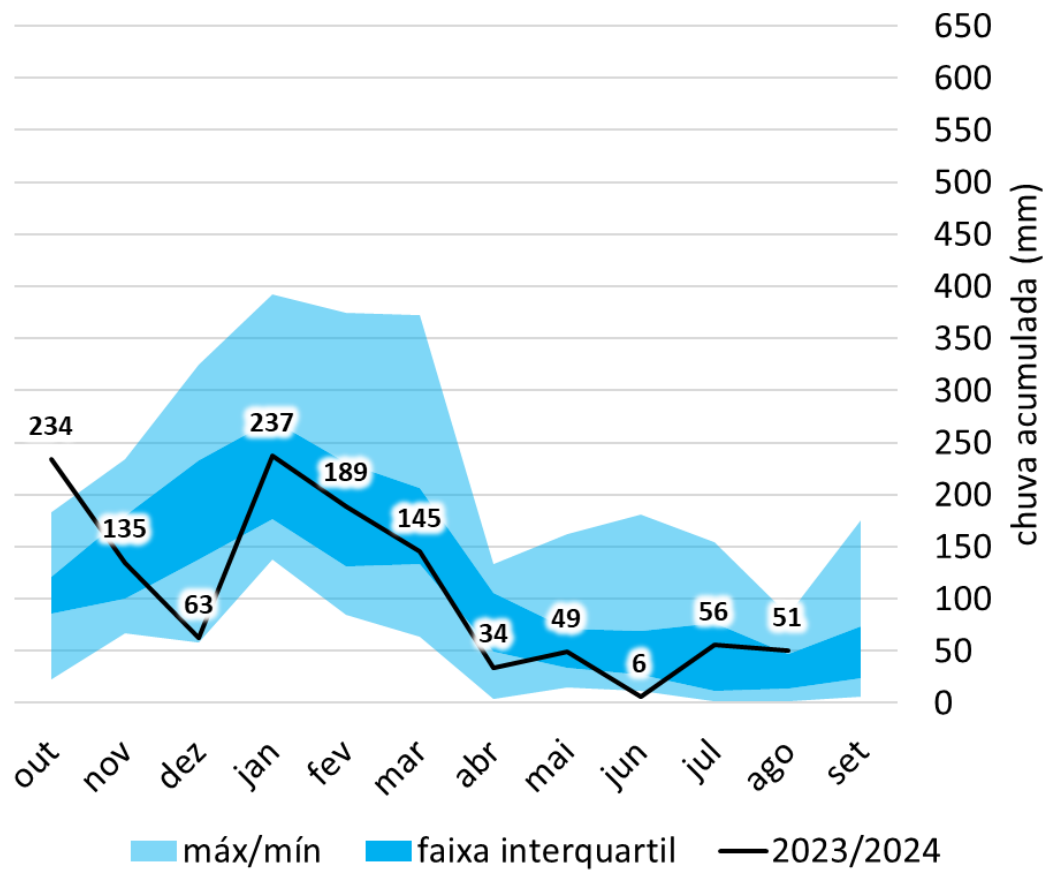
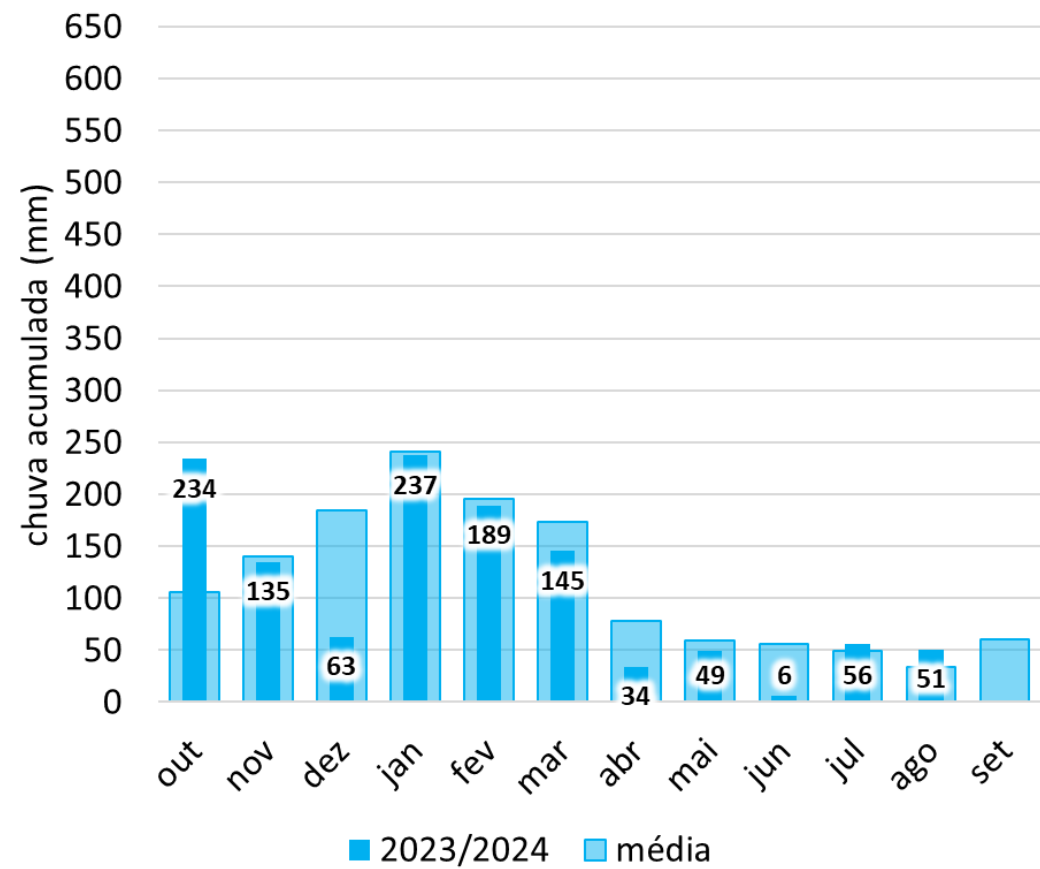
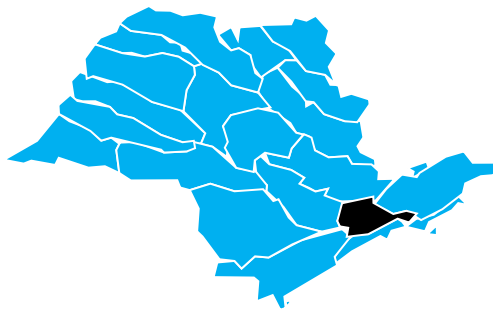
UGRHI 5 - Piracicaba/Capivari/Jundiaí | SPI-12



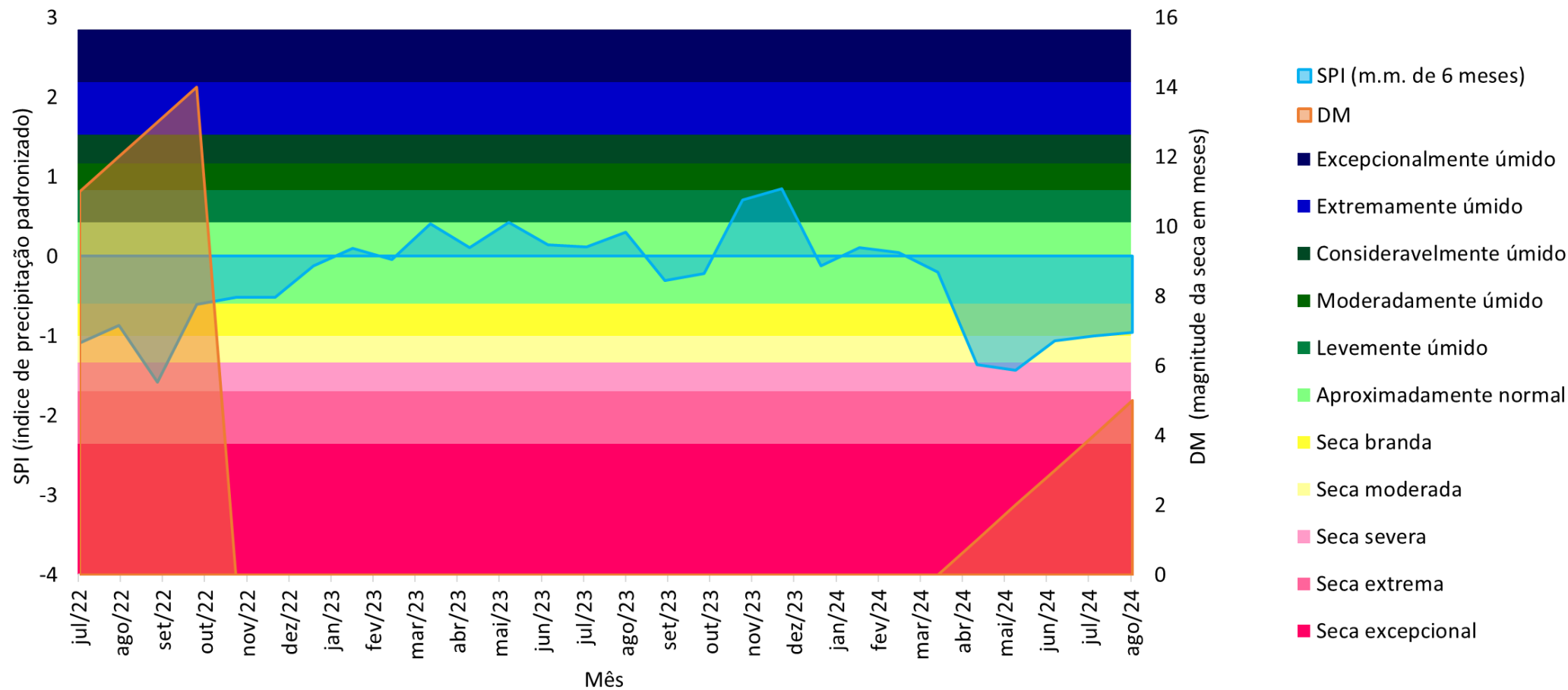
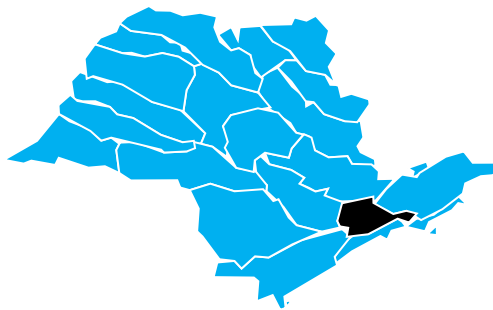
UGRHI 6 - Alto Tietê



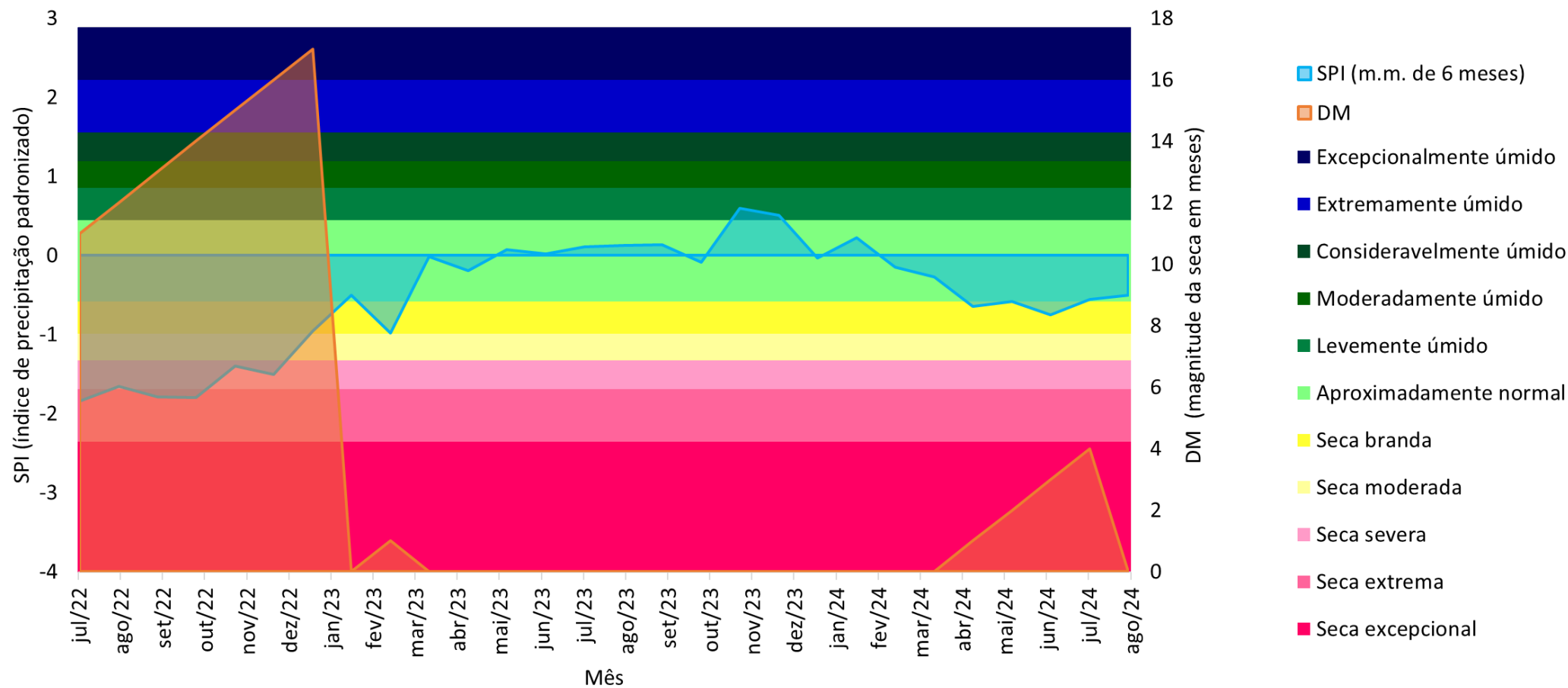
UGRHI 6 - Alto Tietê



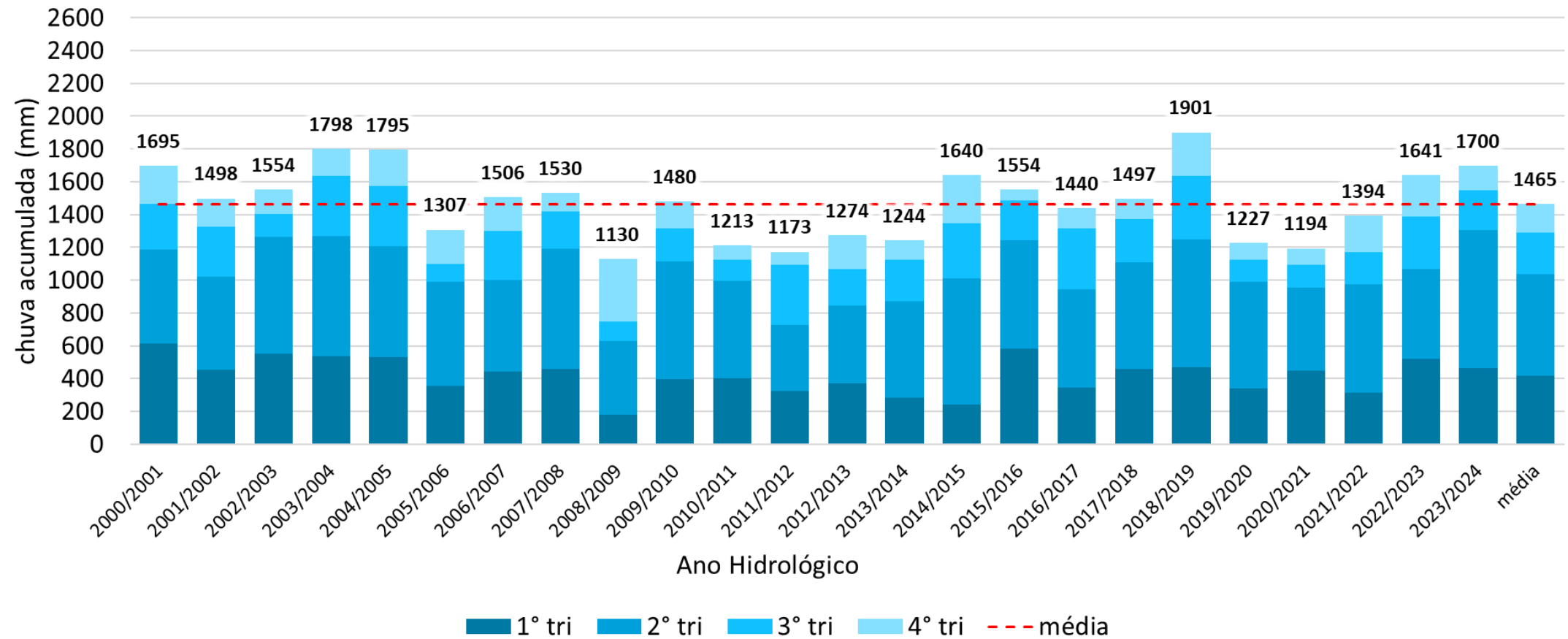
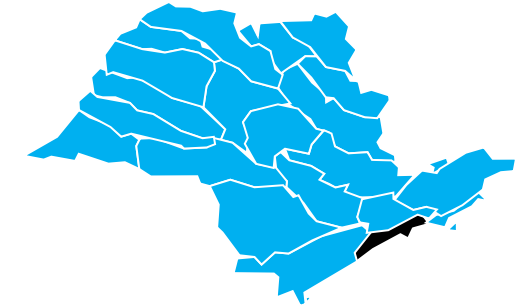
UGRHI 6 - Alto Tietê | SPI-06



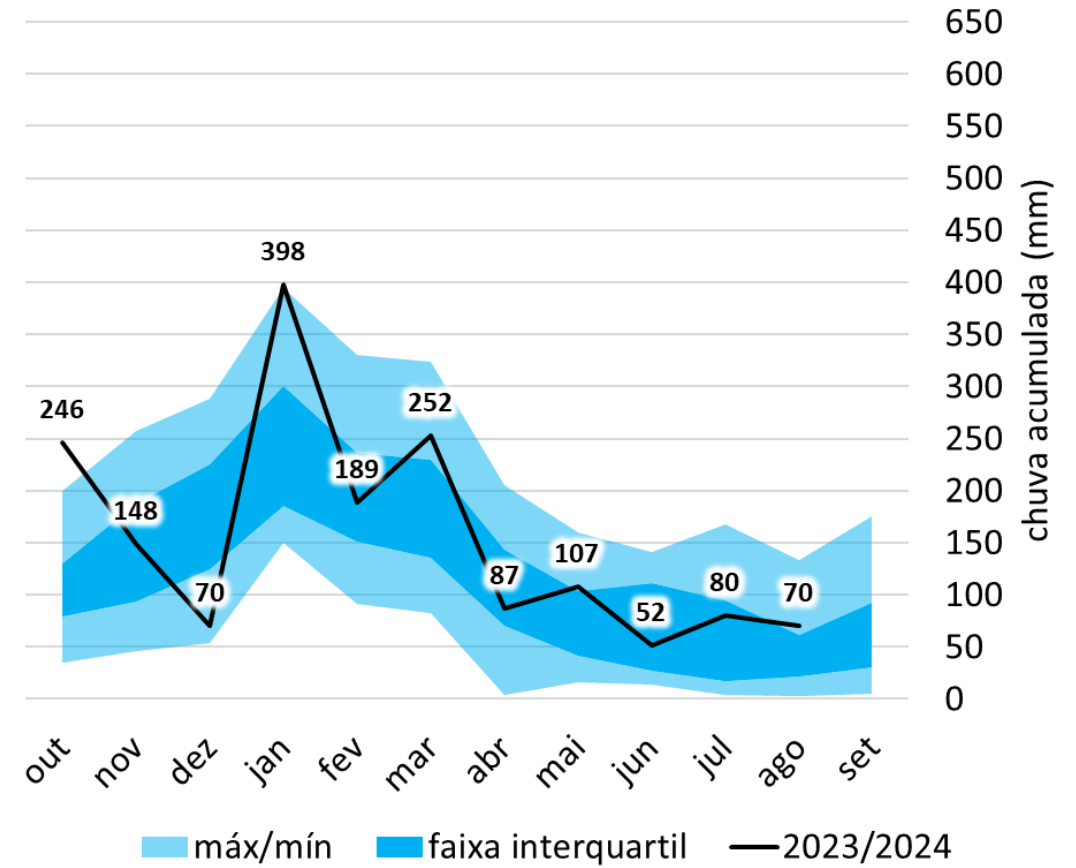
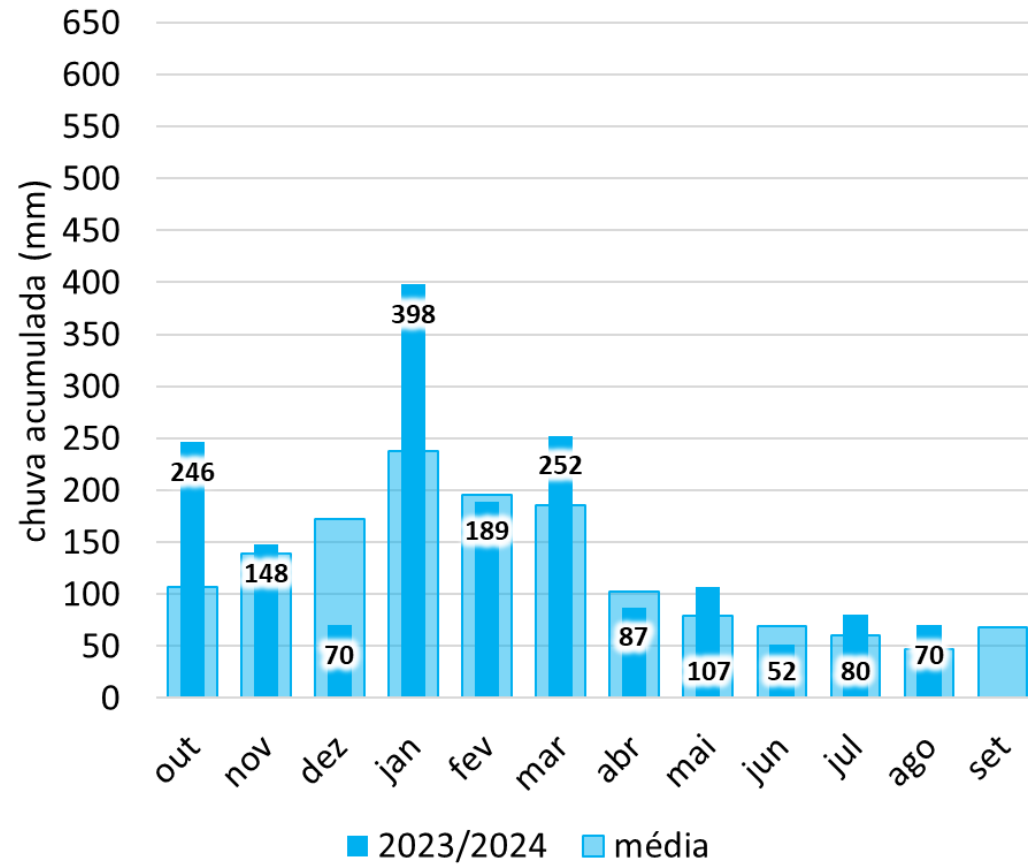
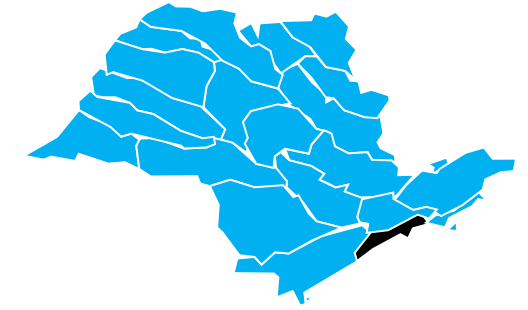
UGRHI 6 - Alto Tietê | SPI-12



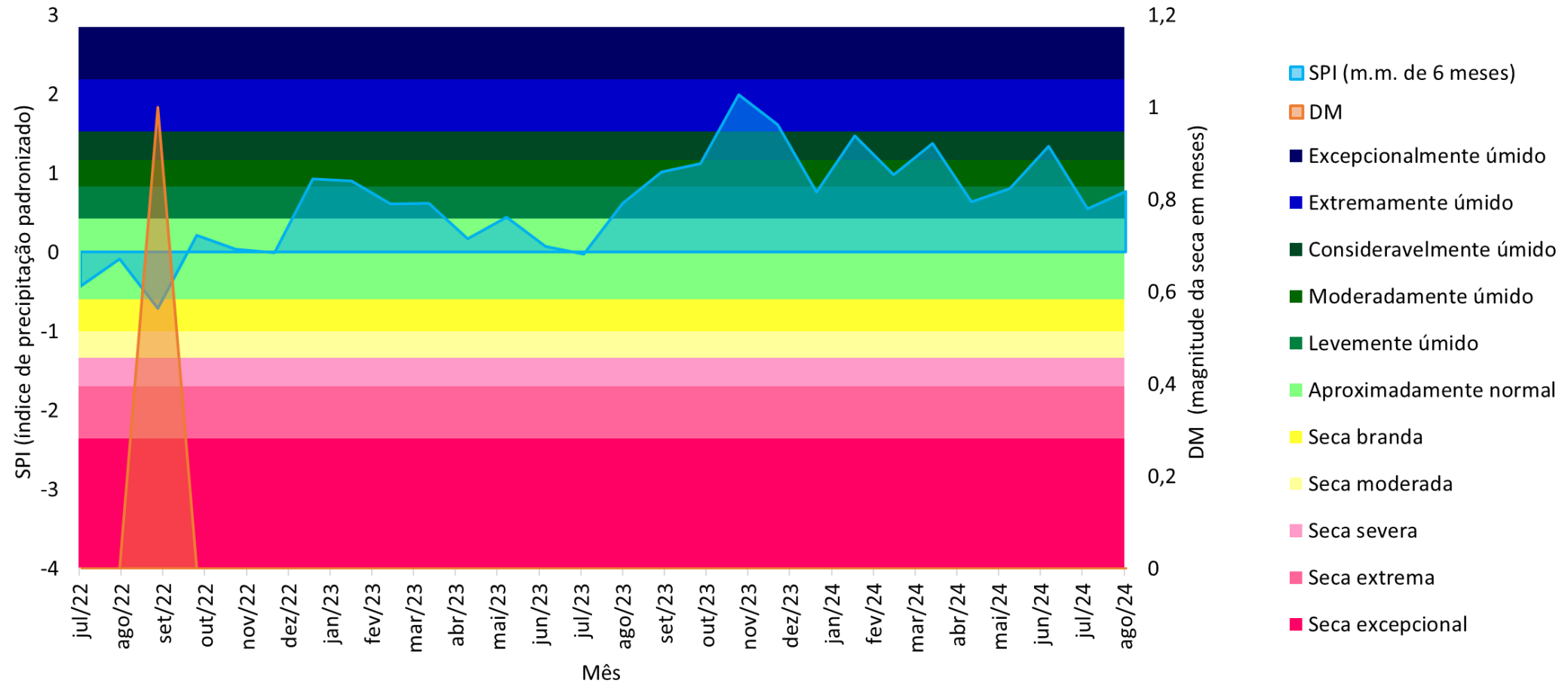
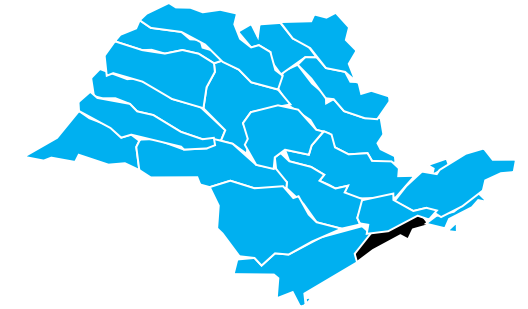
UGRHI 7 - Baixada Santista



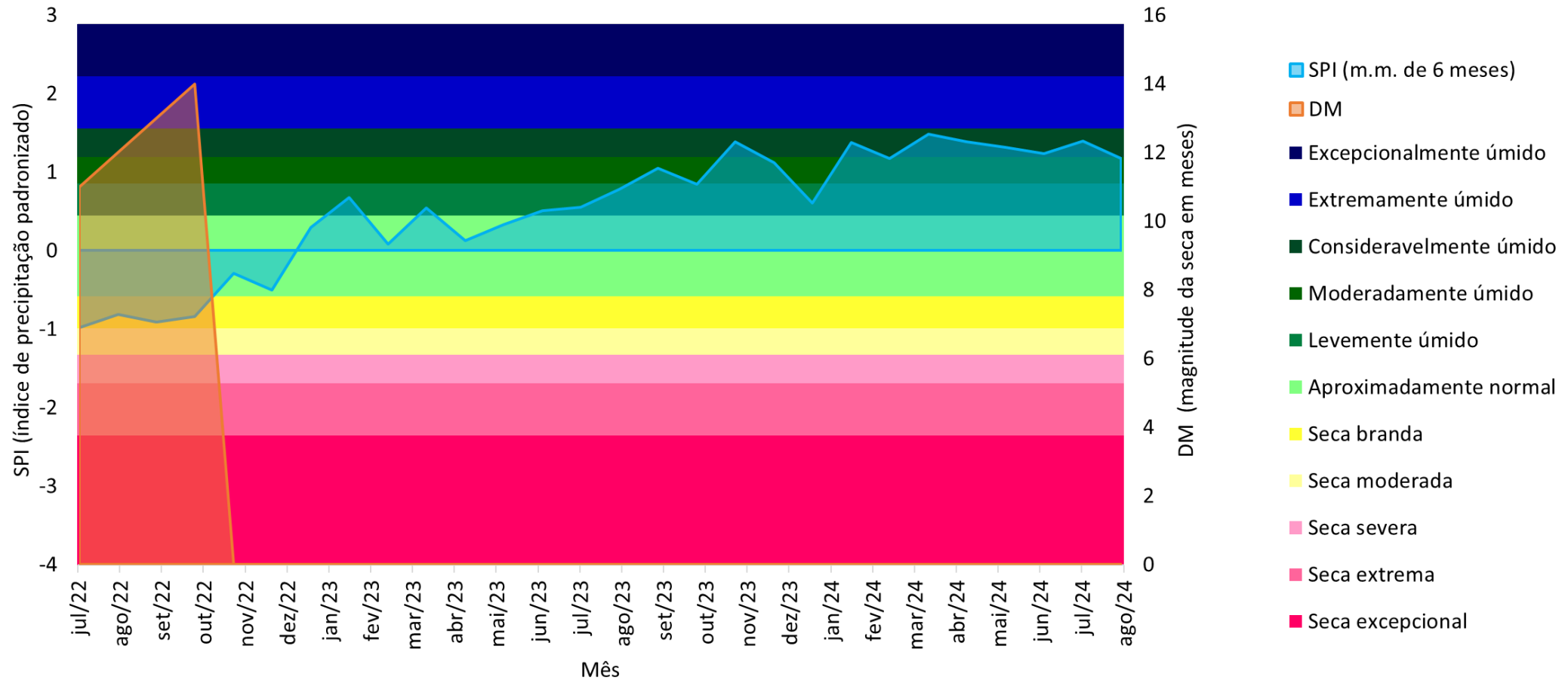
UGRHI 7 - Baixada Santista



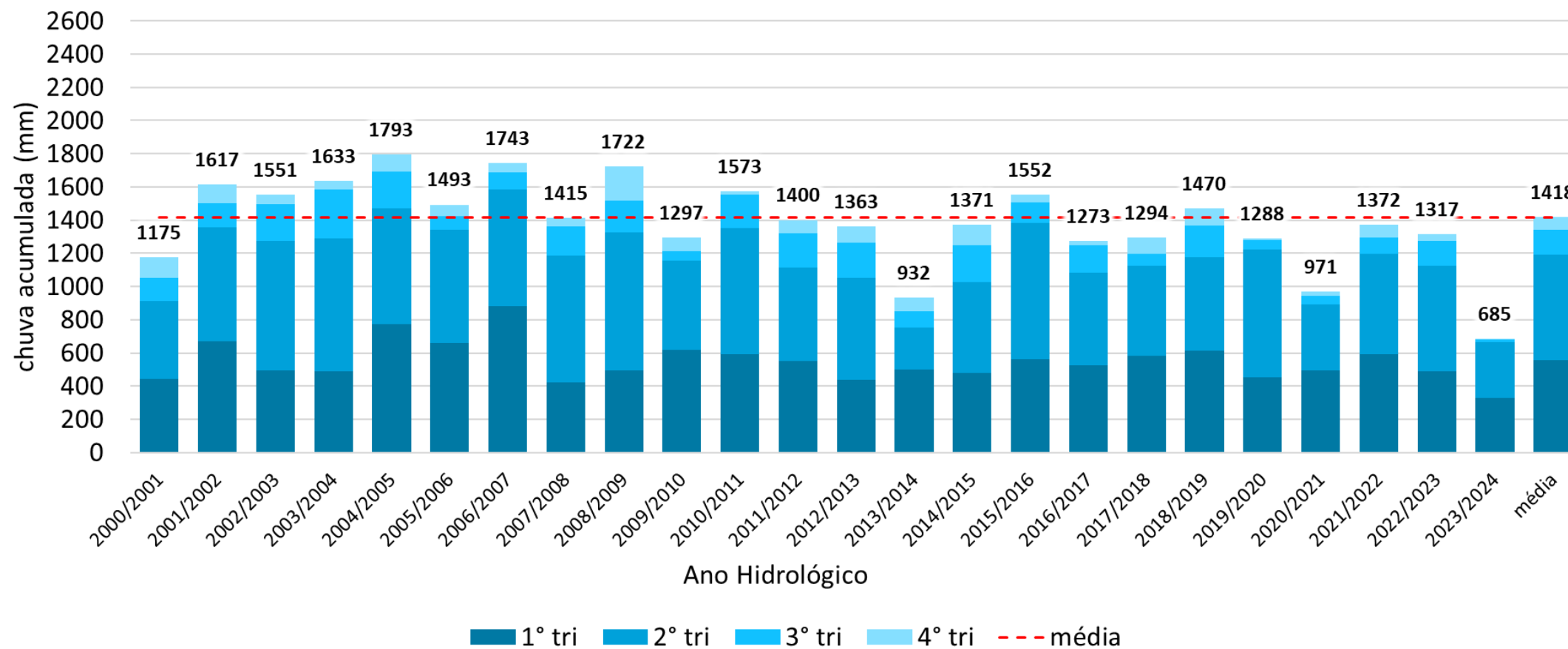
UGRHI 7 - Baixada Santista | SPI-06



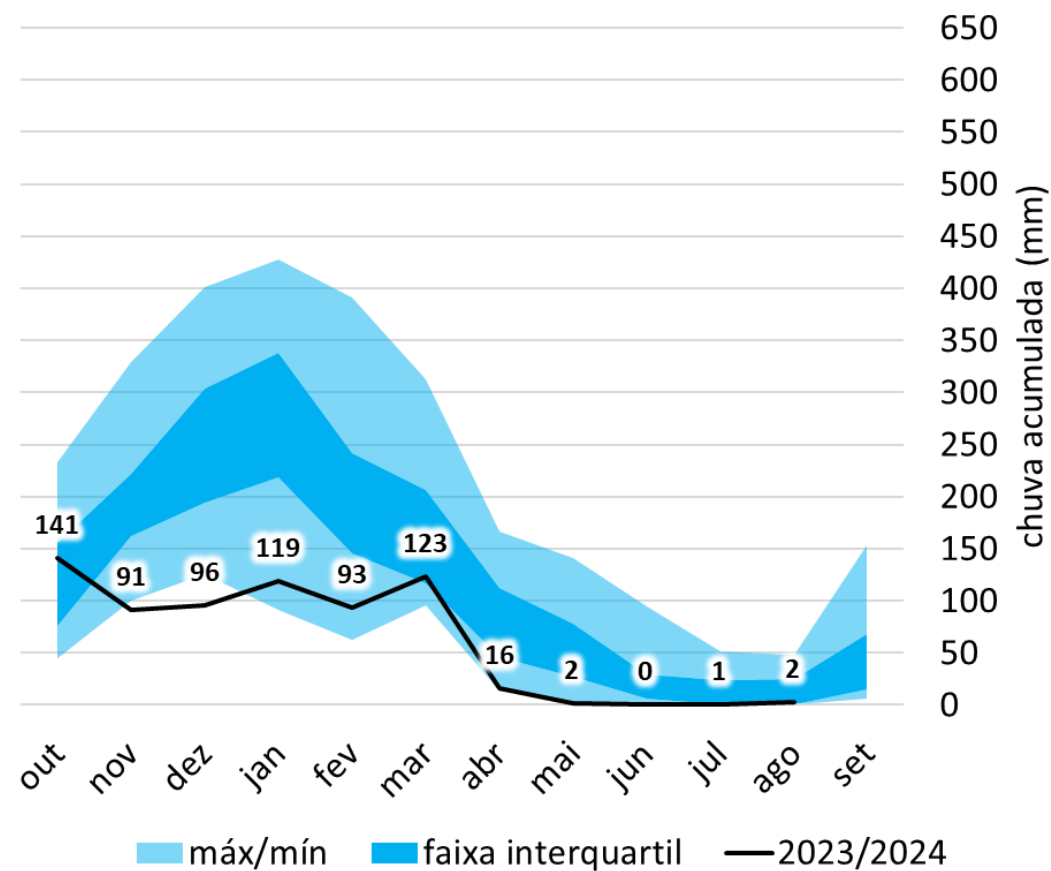
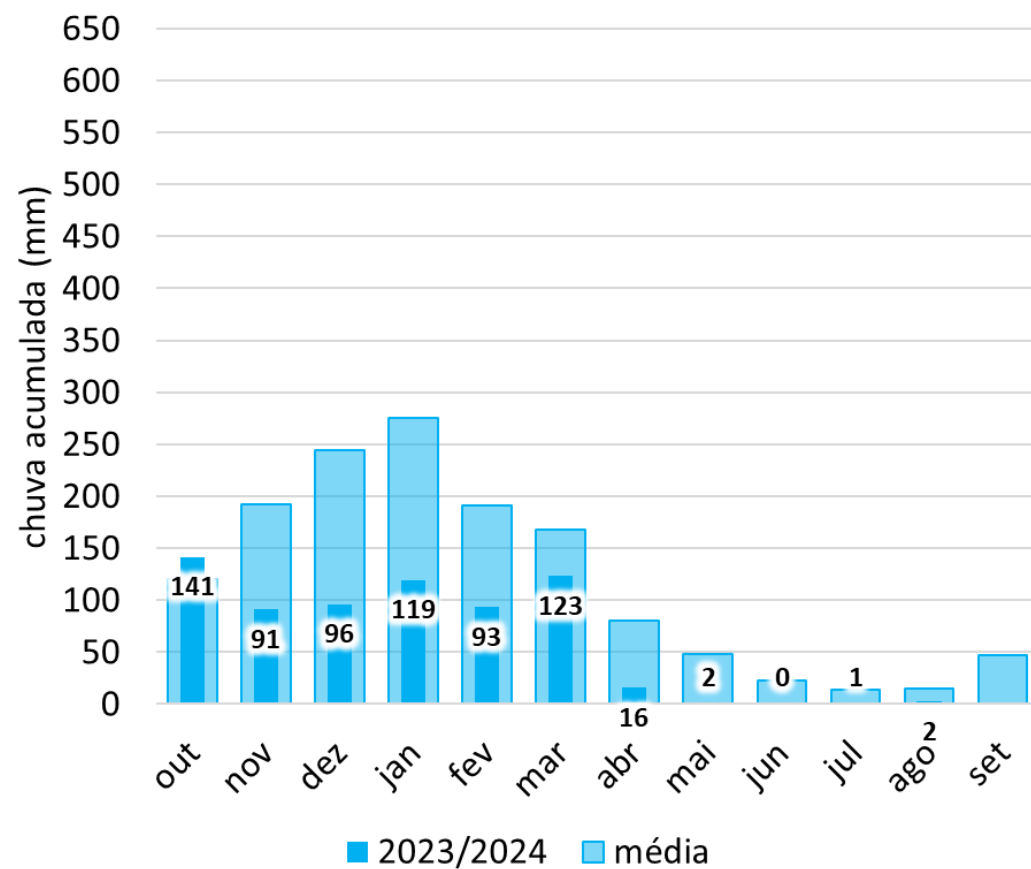
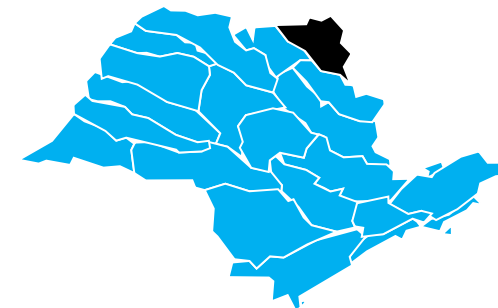
UGRHI 7 - Baixada Santista | SPI-12



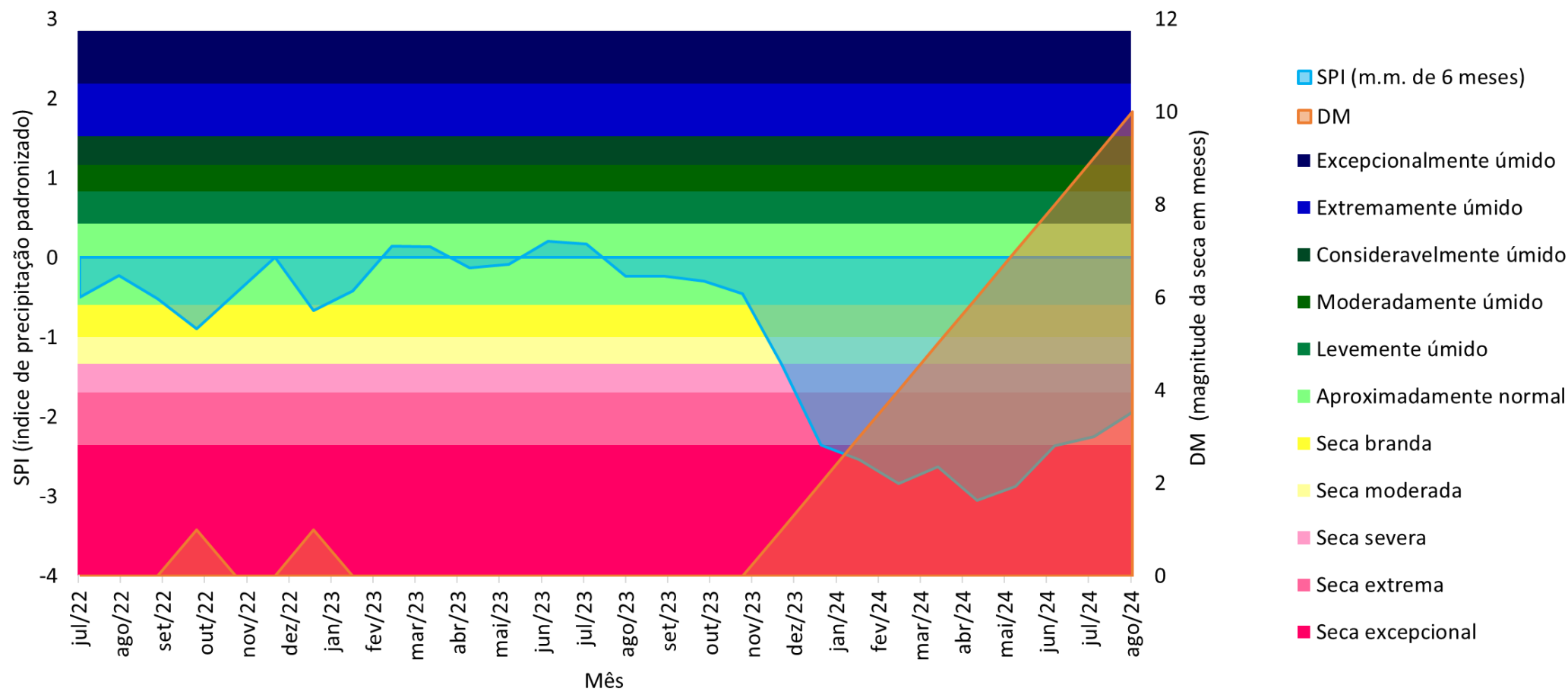
UGRHI 8 - Sapucaí/Grande



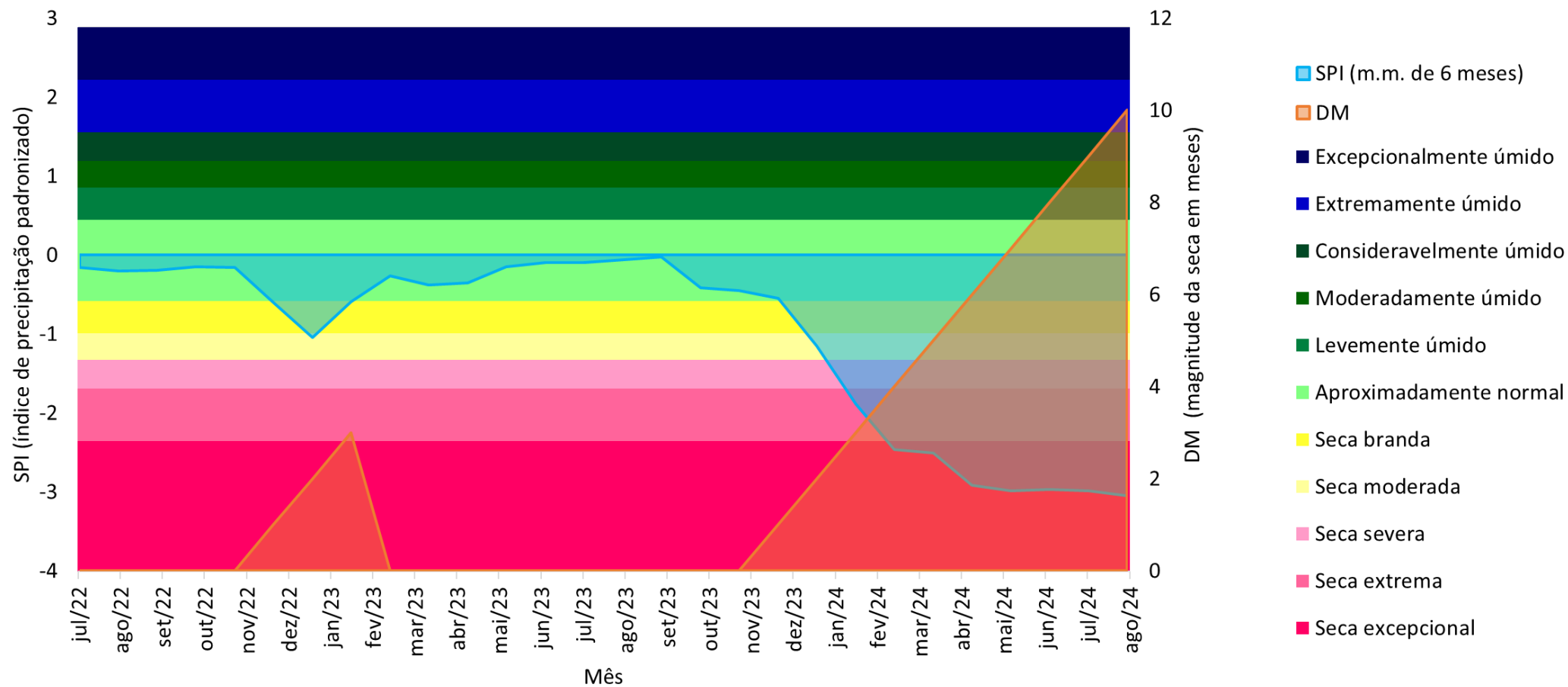
UGRHI 8 - Sapucaí/Grande



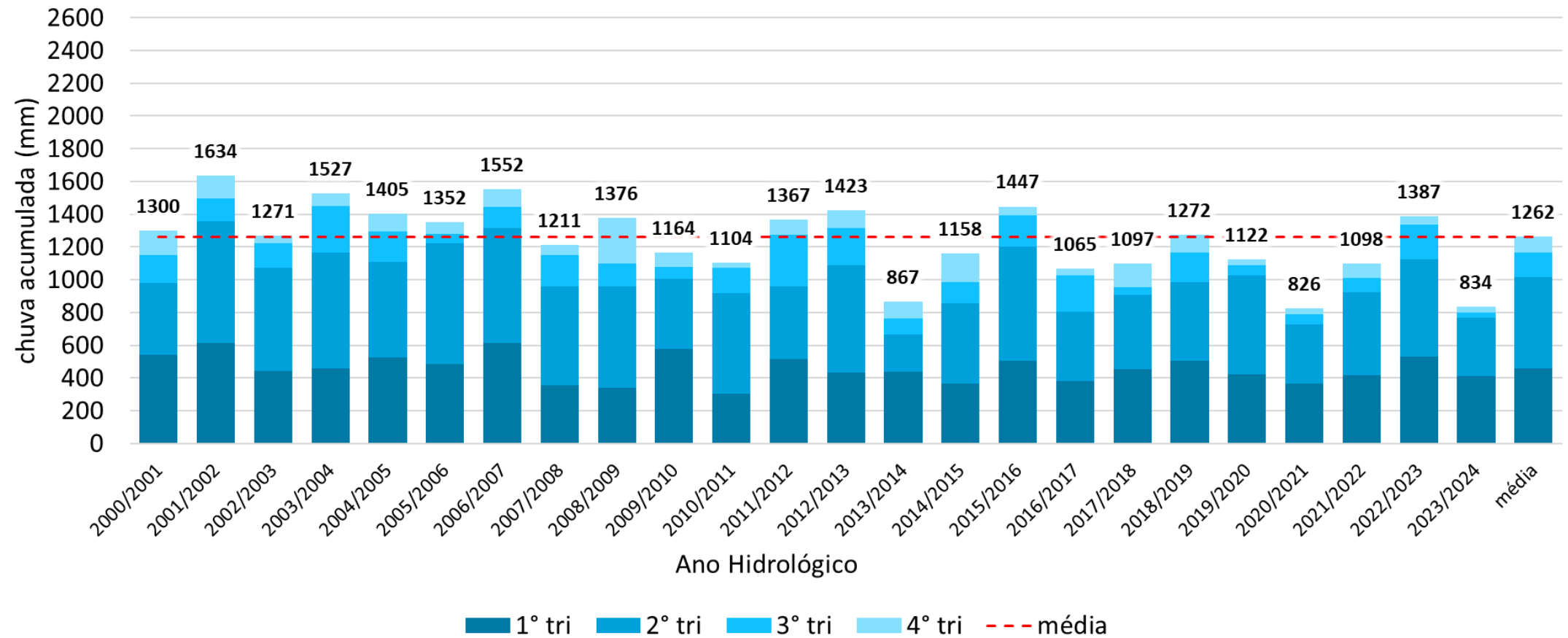
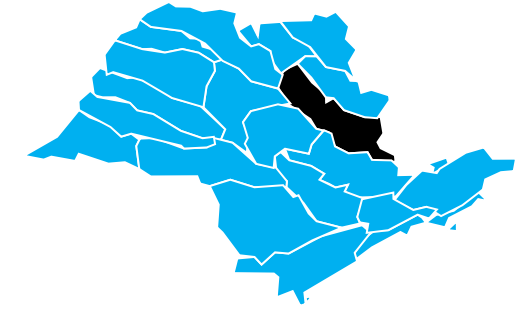
UGRHI 8 - Sapucaí/Grande | SPI-06



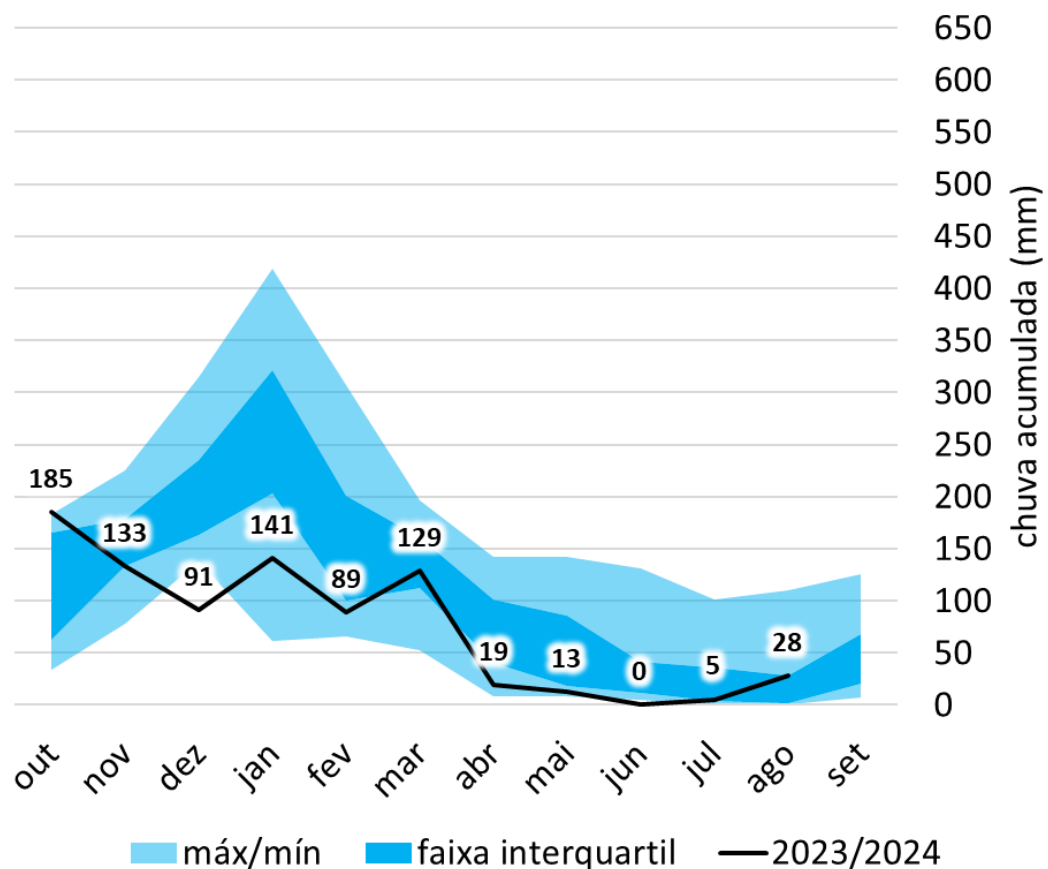
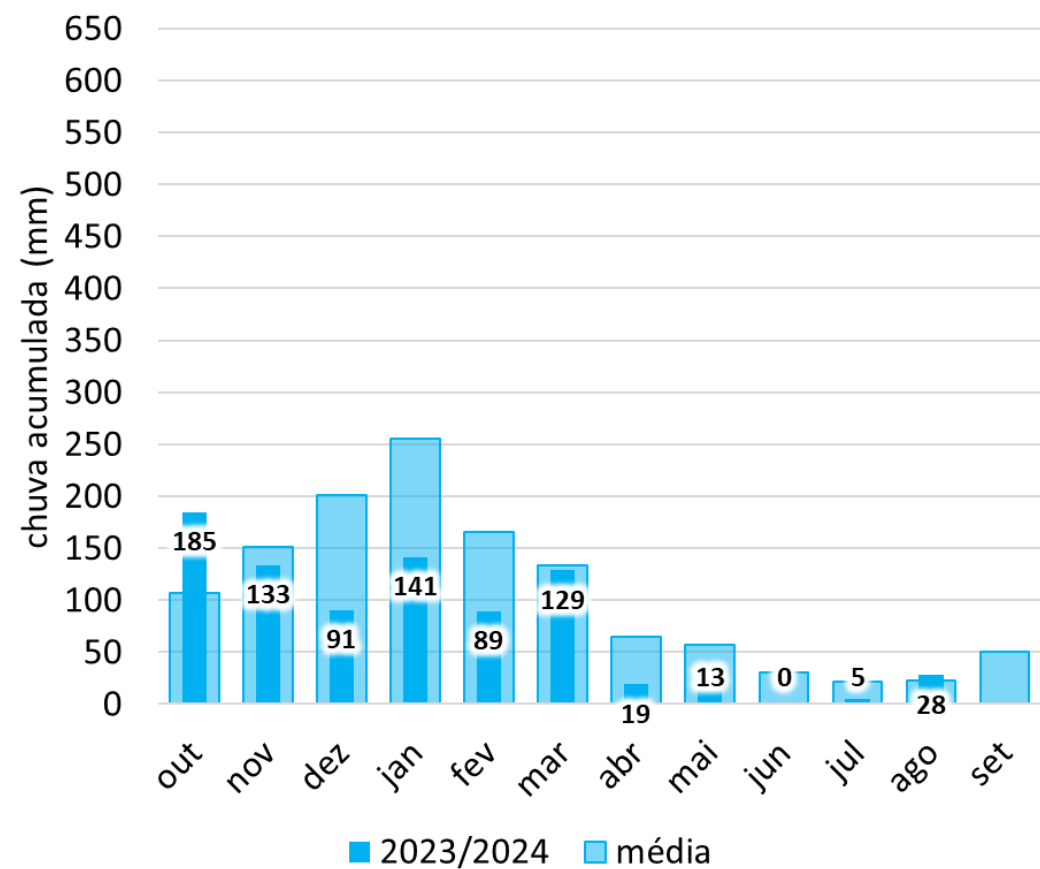
UGRHI 8 - Sapucaí/Grande | SPI-12



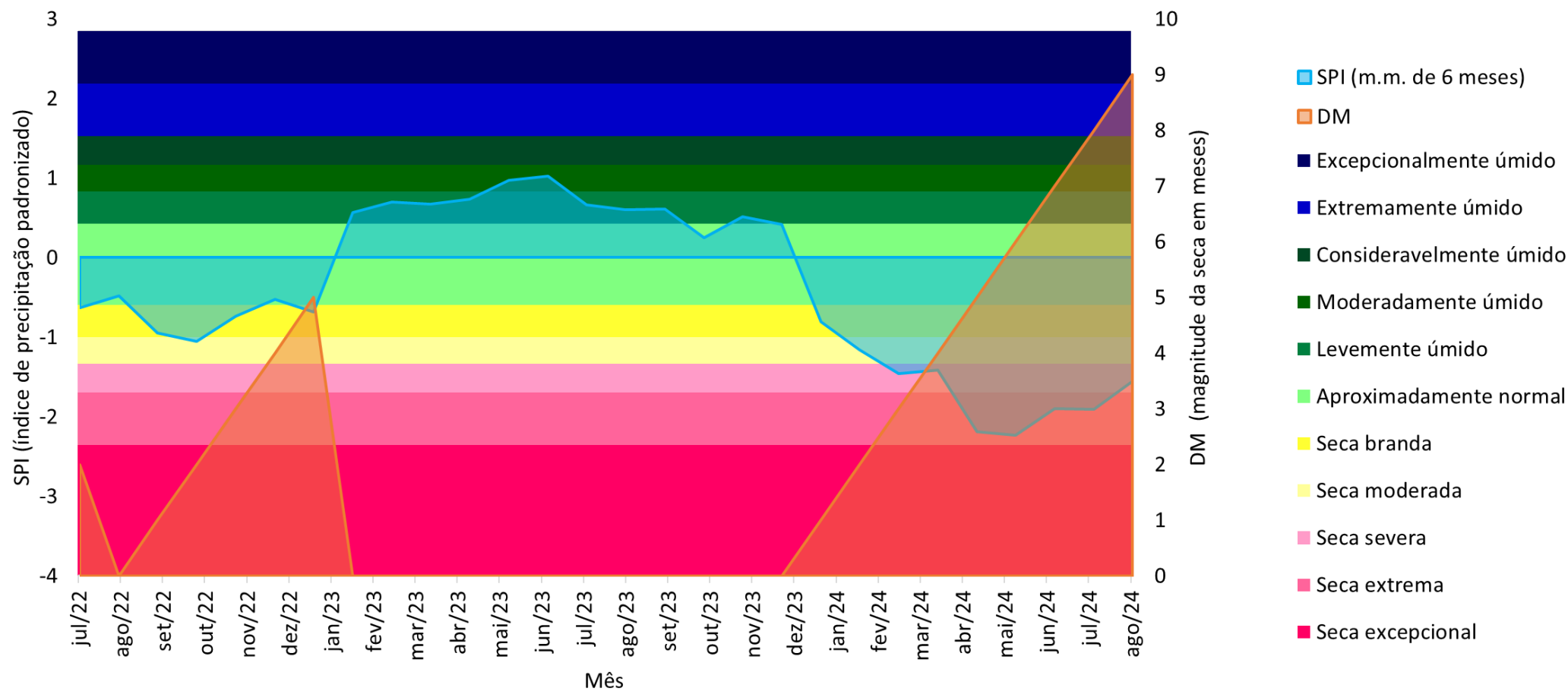
UGRHI 9 - Mogi-Guaçu



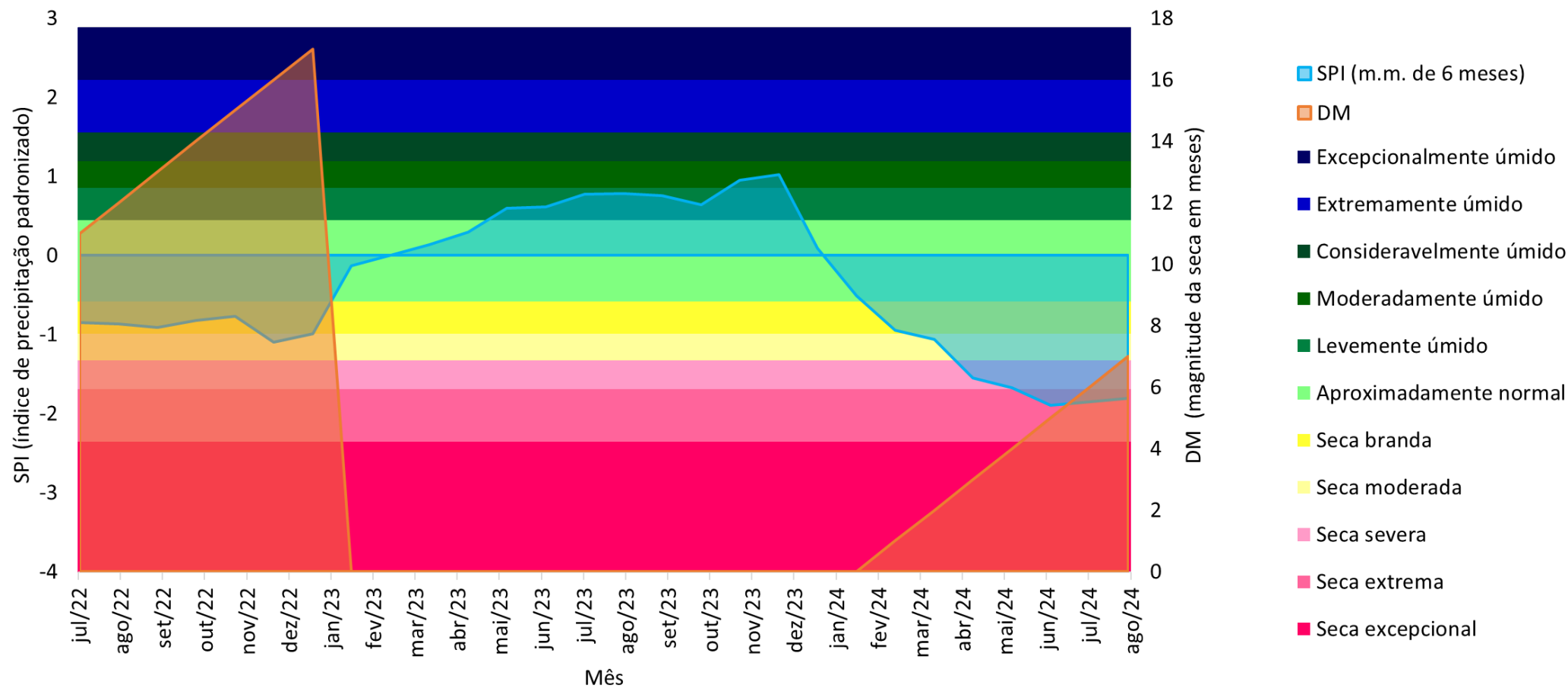
UGRHI 9 - Mogi-Guaçu



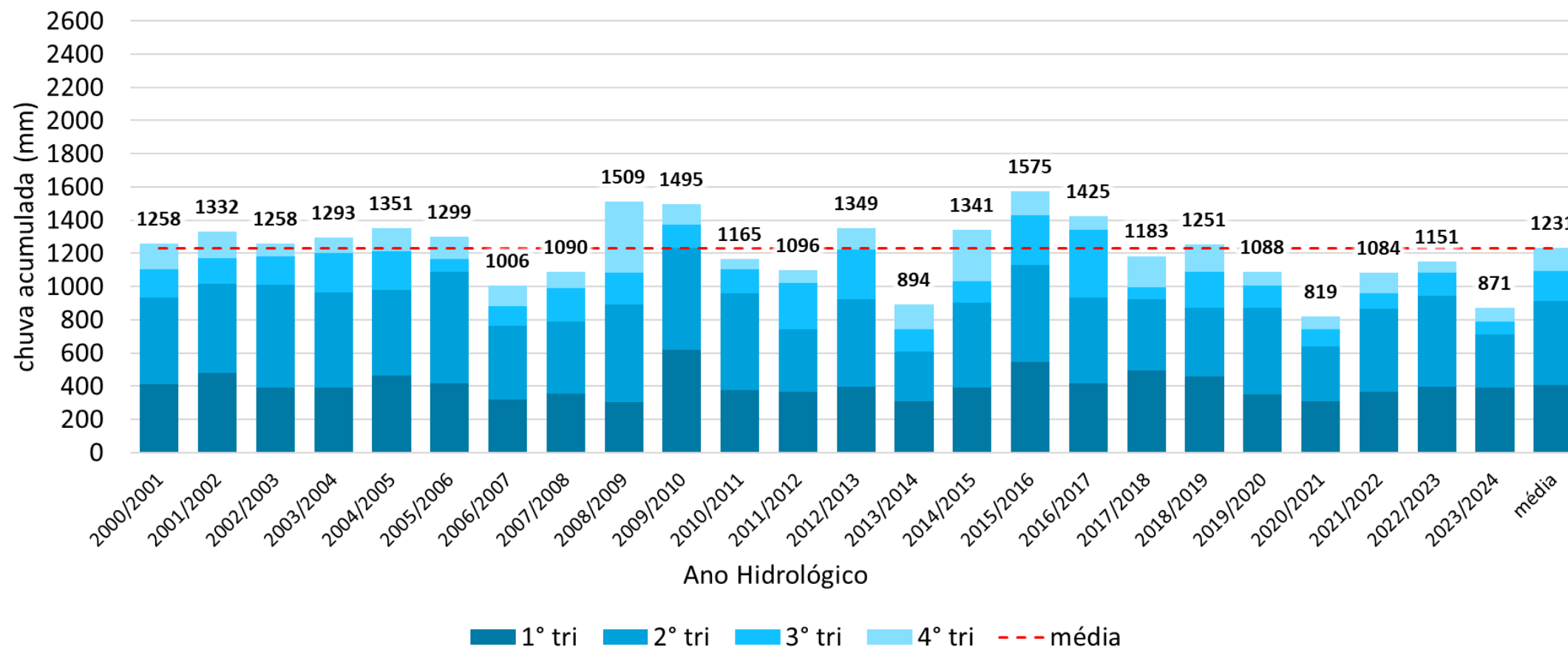
UGRHI 9 - Mogi-Guaçu | SPI-06



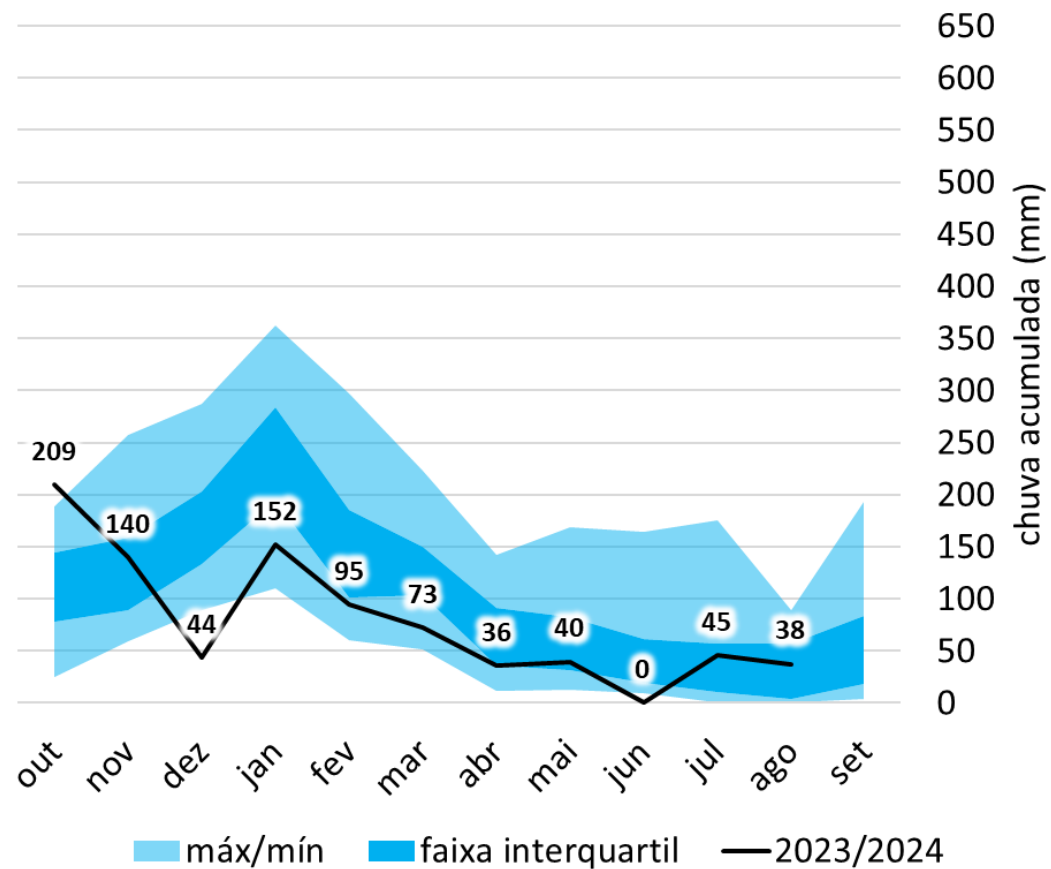
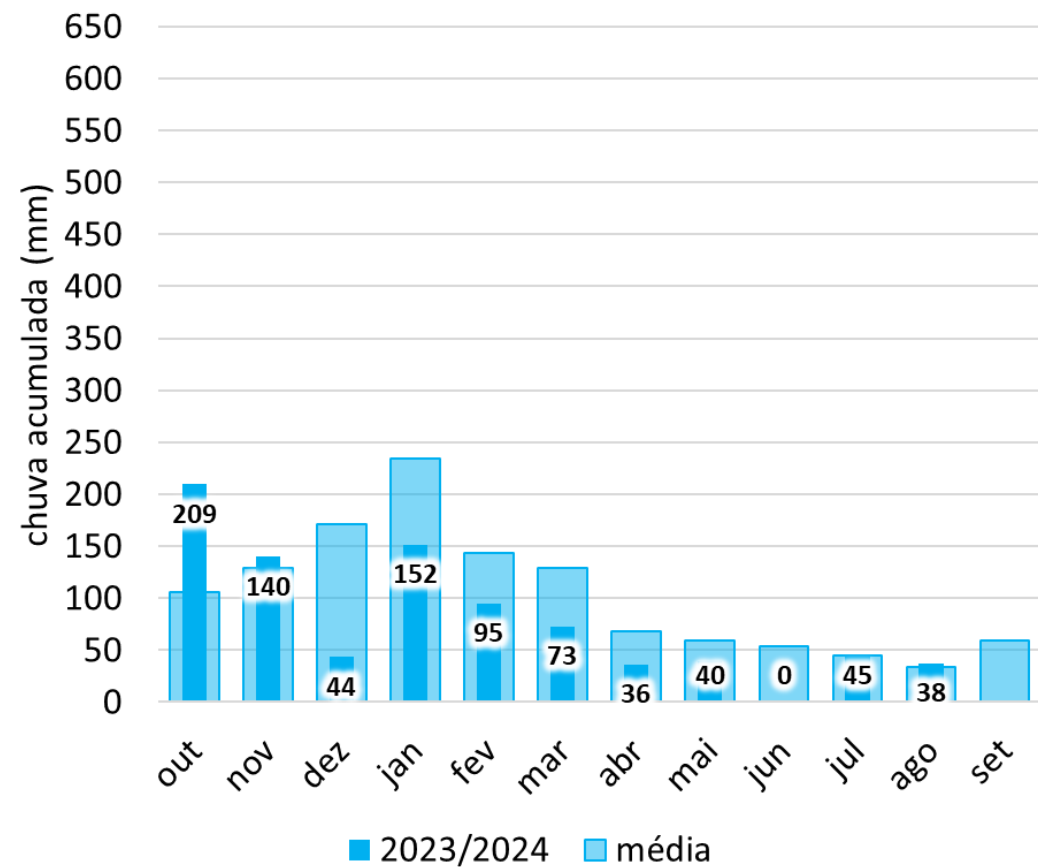
UGRHI 9 - Mogi-Guaçu | SPI-12



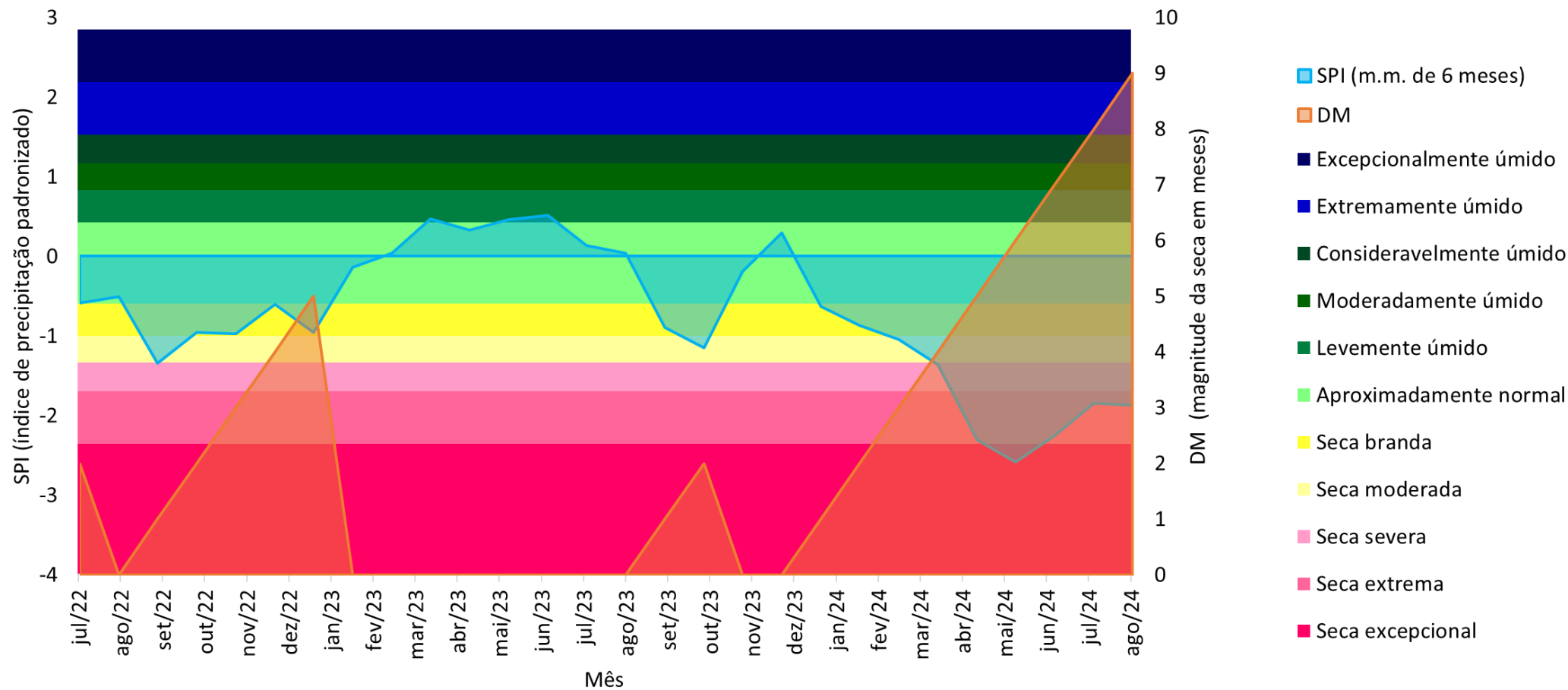
UGRHI 10 - Tietê/Sorocaba



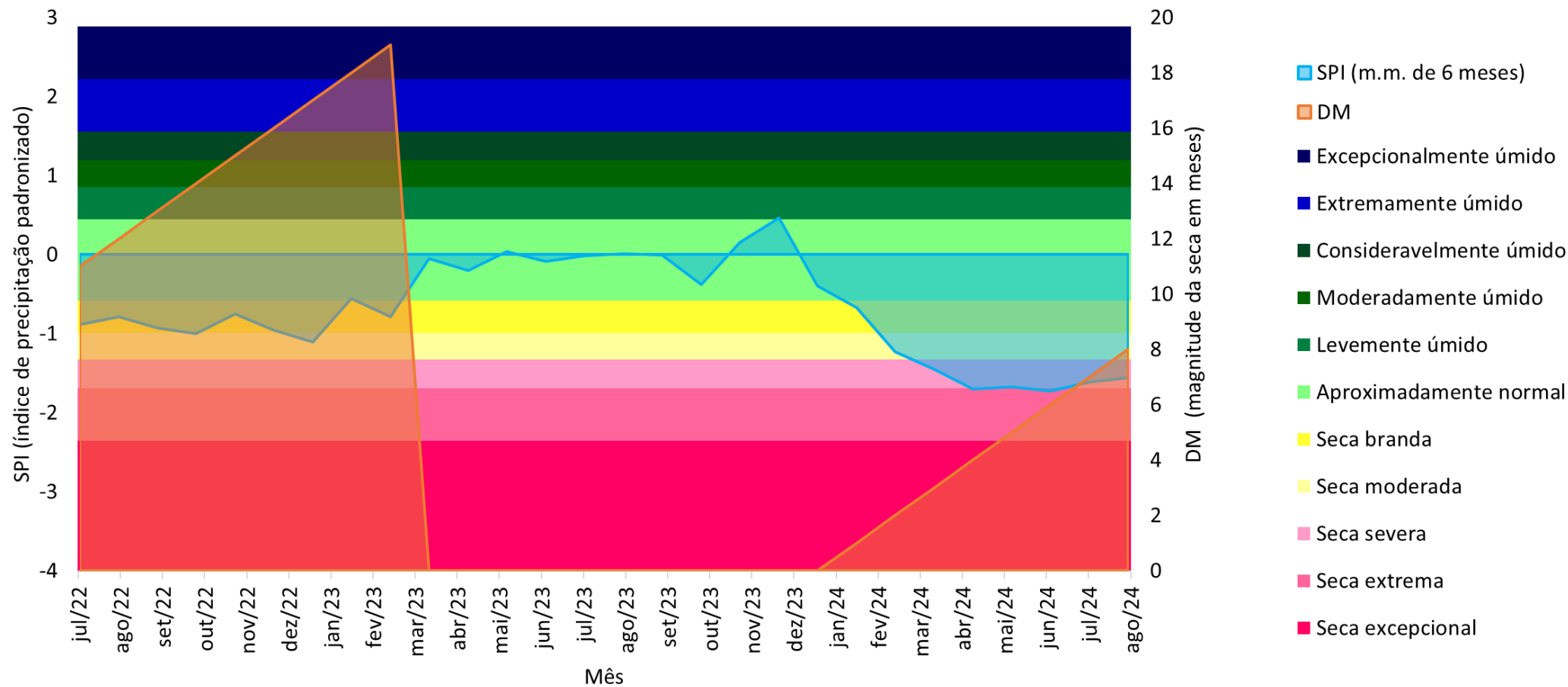
UGRHI 10 - Tietê/Sorocaba



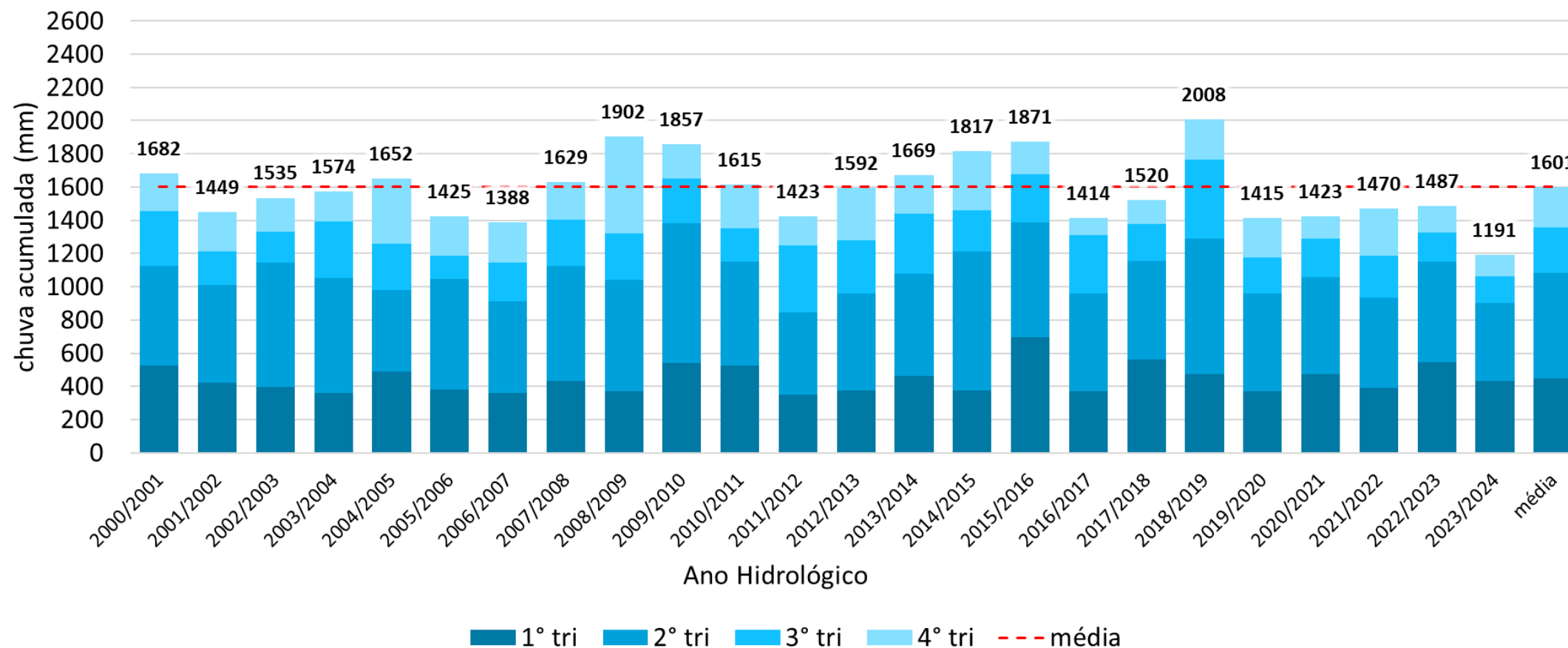
UGRHI 10 - Tietê/Sorocaba | SPI-06



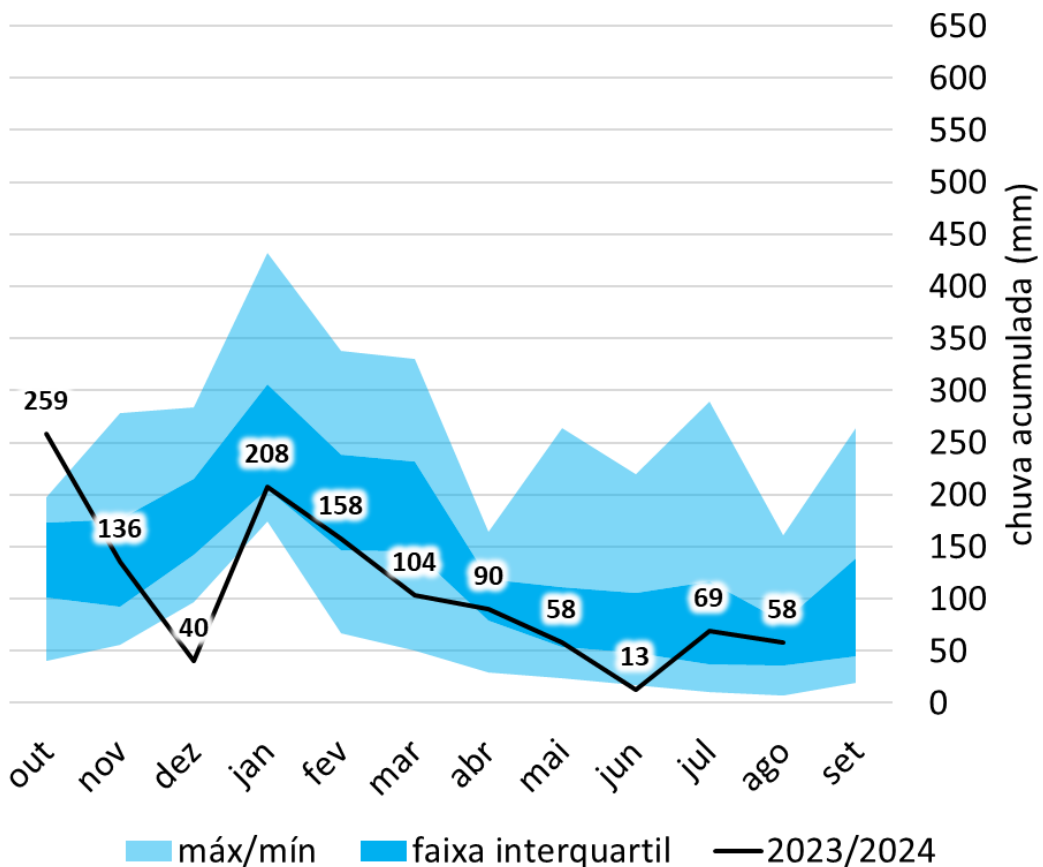
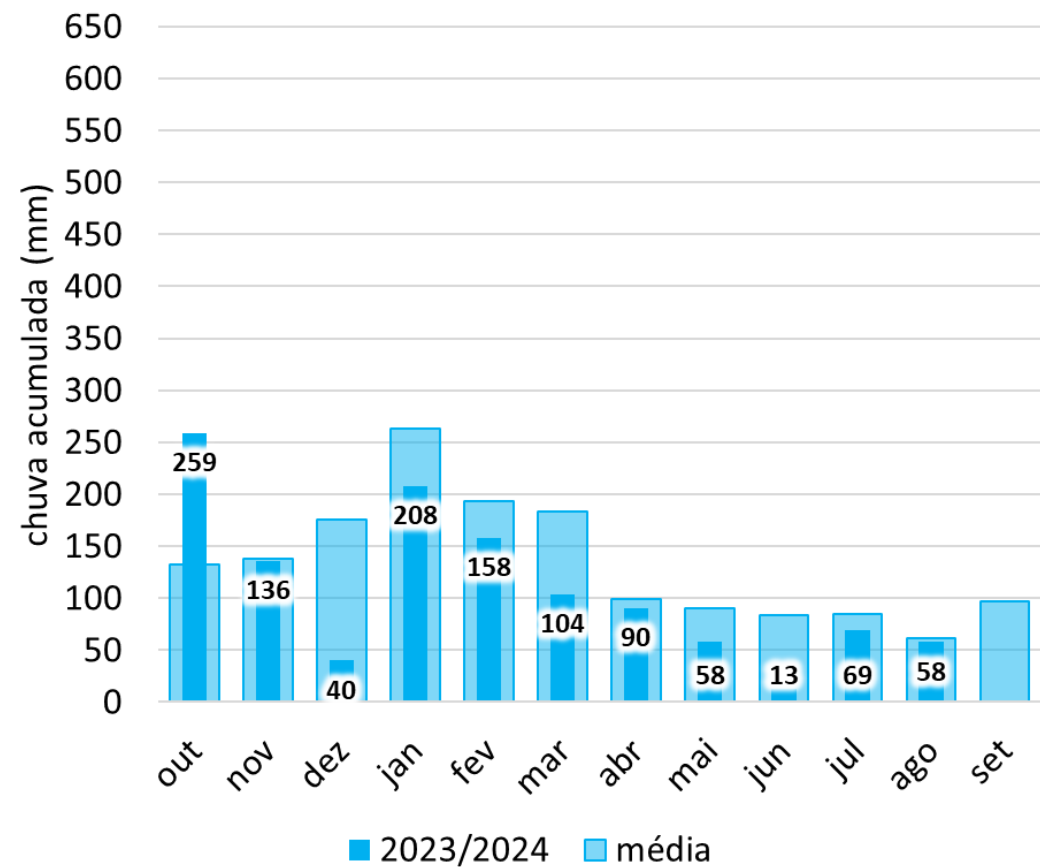
UGRHI 10 - Tietê/Sorocaba | SPI-12



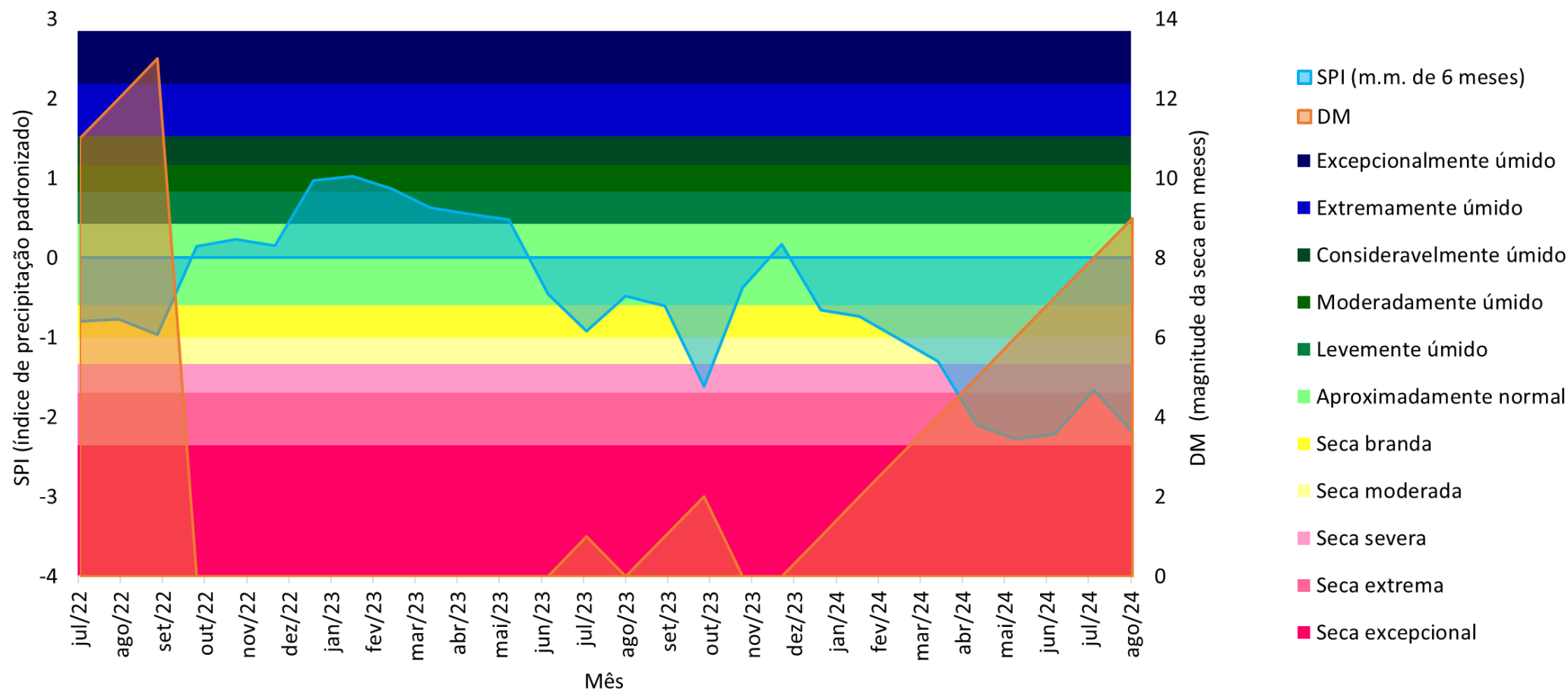
UGRHI 11 - Ribeira de Iguape/Litoral Sul



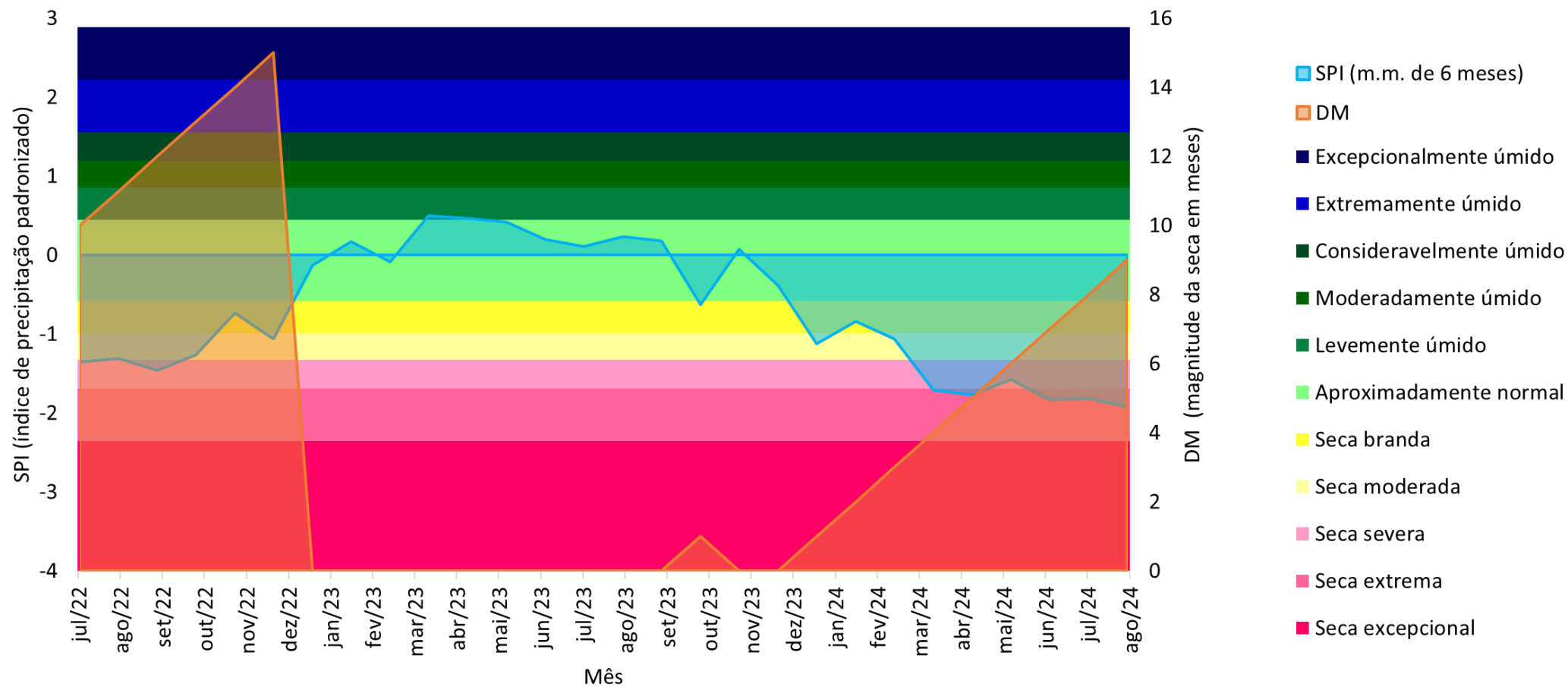
UGRHI 11 - Ribeira de Iguape/Litoral Sul



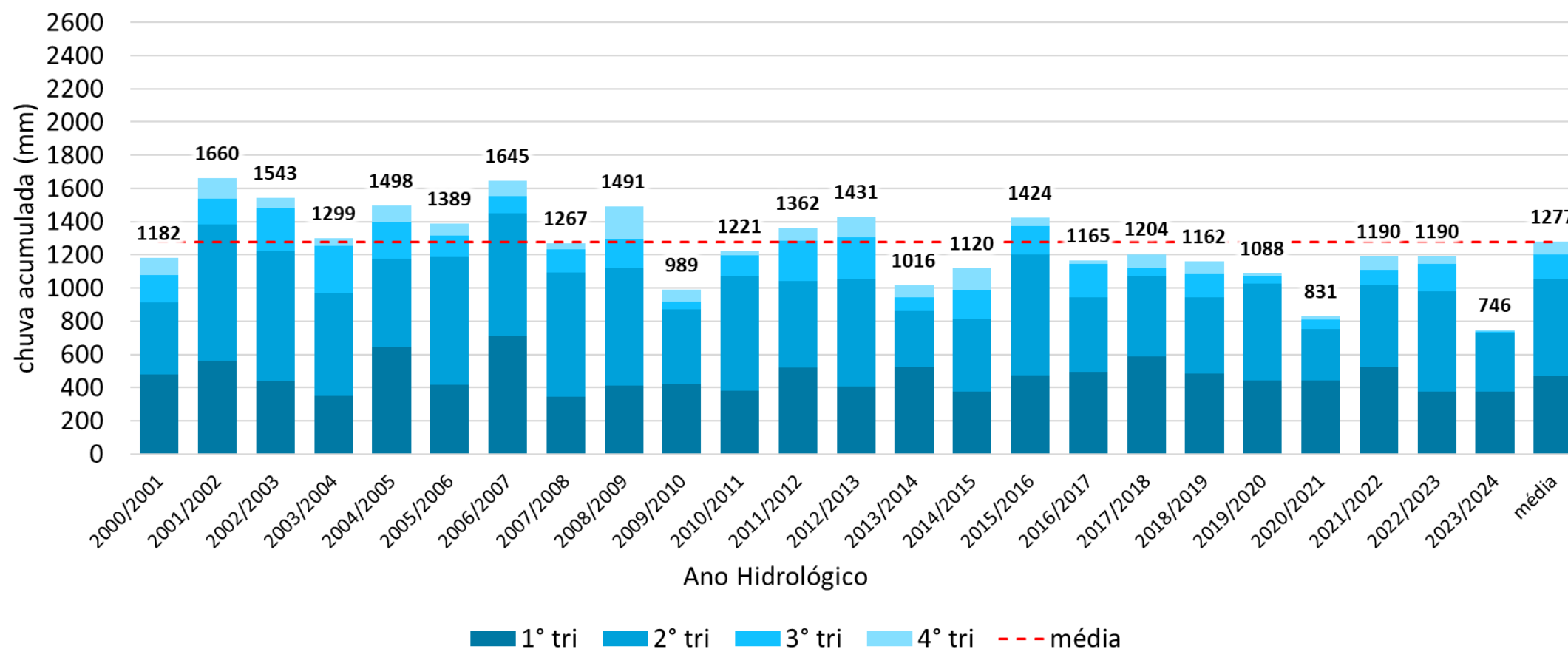
UGRHI 11 Ribeira de Iguape/Litoral Sul | SPI-06



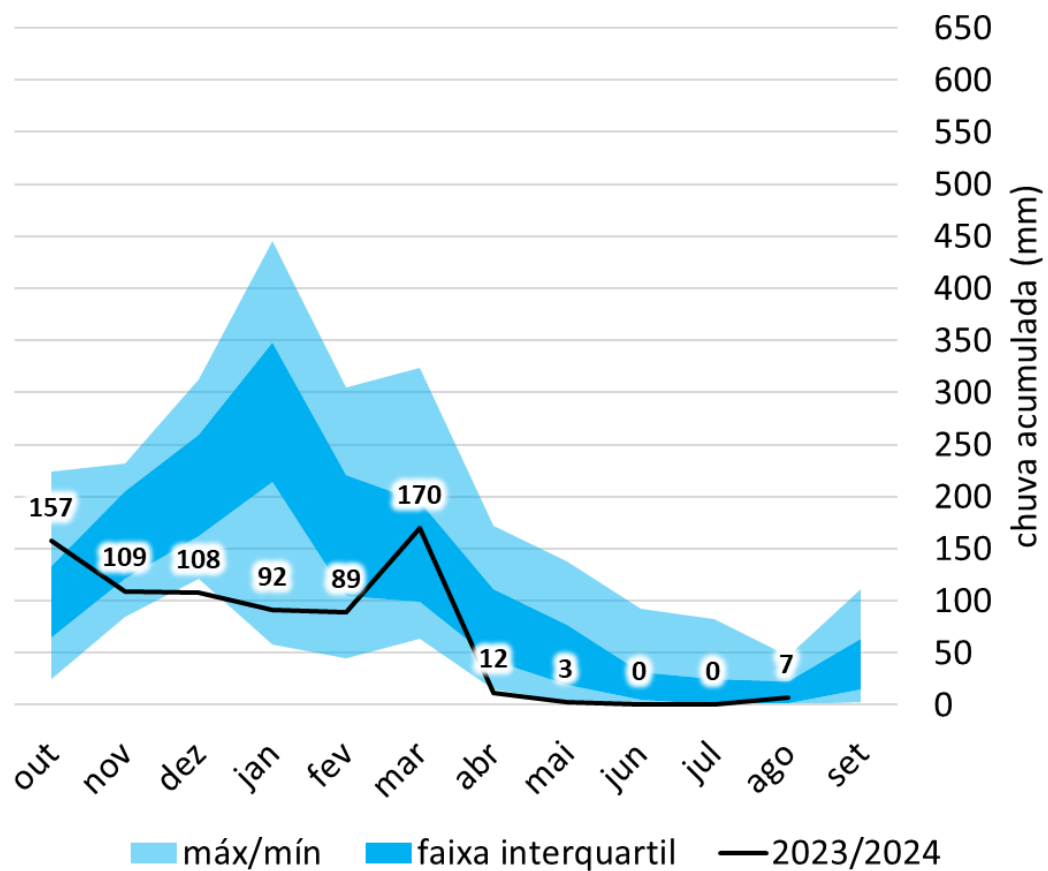
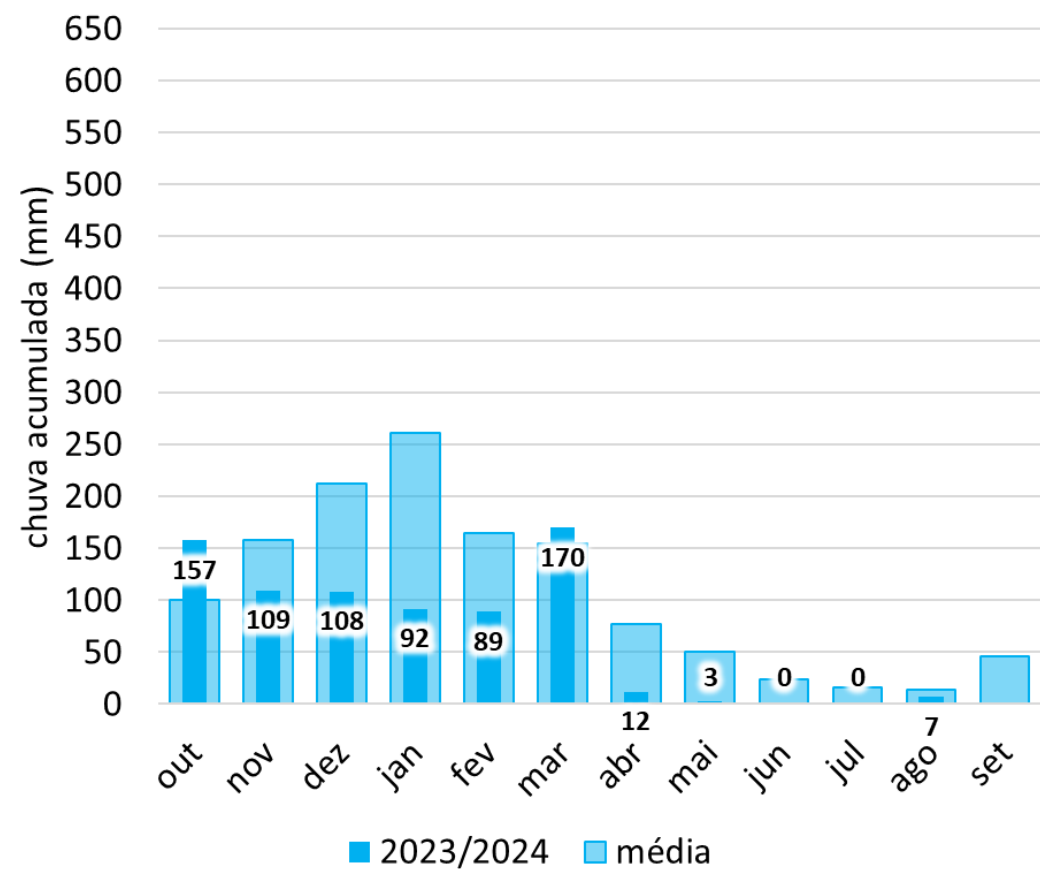
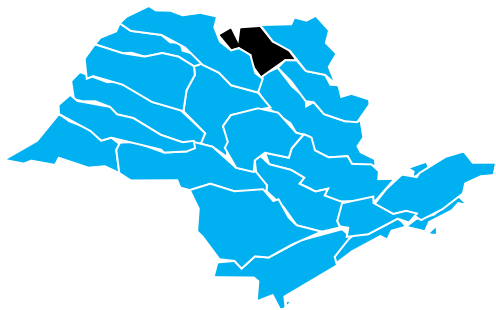
UGRHI 11 Ribeira de Iguape/Litoral Sul | SPI-12



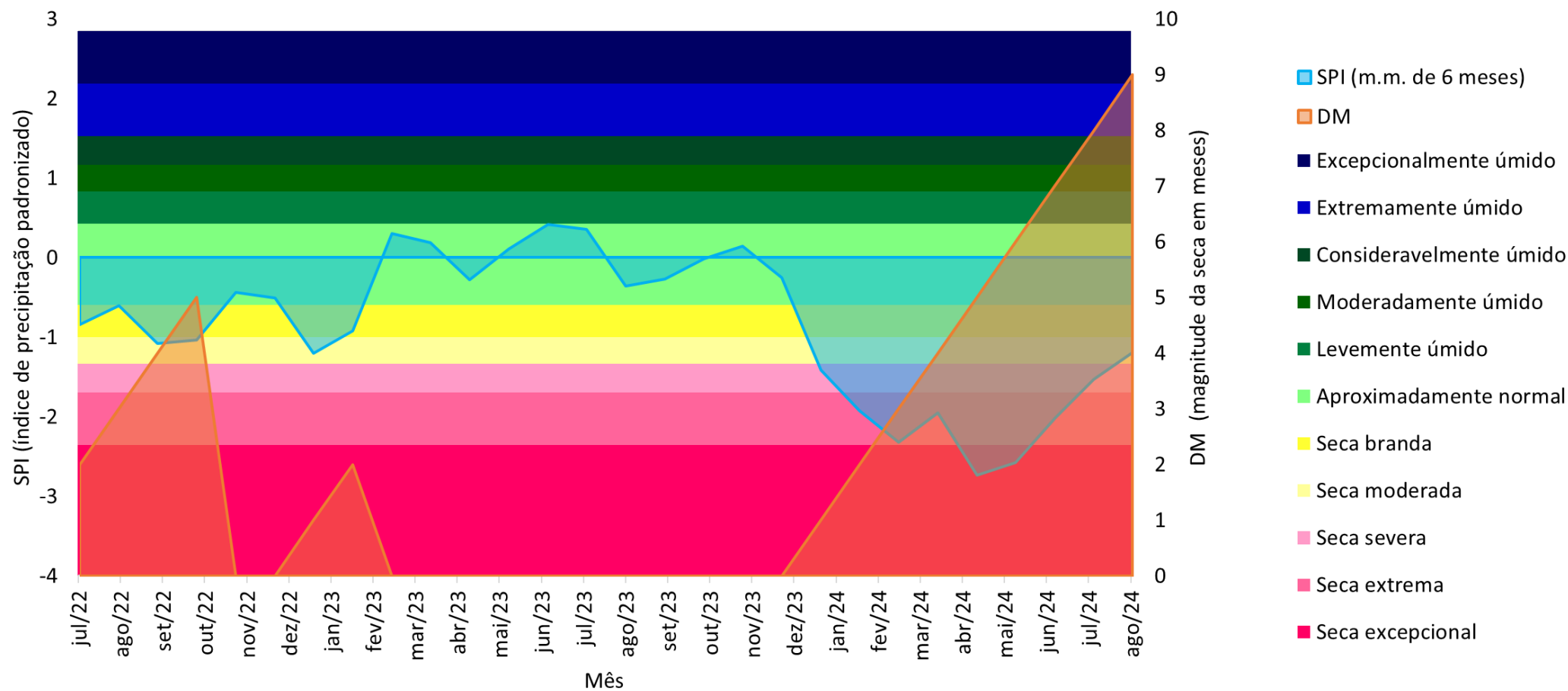
UGRHI 12 - Baixo Pardo/Grande



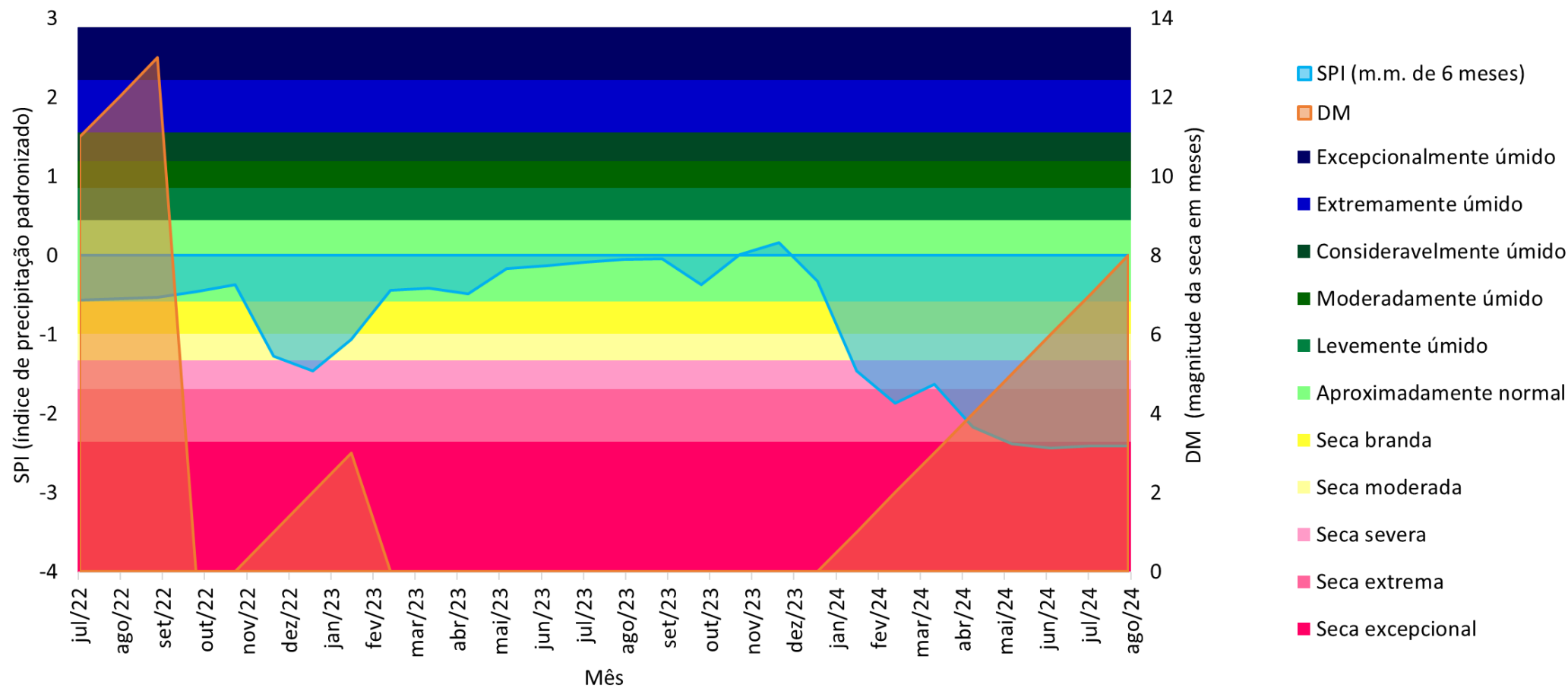
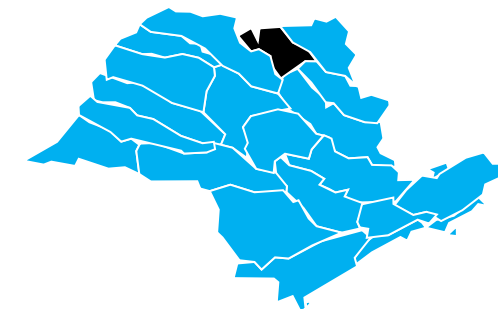
UGRHI 12 - Baixo Pardo/Grande



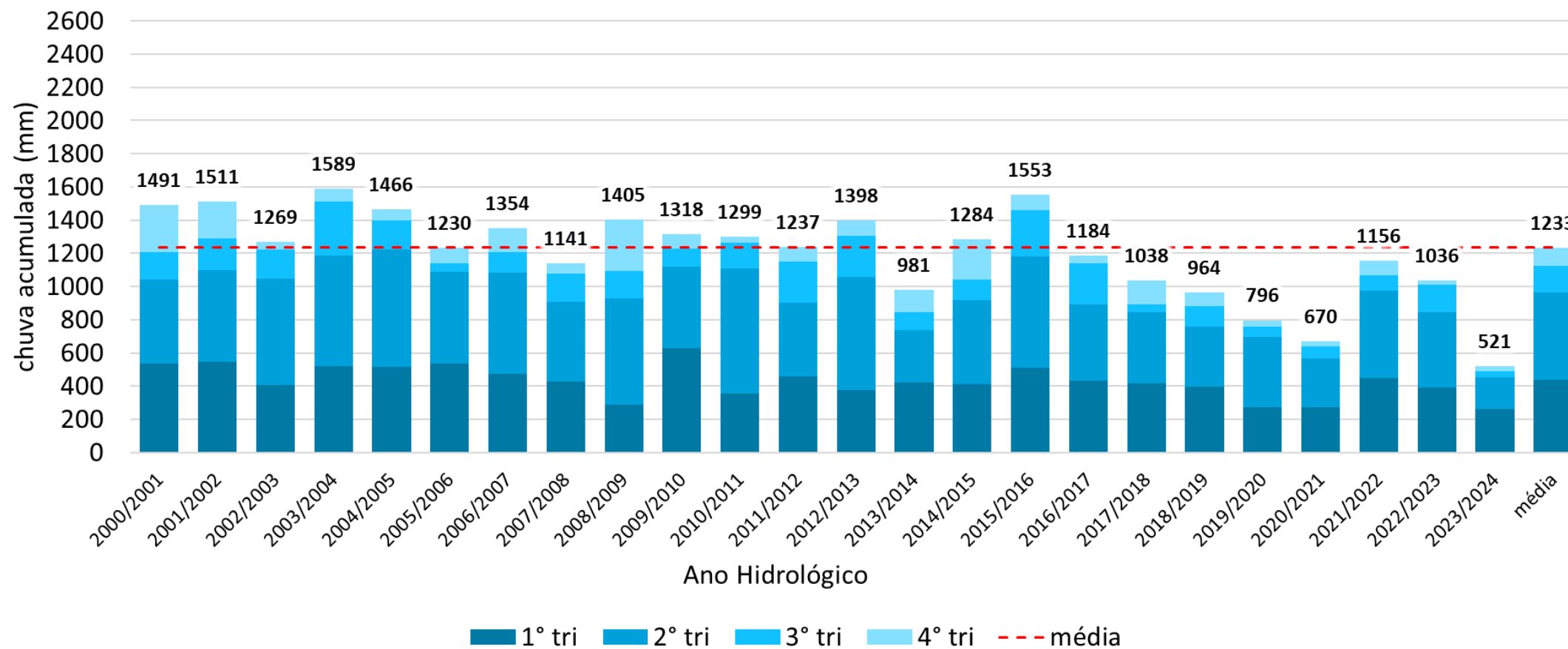
UGRHI 12 - Baixo Pardo/Grande | SPI-06



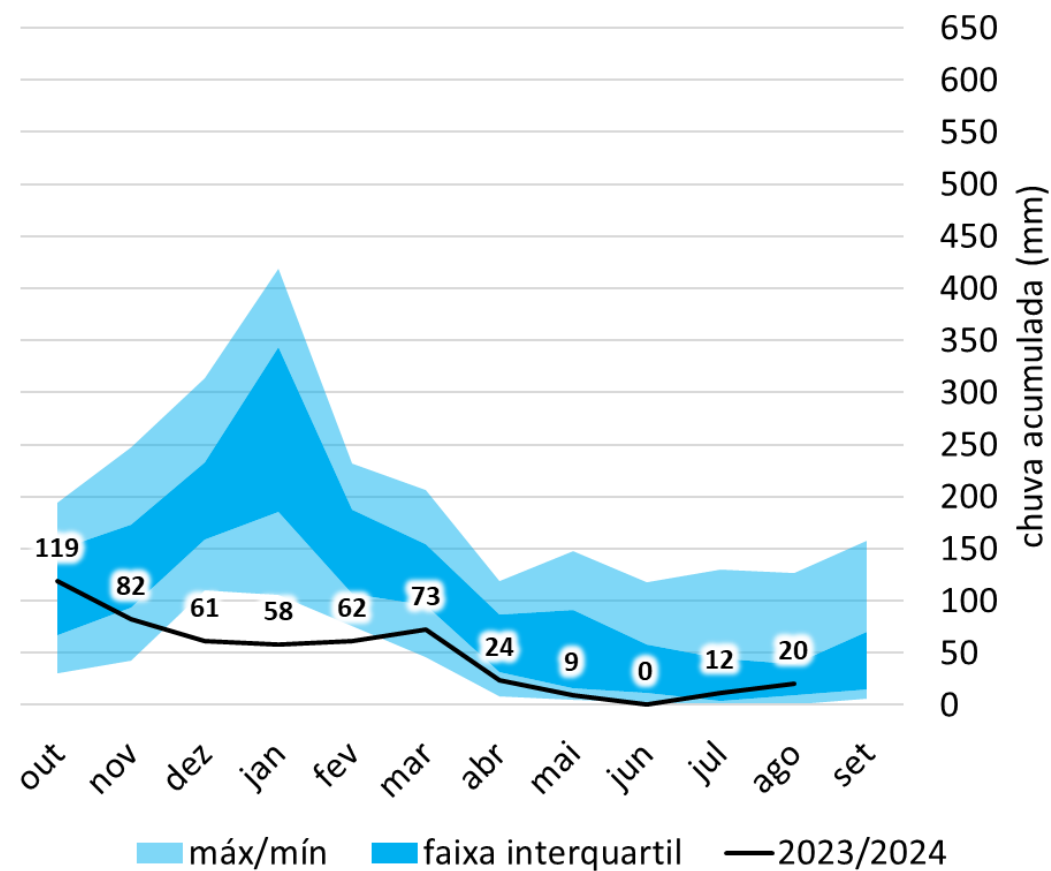
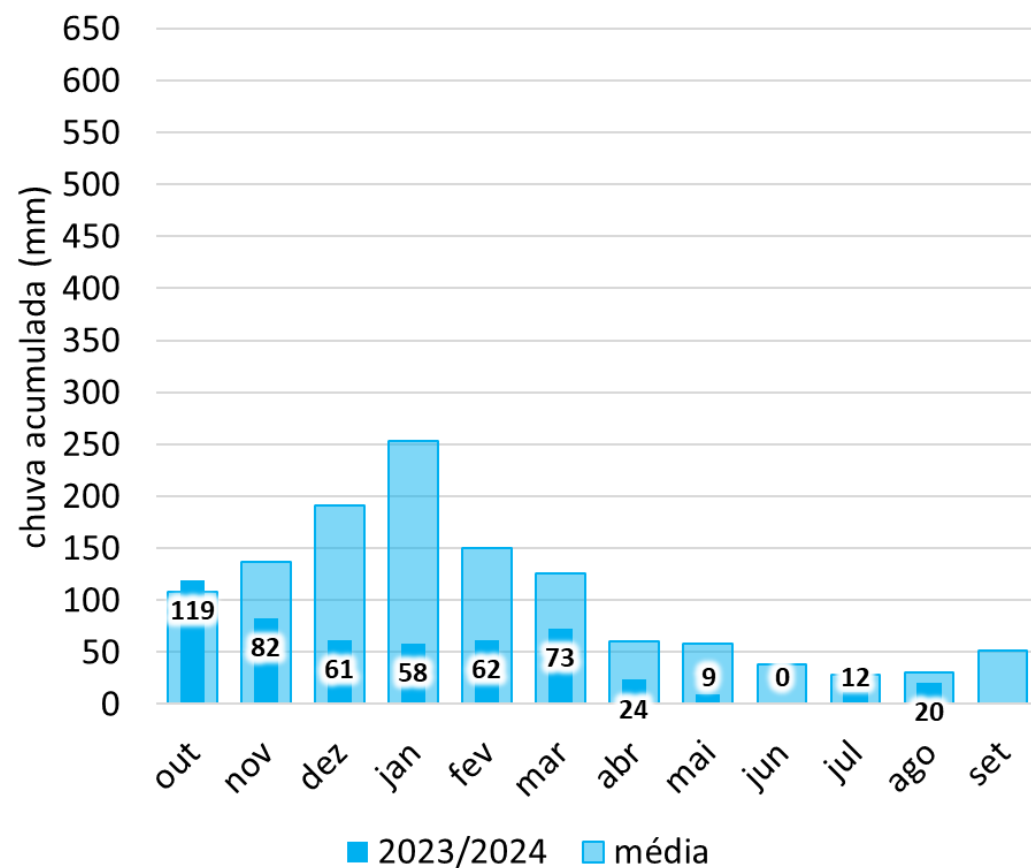
UGRHI 12 - Baixo Pardo/Grande | SPI-12



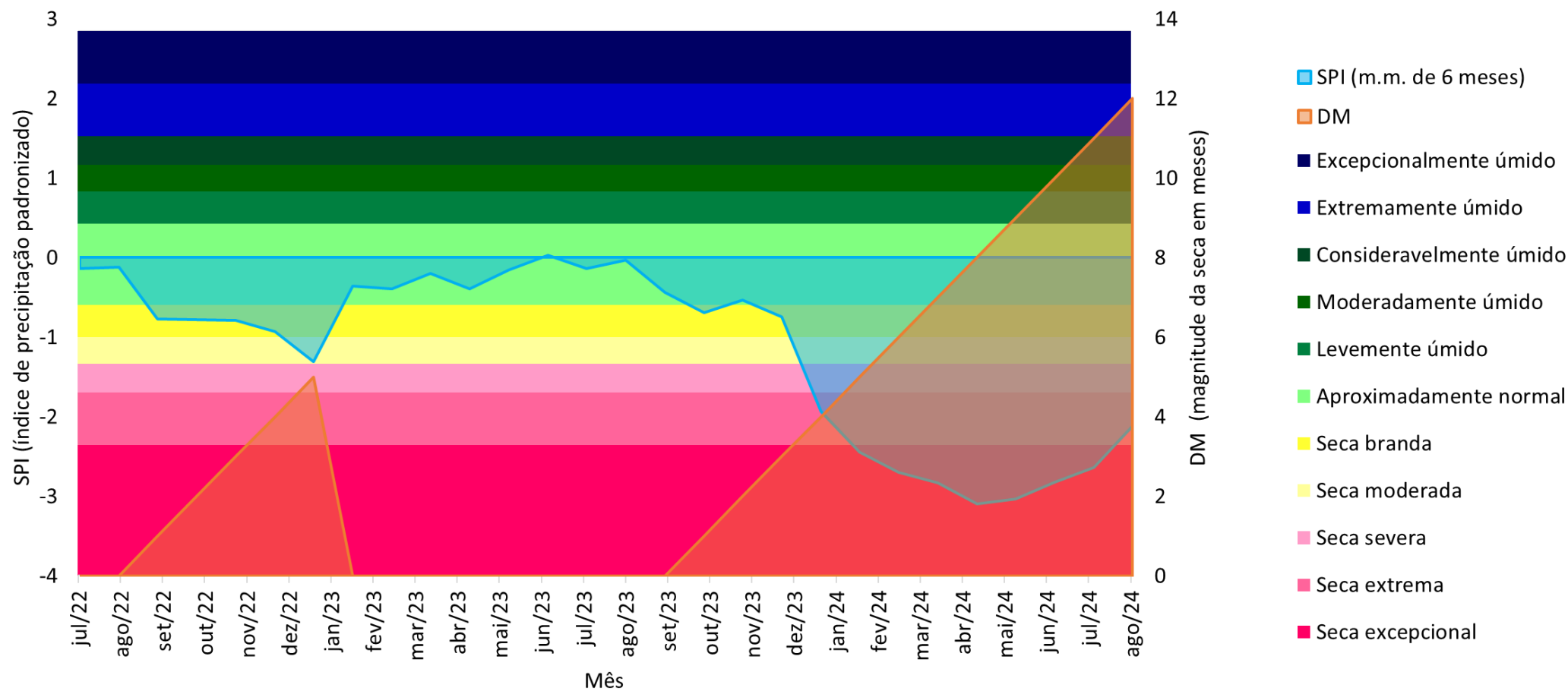
UGRHI 13 - Tietê/Jacararé



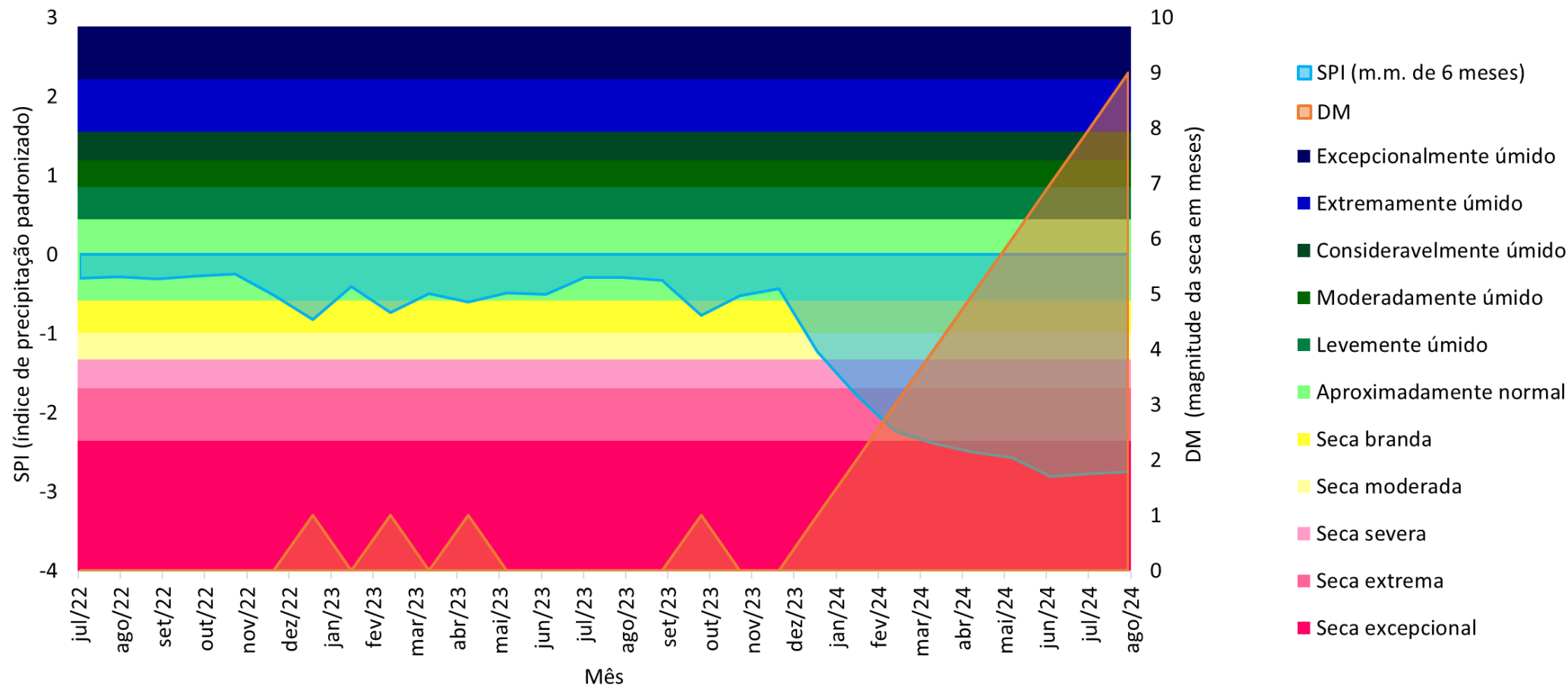
UGRHI 13 - Tietê/Jacaré



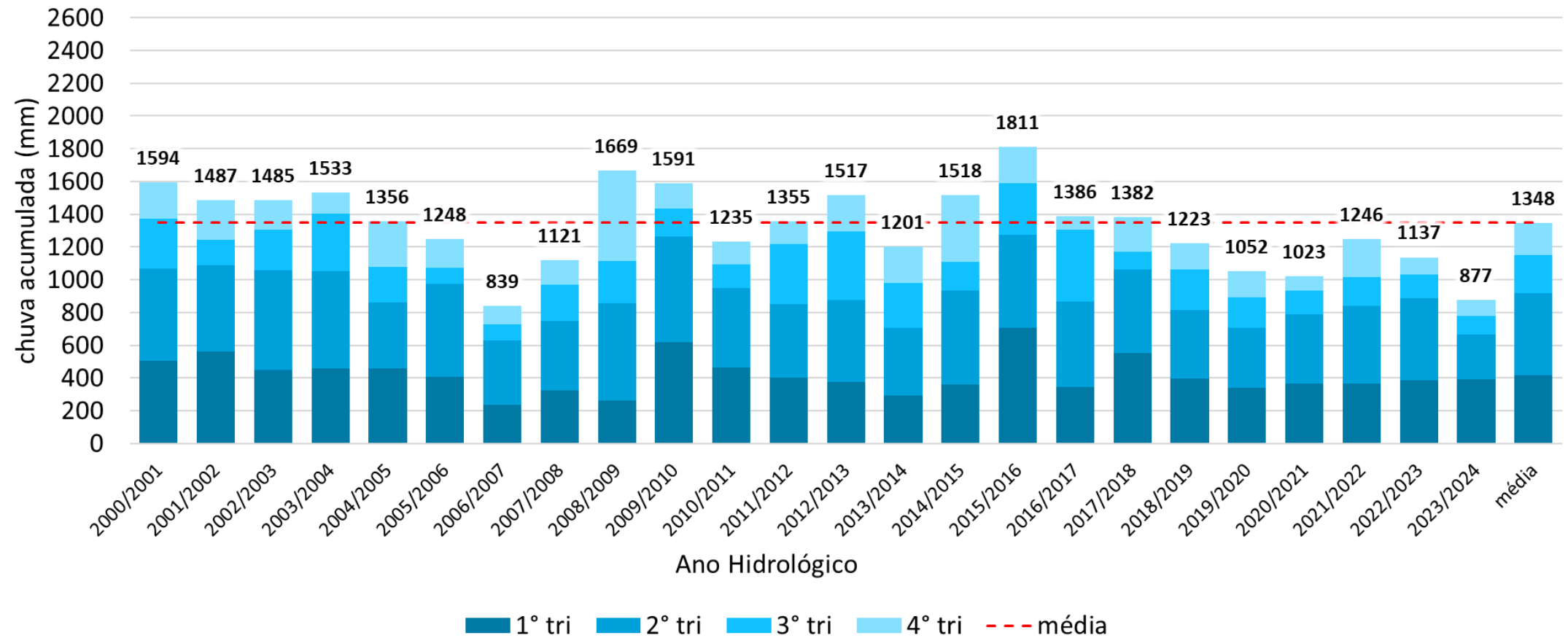
UGRHI 13 - Tietê/Jacararé | SPI-06



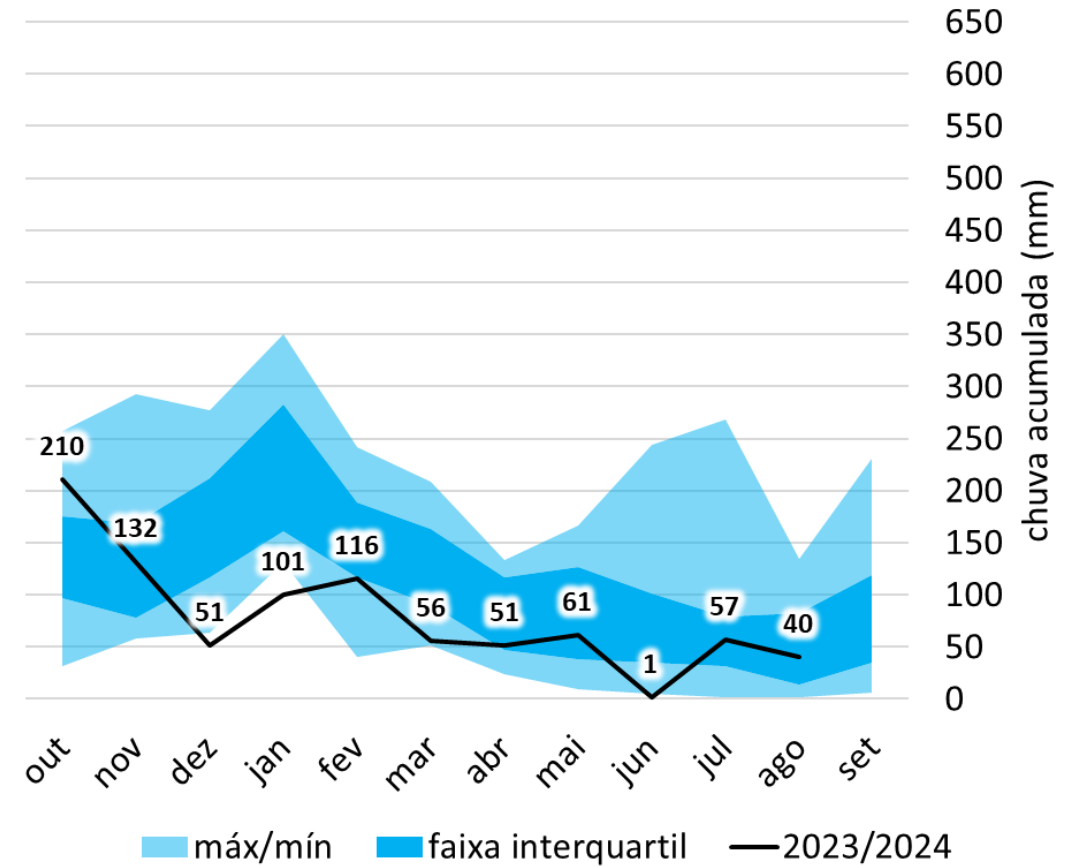
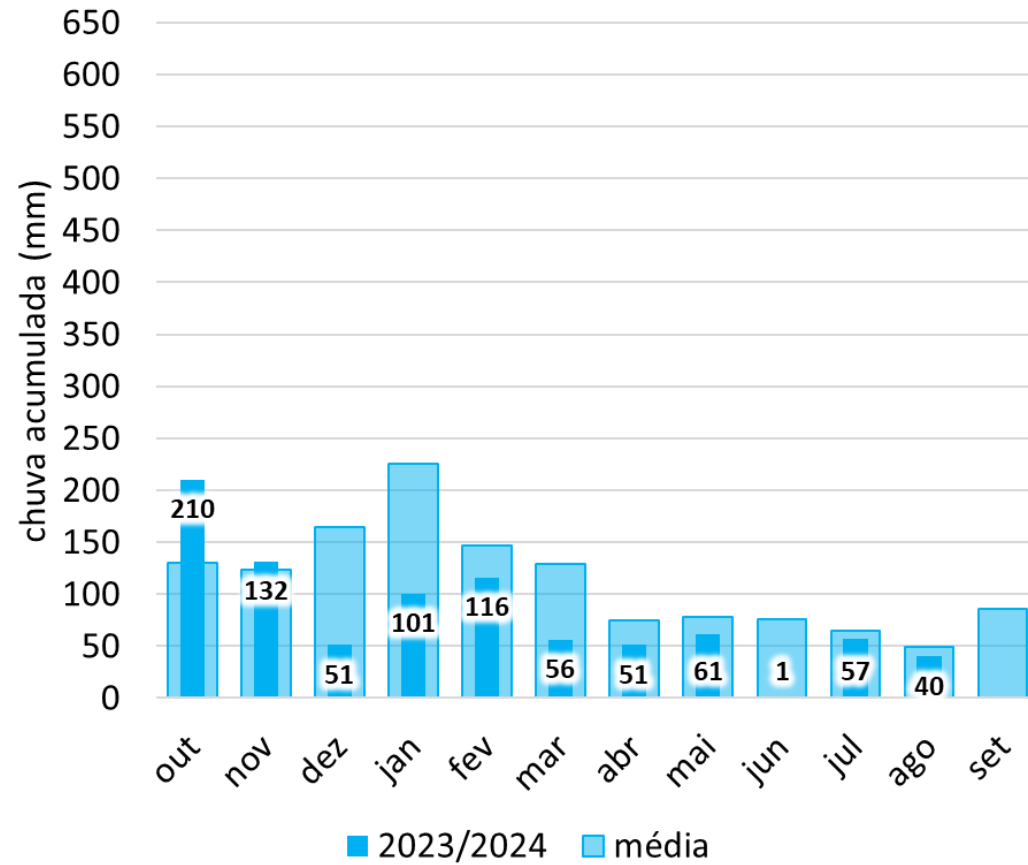
UGRHI 13 - Tietê/Jacararé | SPI-12



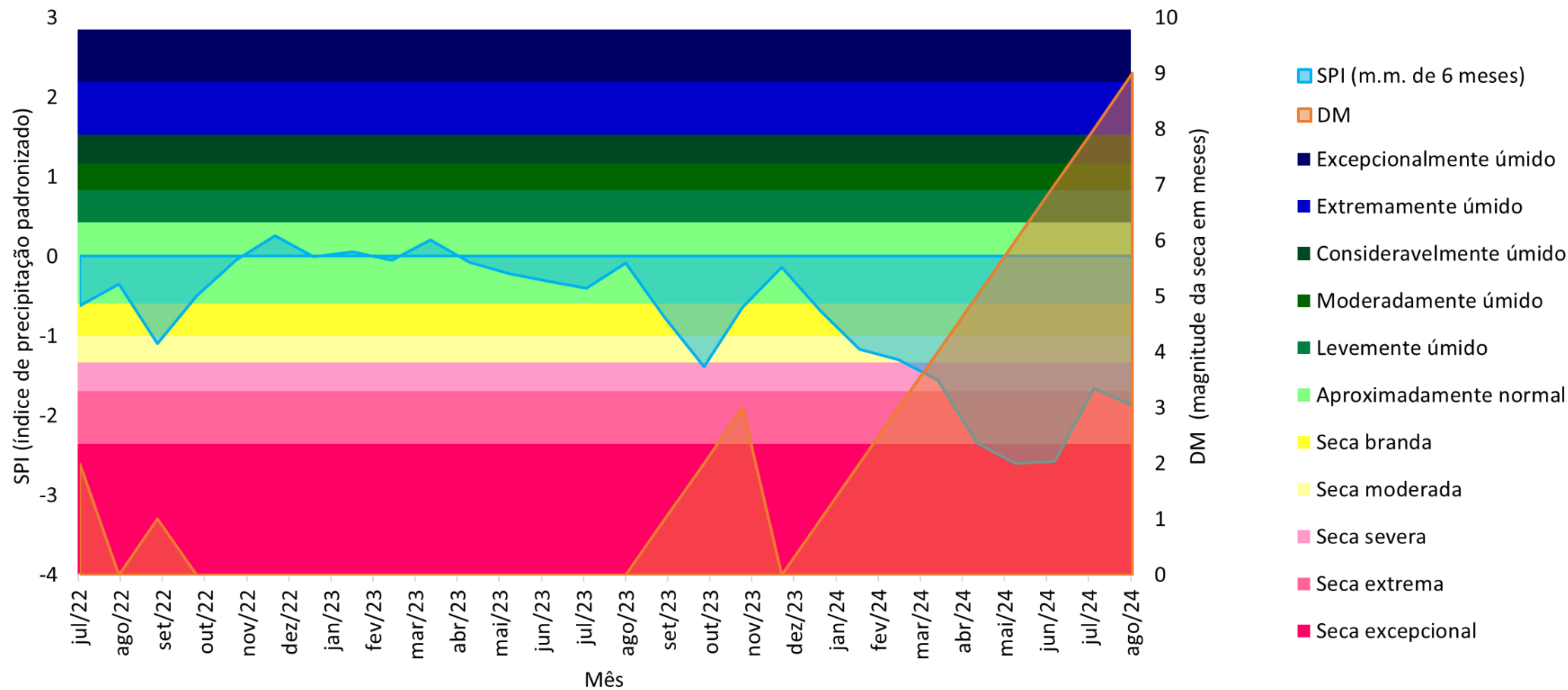
UGRHI 14 - Alto Paranapanema



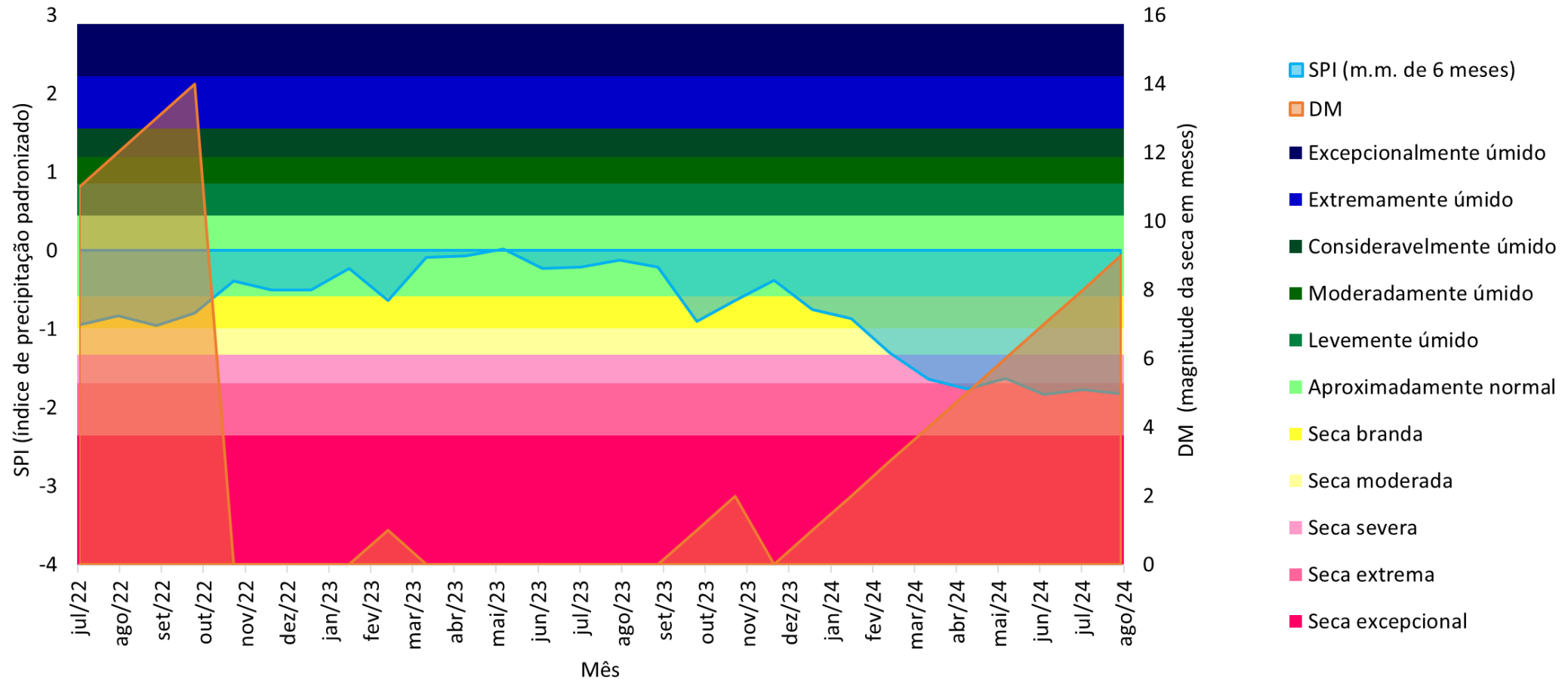
UGRHI 14 - Alto Paranapanema



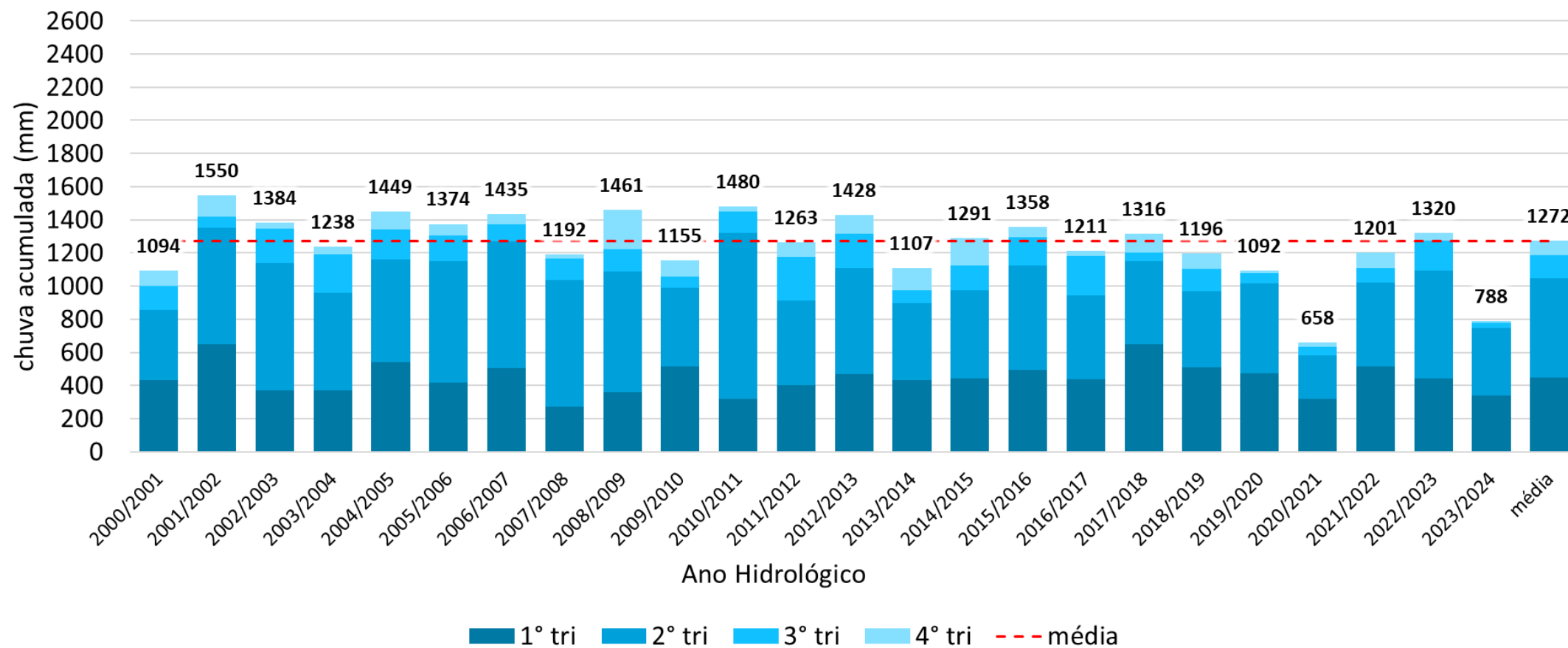
UGRHI 14 - Alto Paranapanema | SPI-06



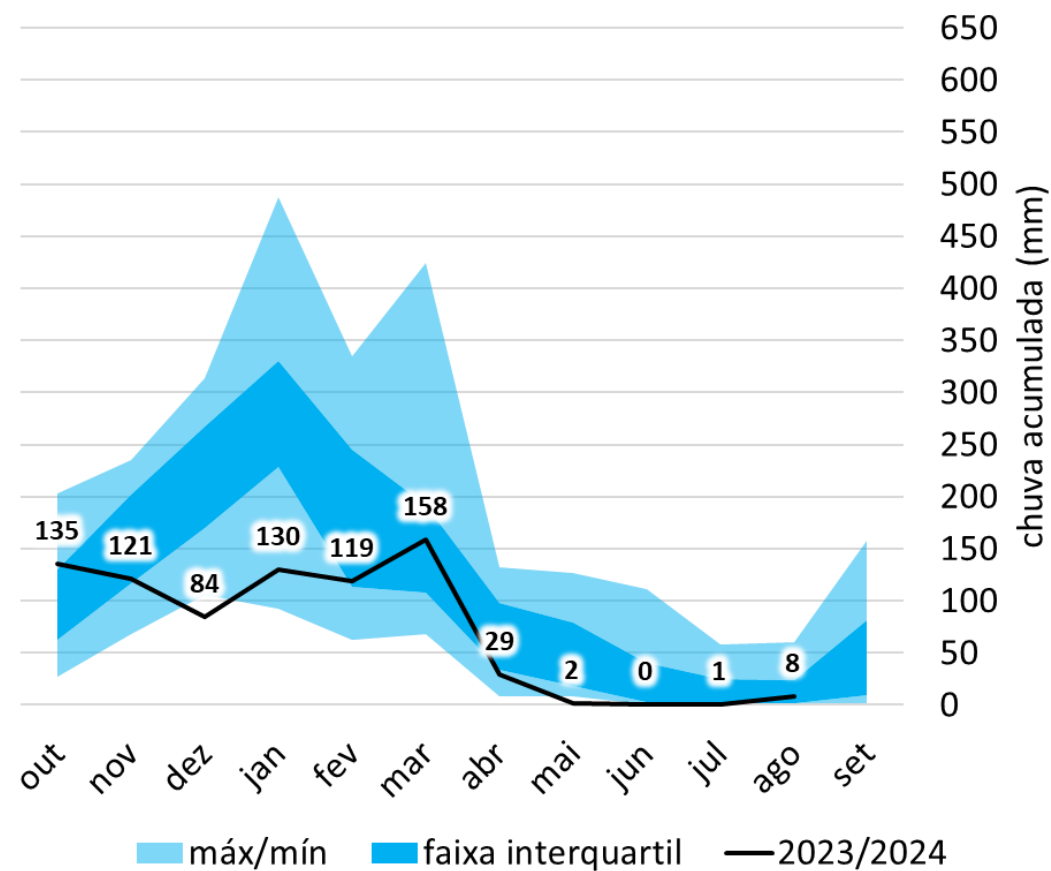
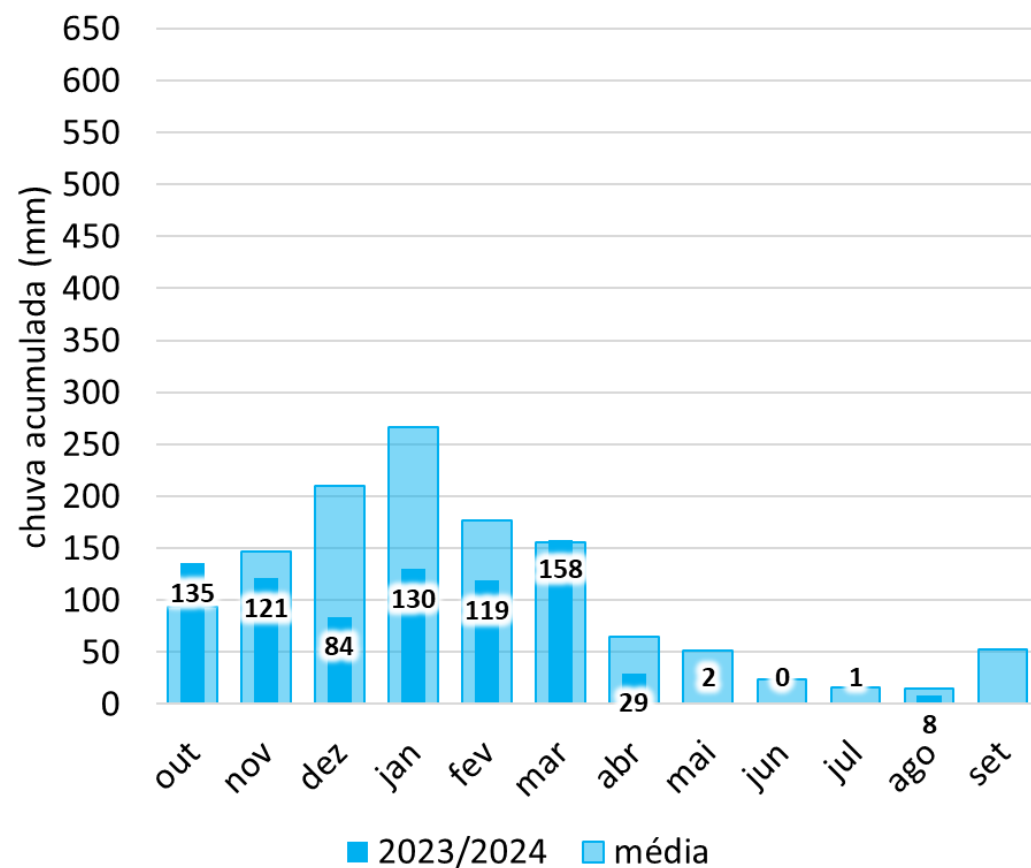
UGRHI 14 - Alto Paranapanema | SPI-12



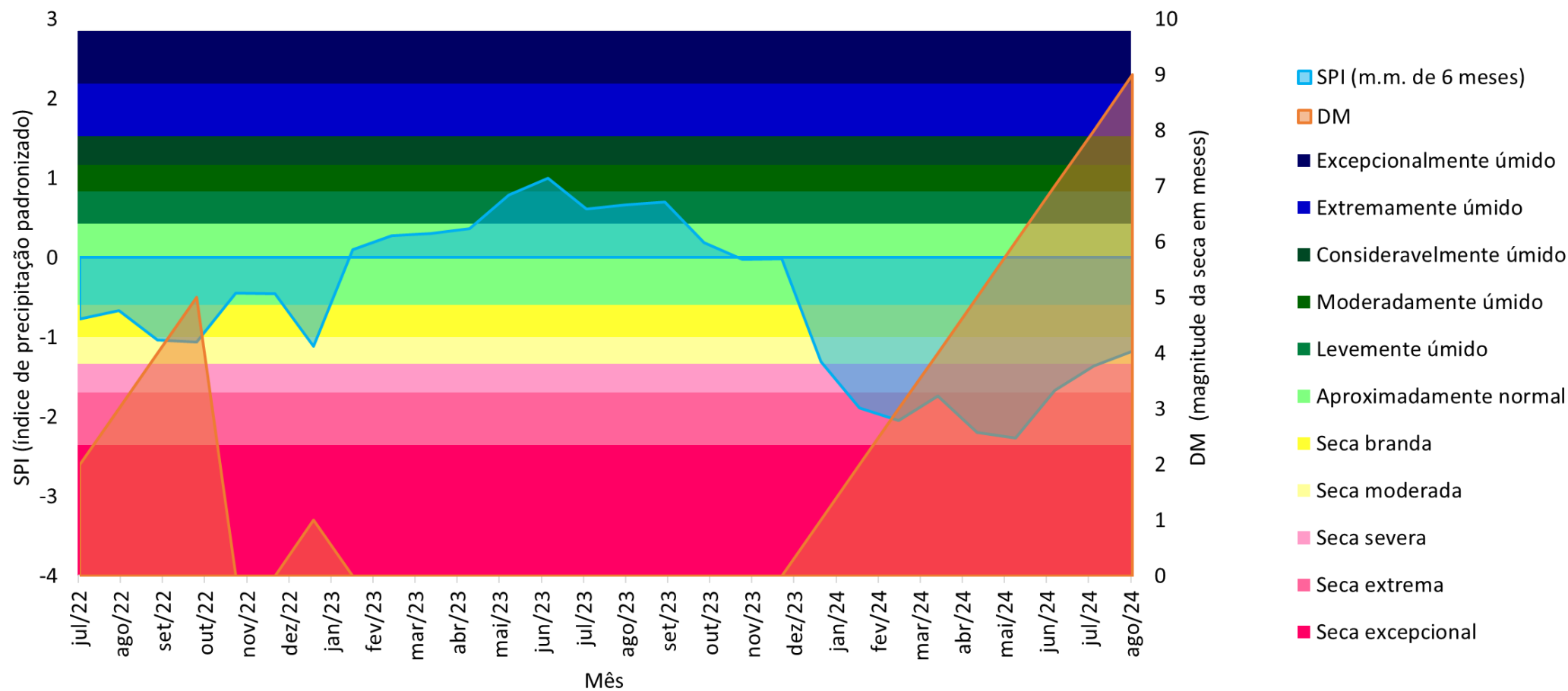
UGRHI 15 - Turvo/Grande



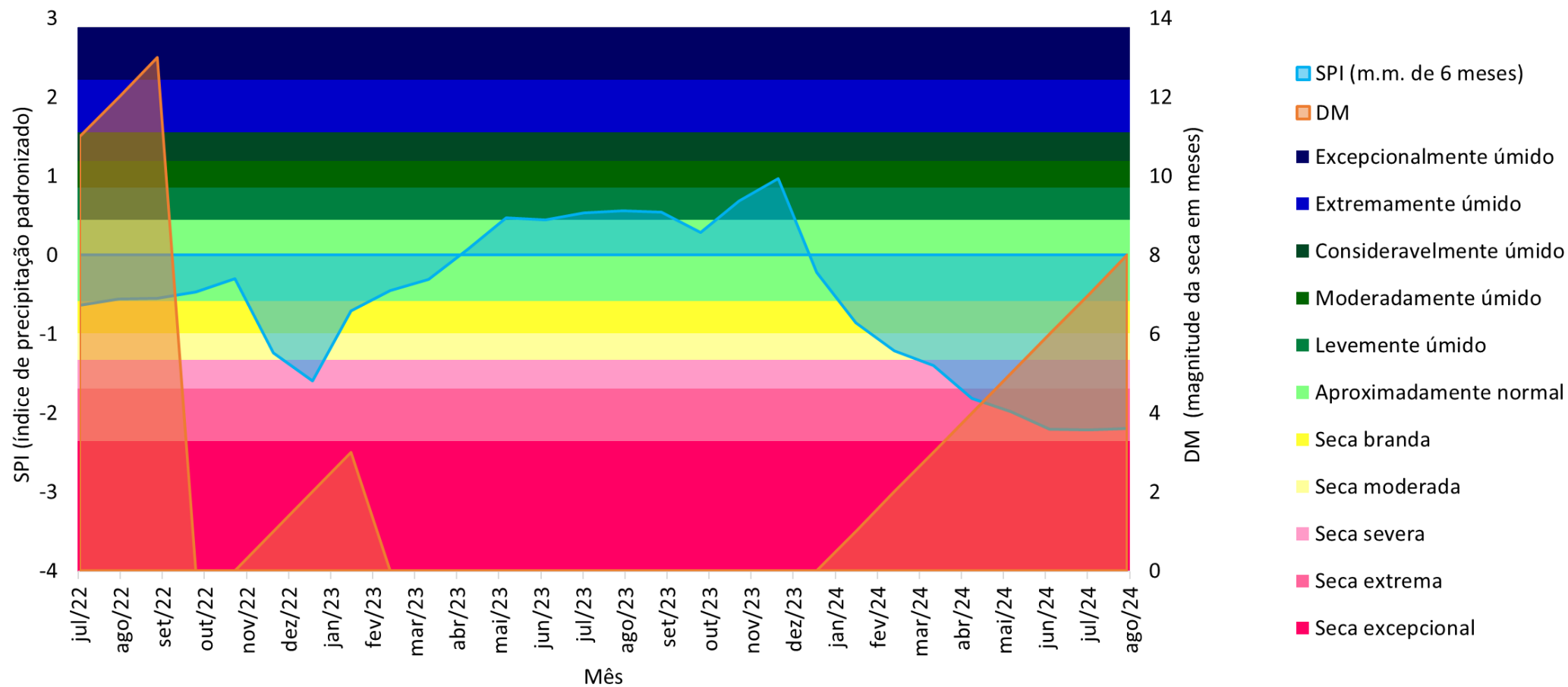
UGRHI 15 - Turvo/Grande



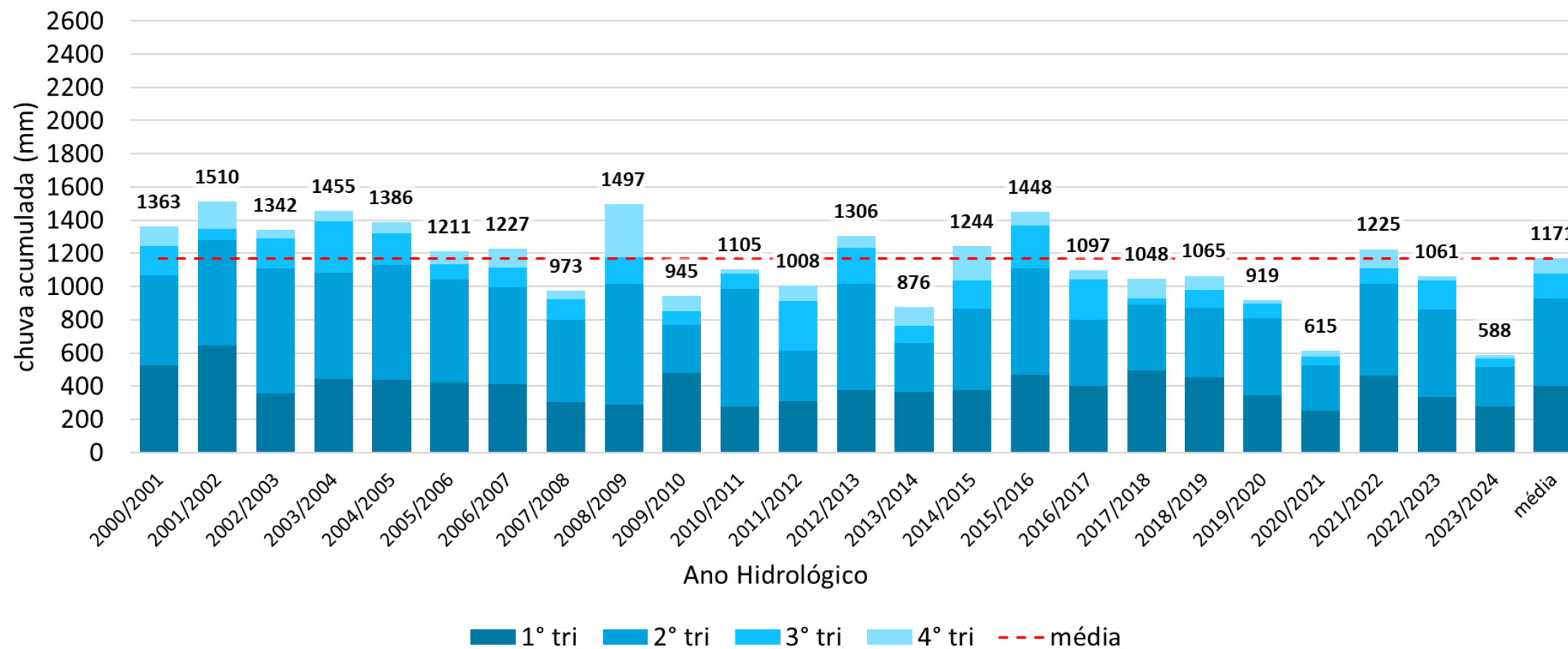
UGRHI 15 - Turvo/Grande | SPI-06



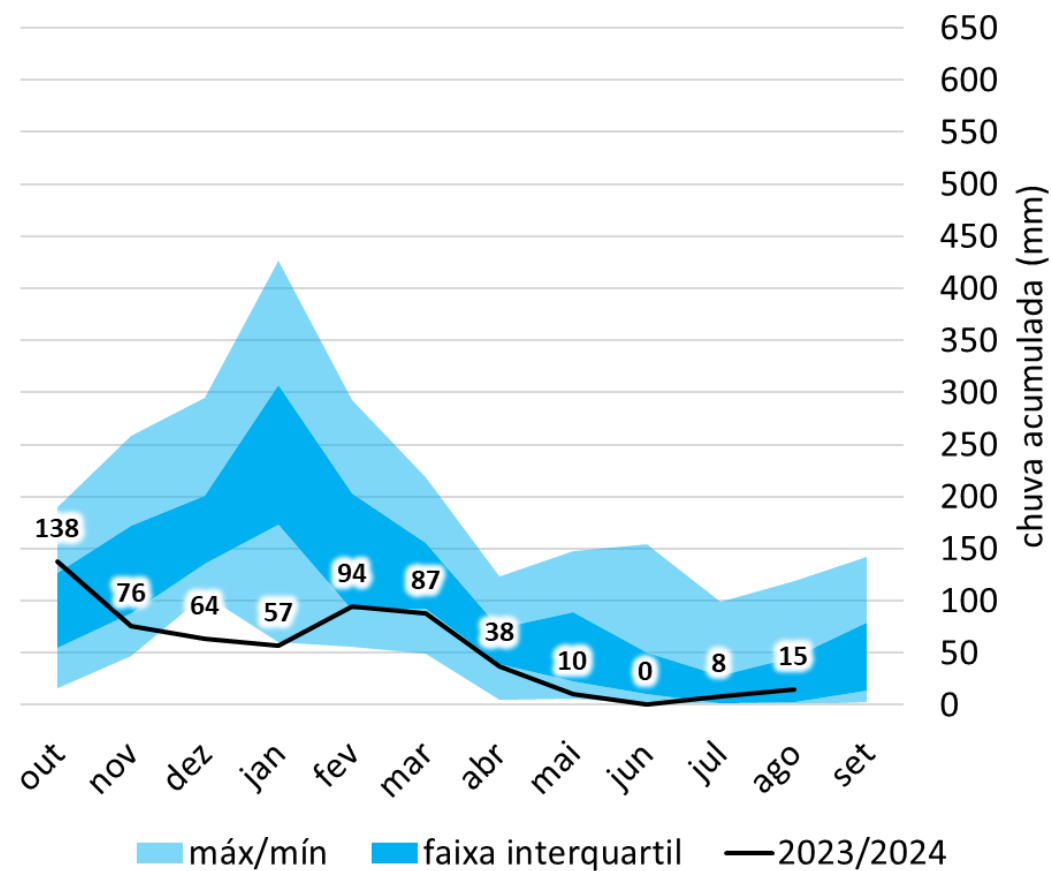
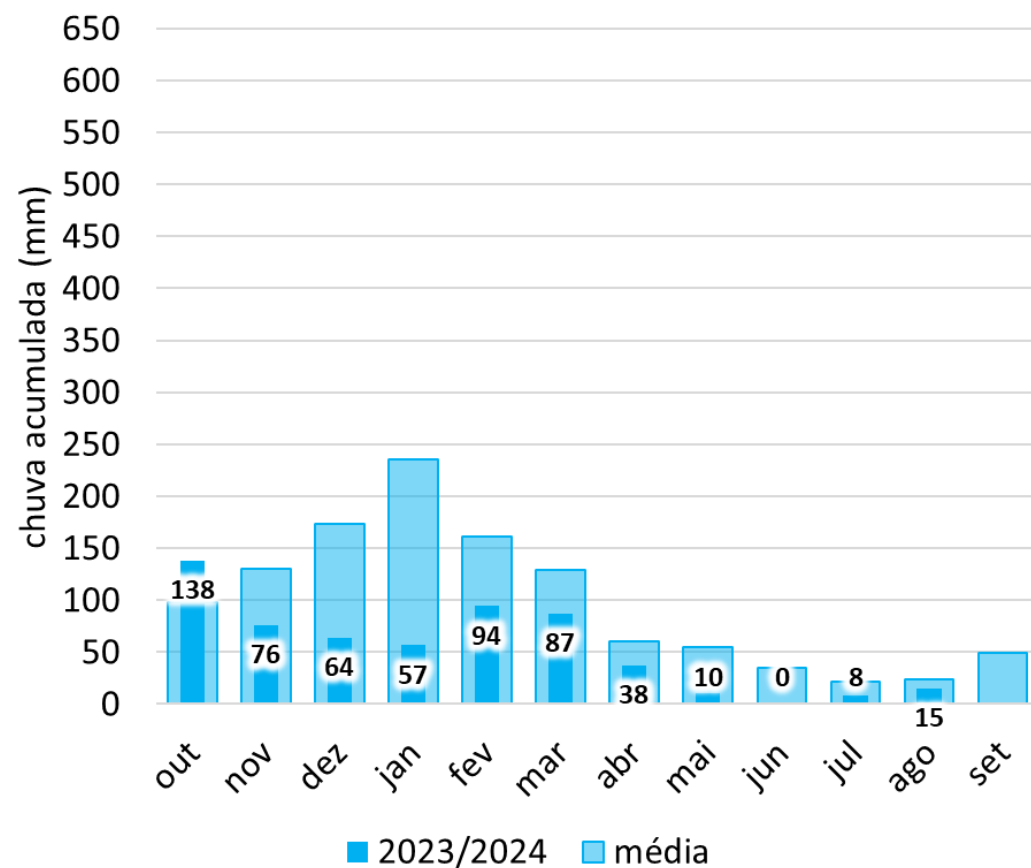
UGRHI 15 - Turvo/Grande | SPI-12



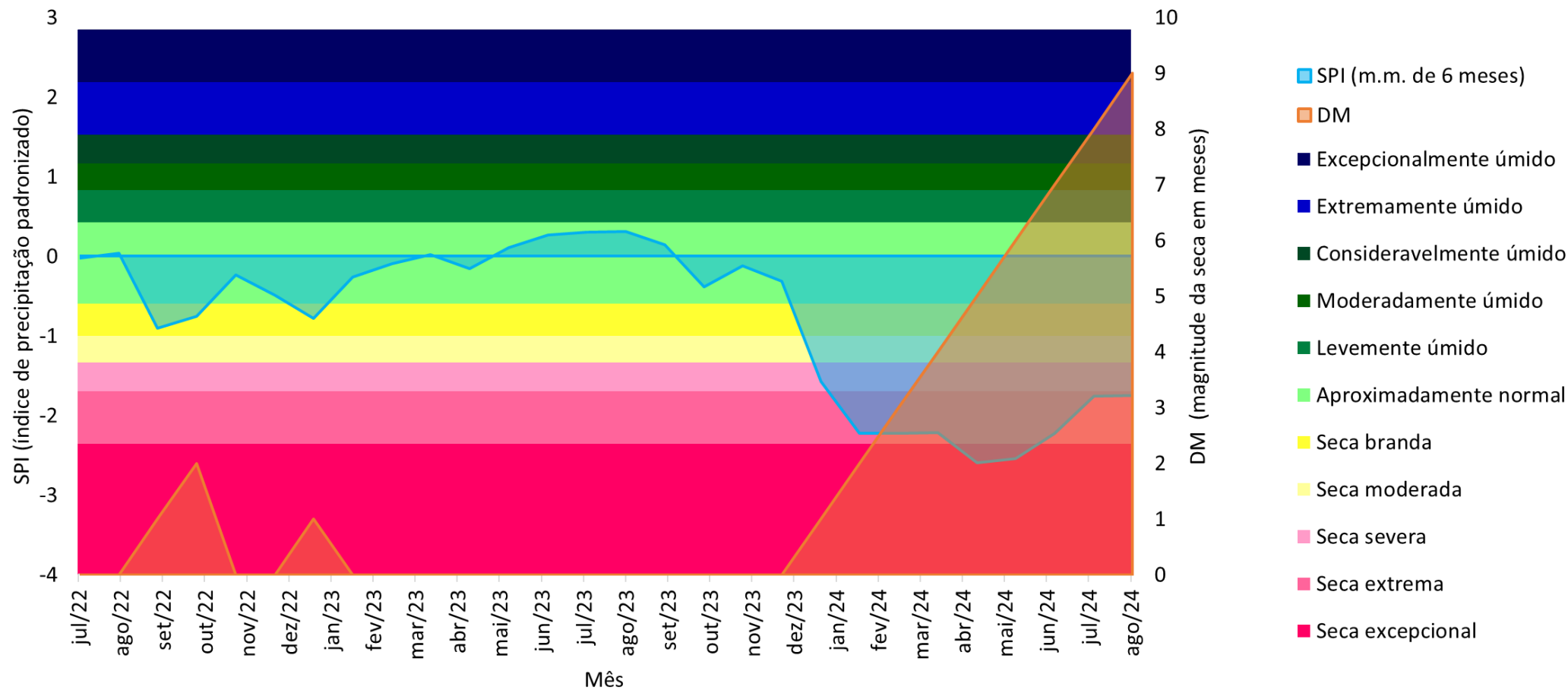
UGRHI 16 - Tietê/Batalha



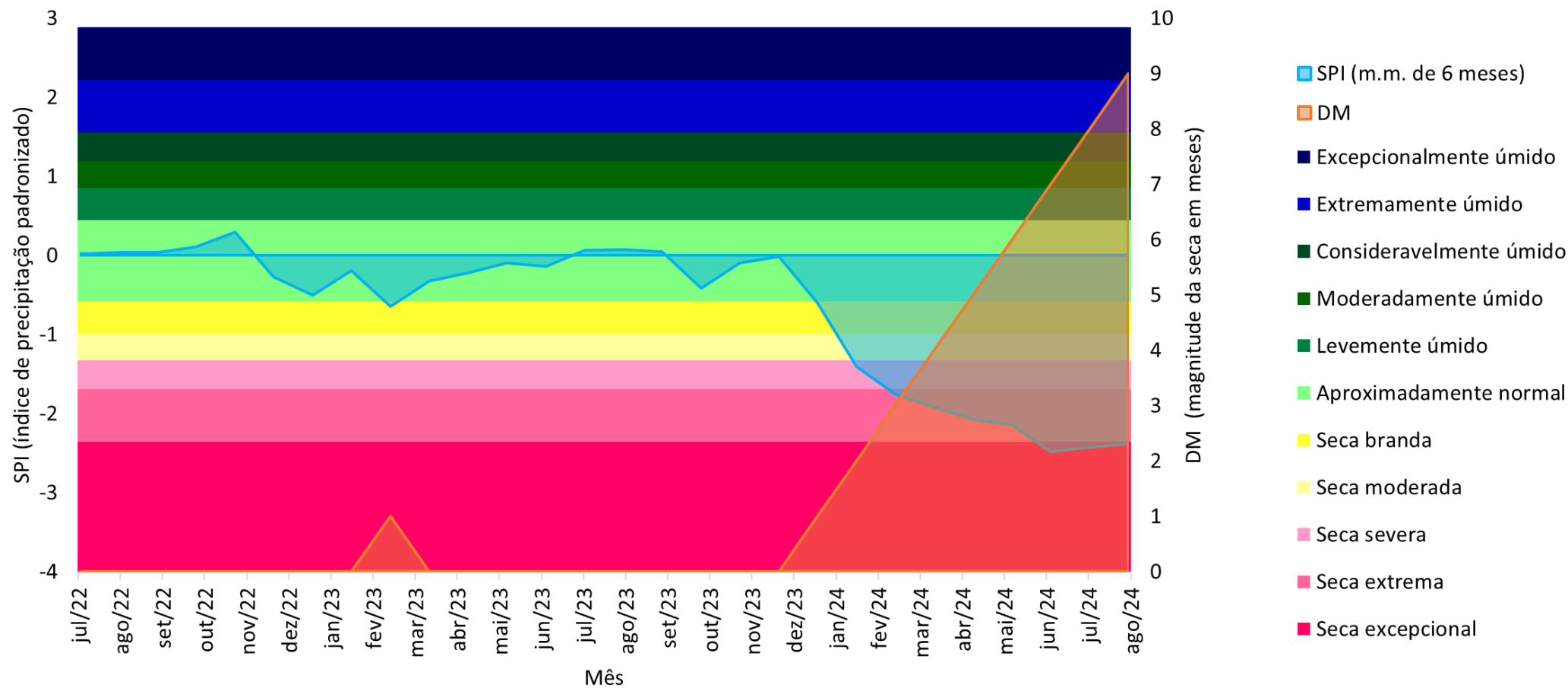
UGRHI 16 - Tietê/Batalha



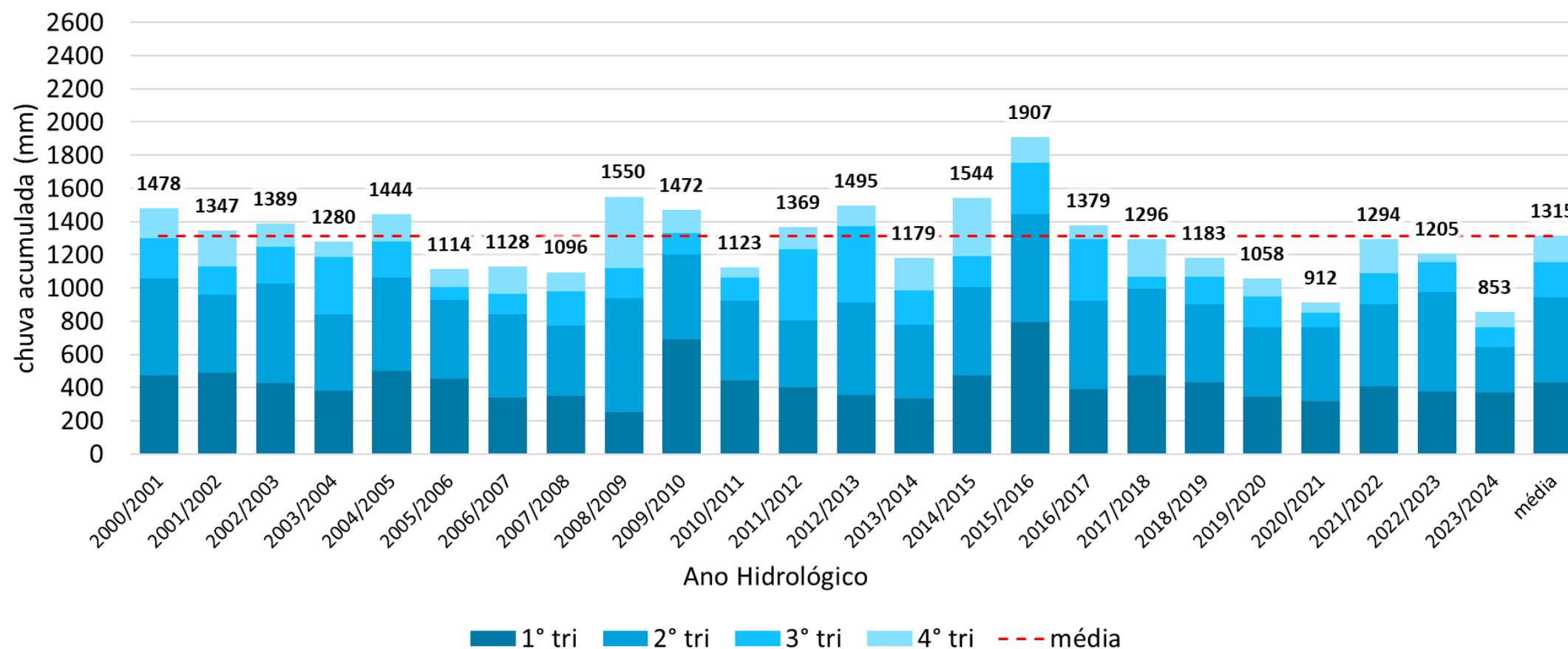
UGRHI 16 - Tietê/Batalha | SPI-06



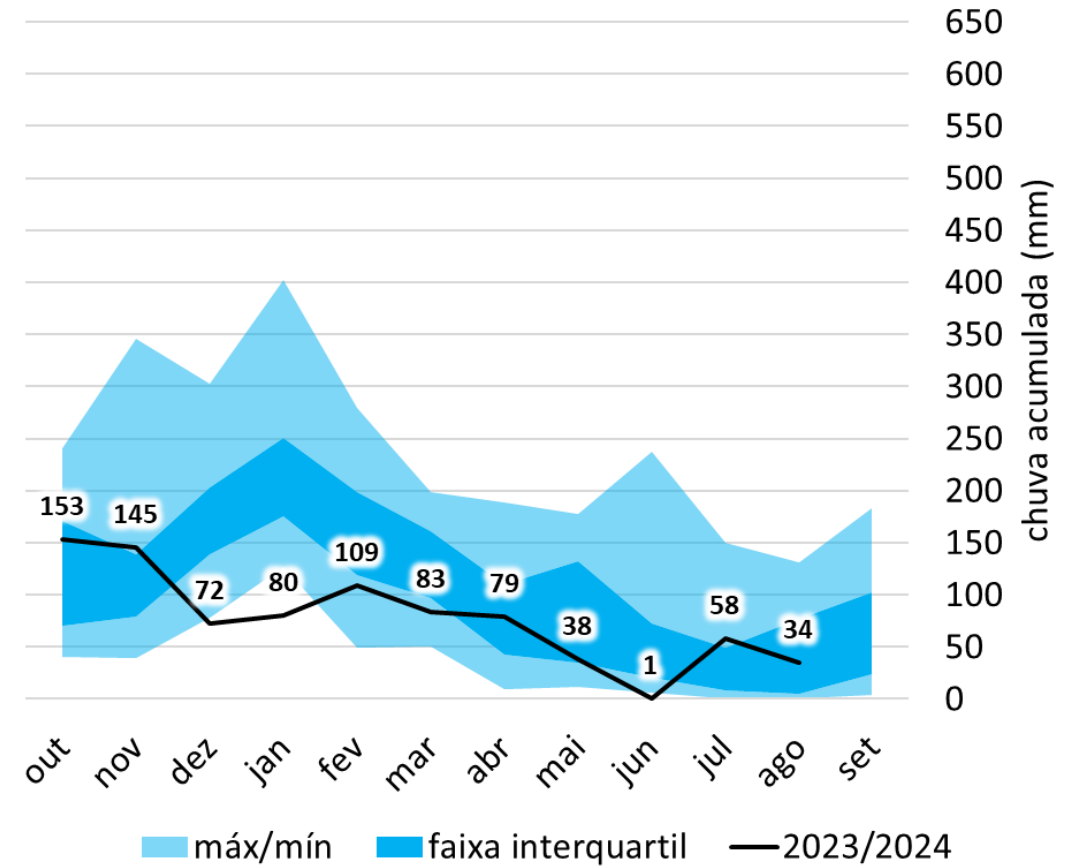
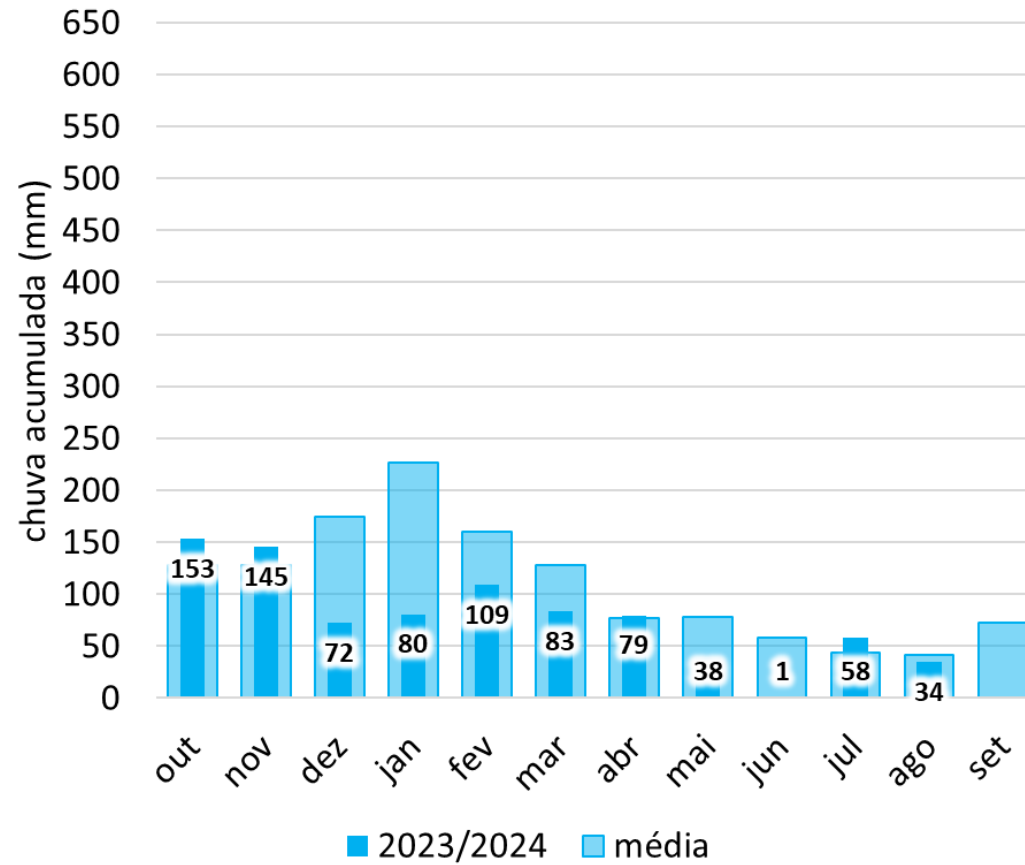
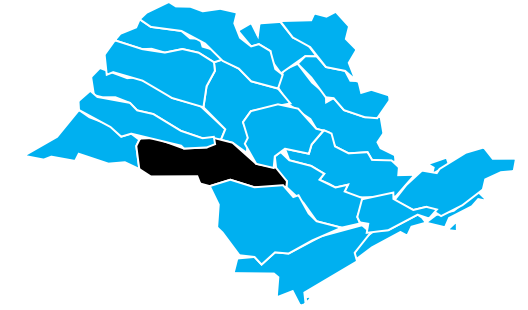
UGRHI 16 - Tietê/Batalha | SPI-12



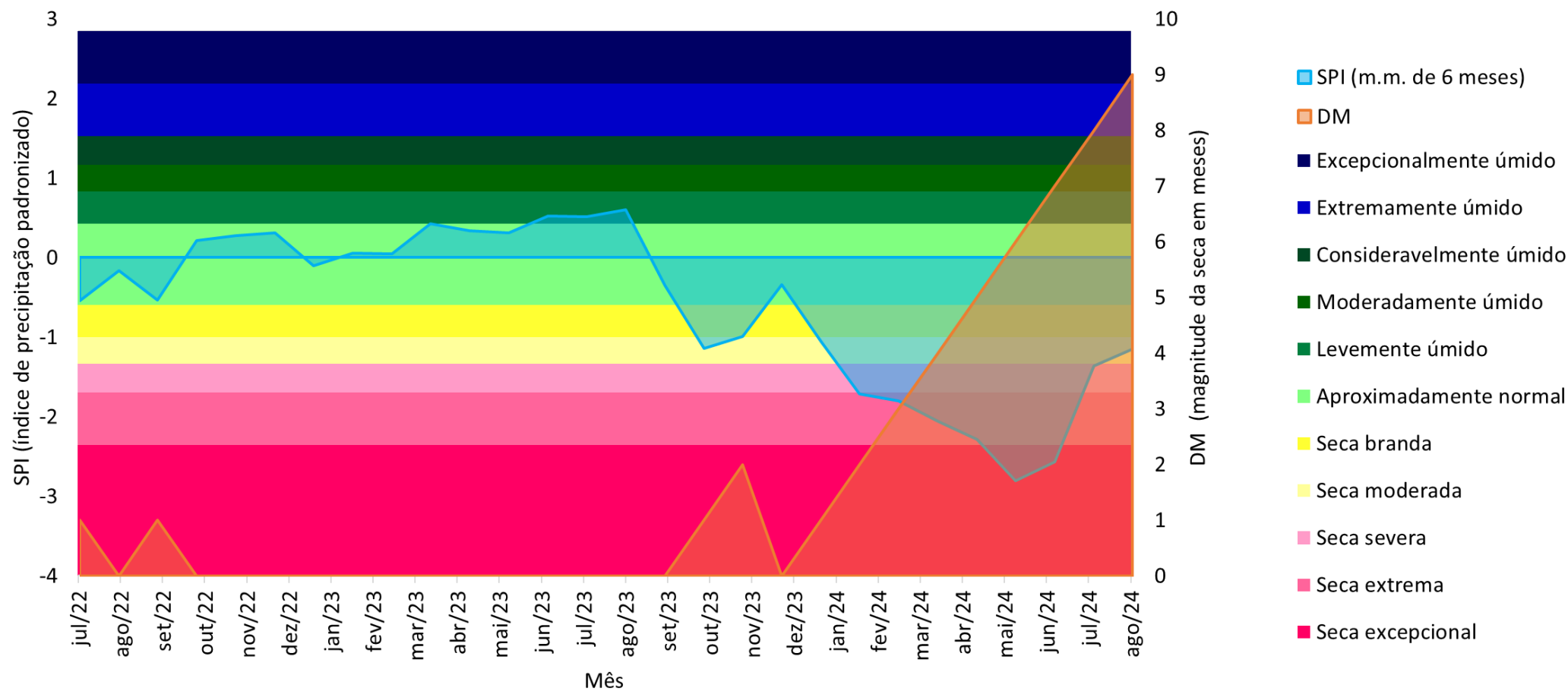
UGRHI 17 - Médio Paranapanema



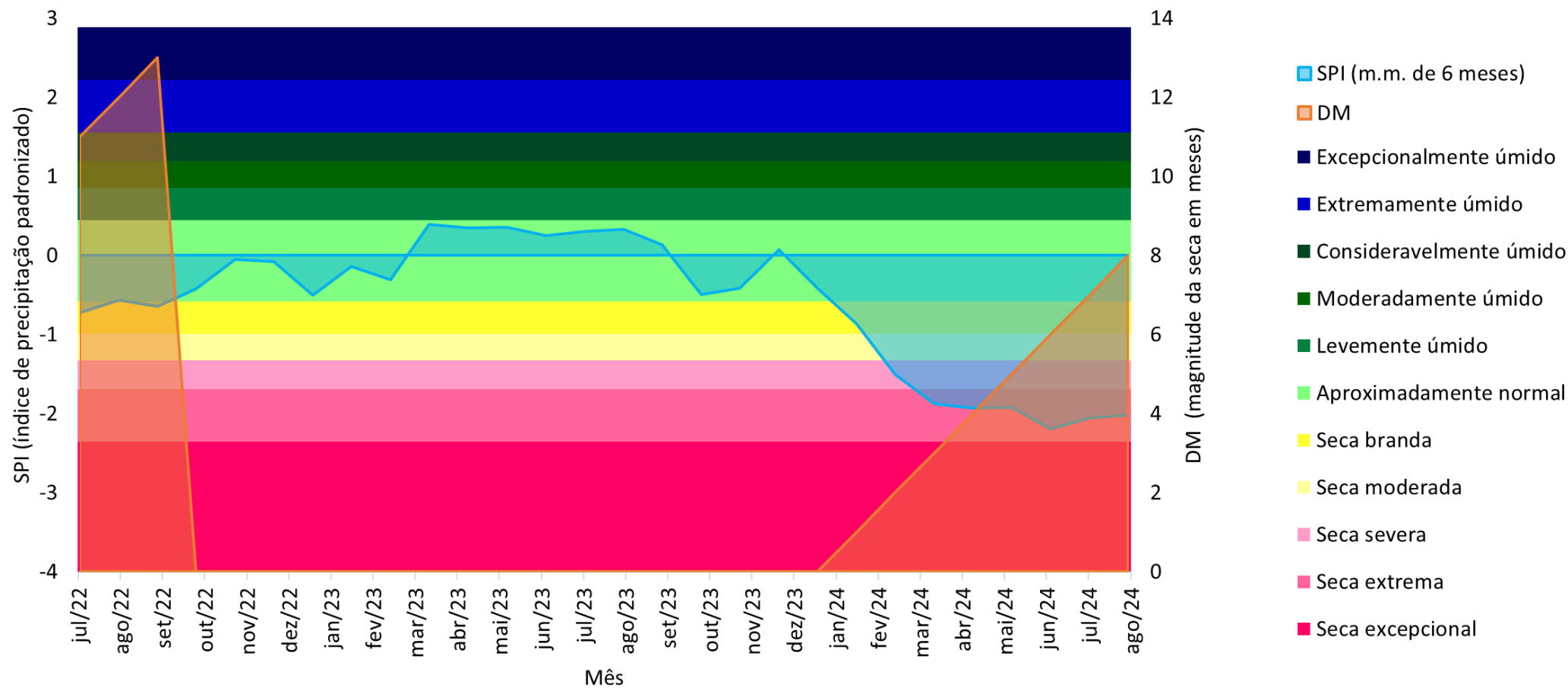
UGRHI 17 - Médio Paranapanema



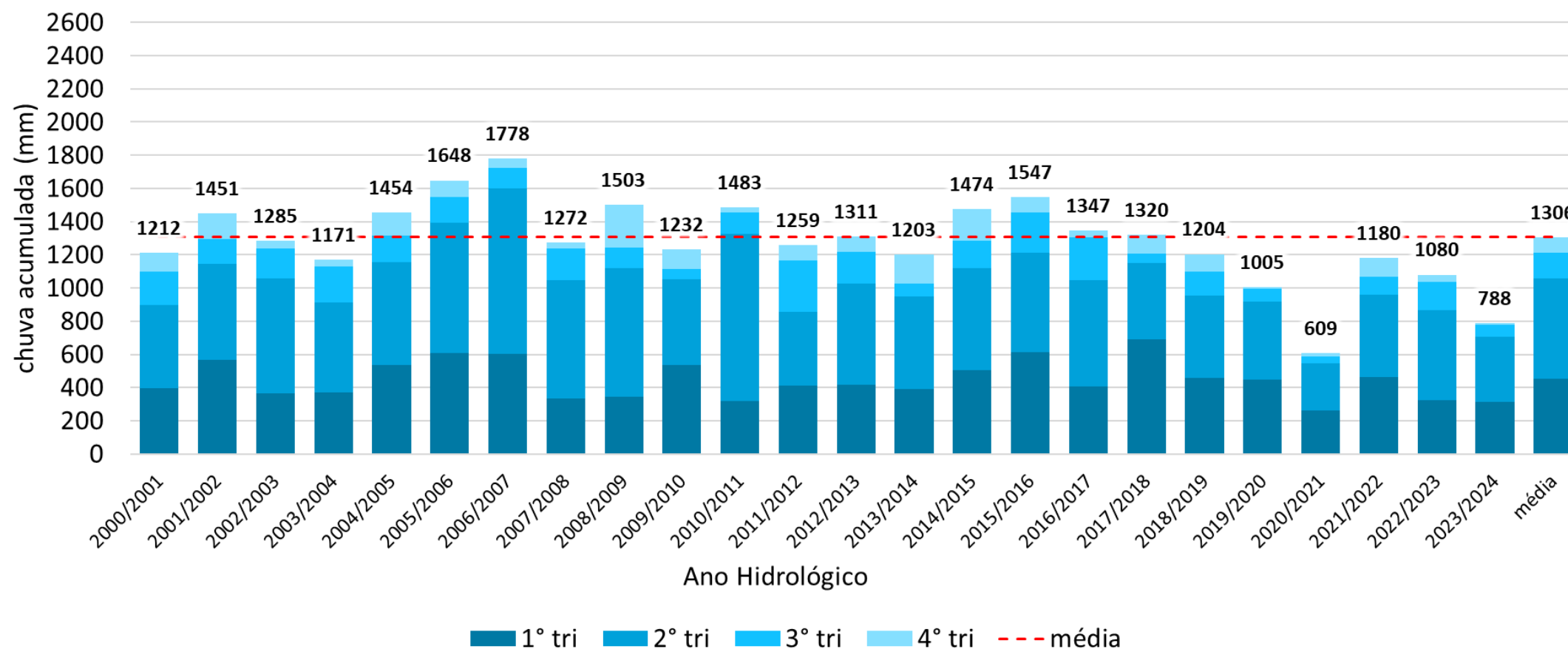
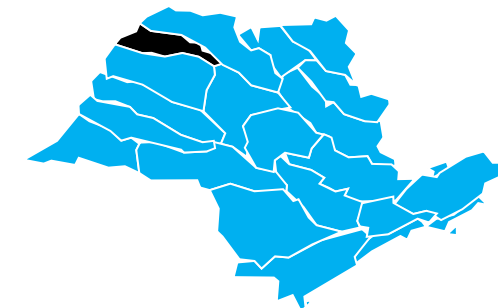
UGRHI 17 - Médio Paranapanema | SPI-06



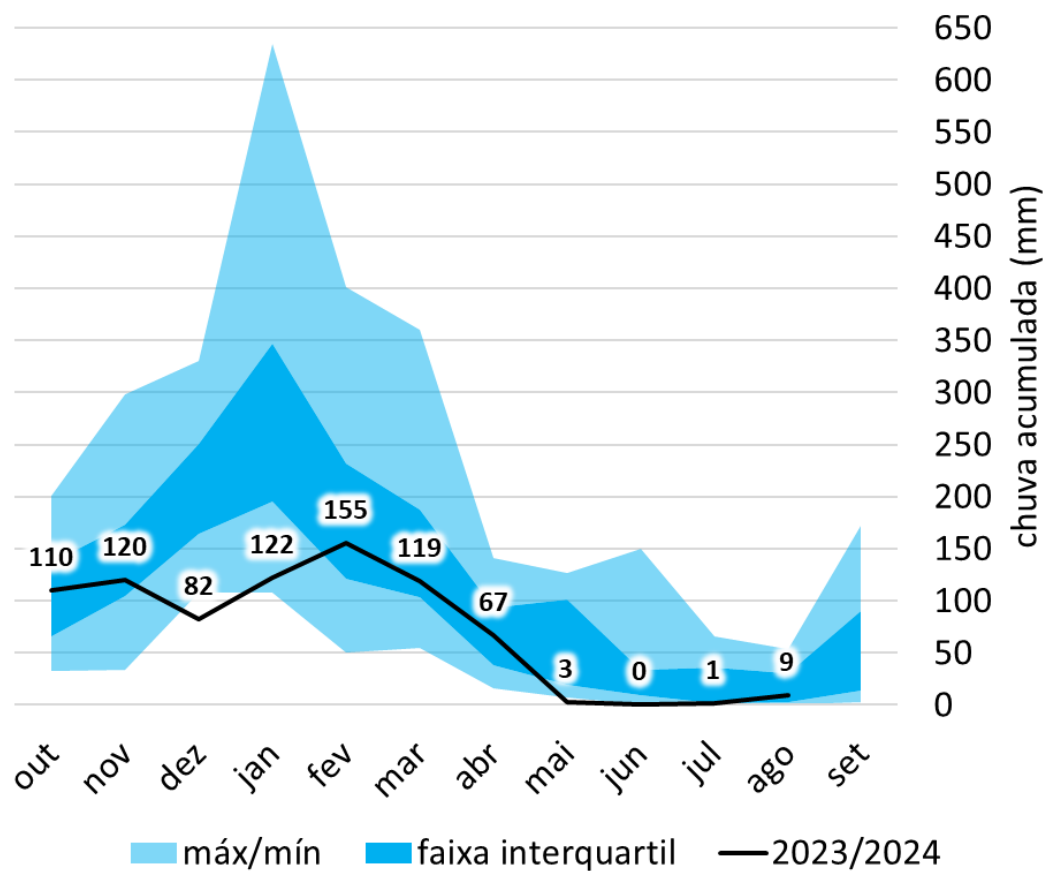
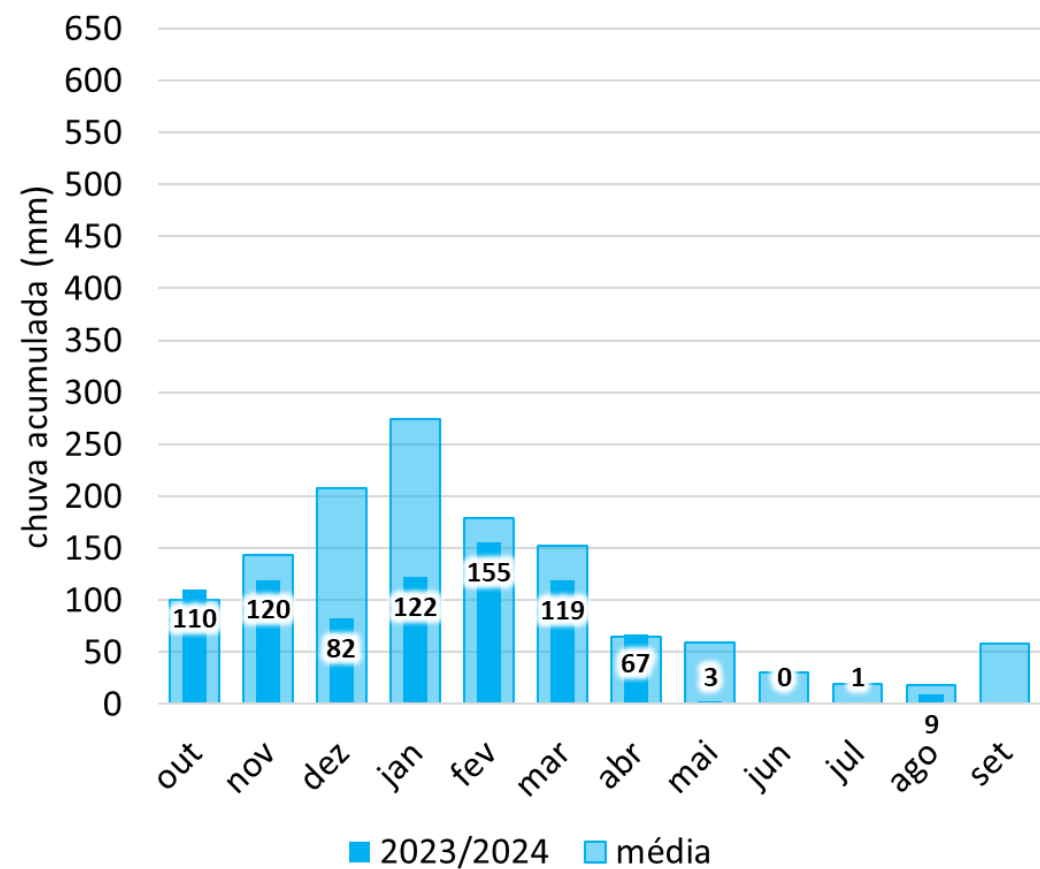
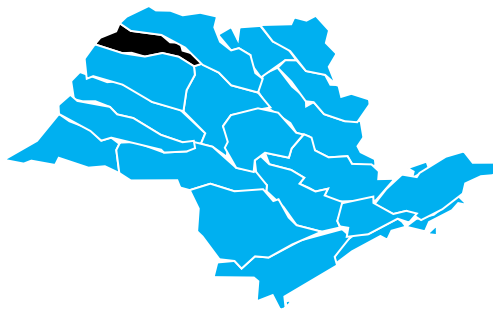
UGRHI 17 - Médio Paranapanema | SPI-12



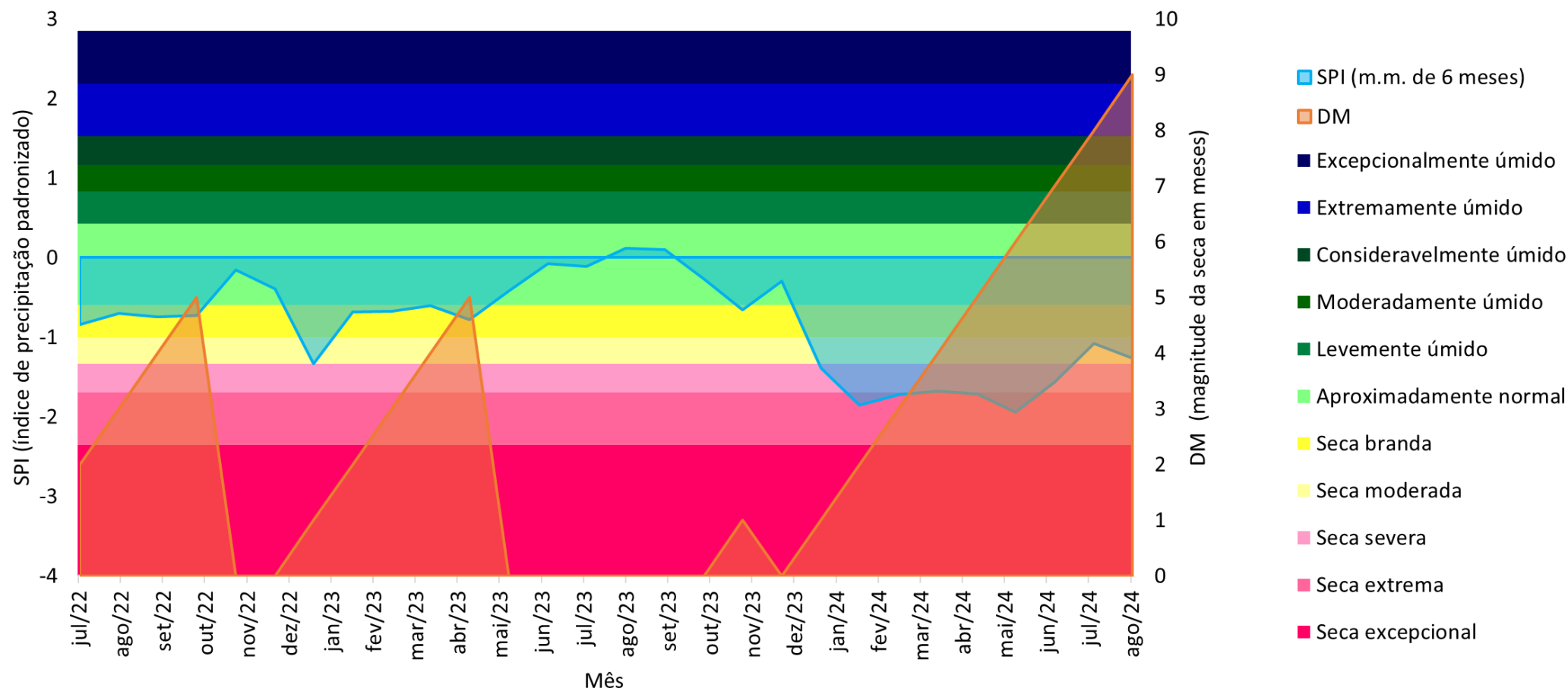
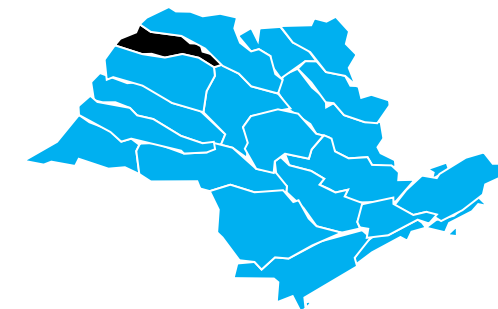
UGRHI 18 - São José dos Dourados



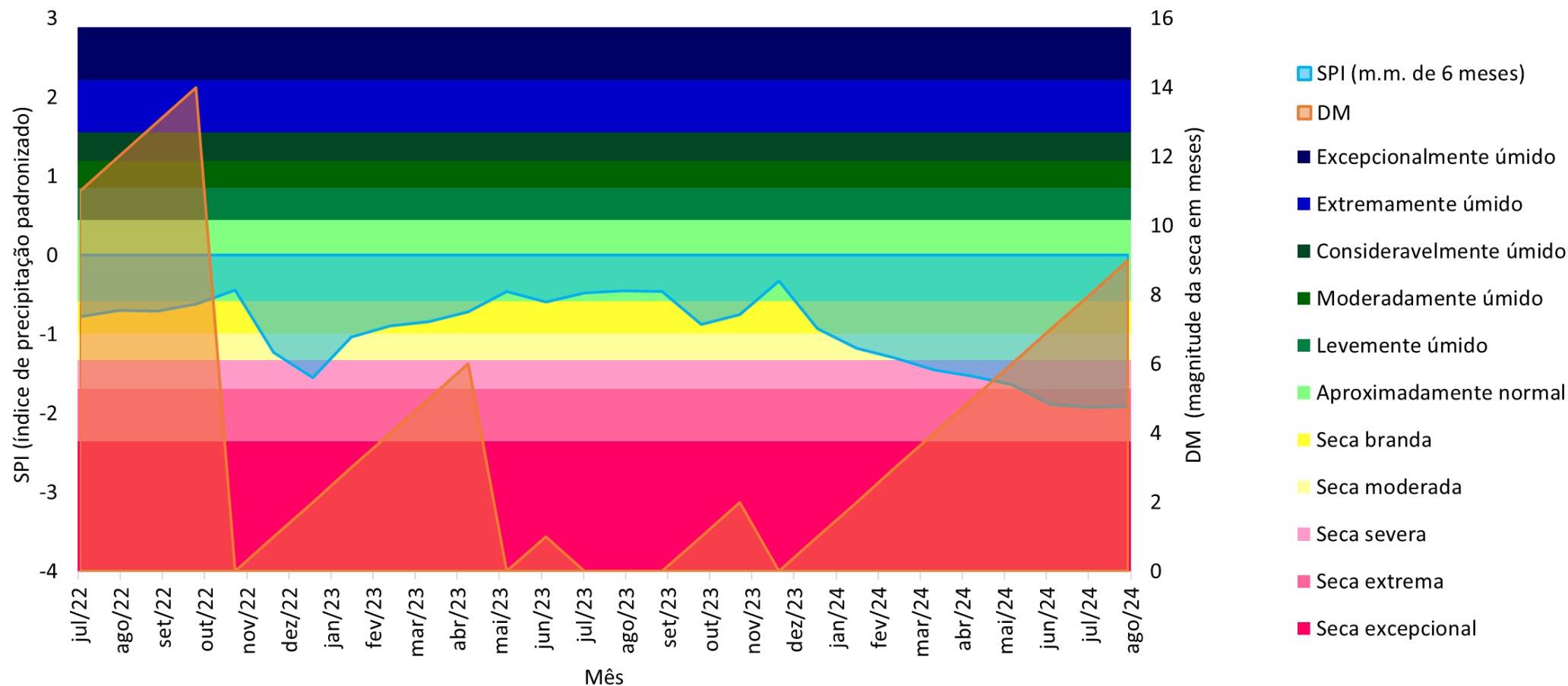
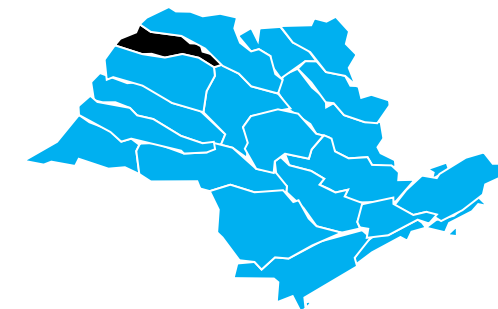
UGRHI 18 - São José dos Dourados



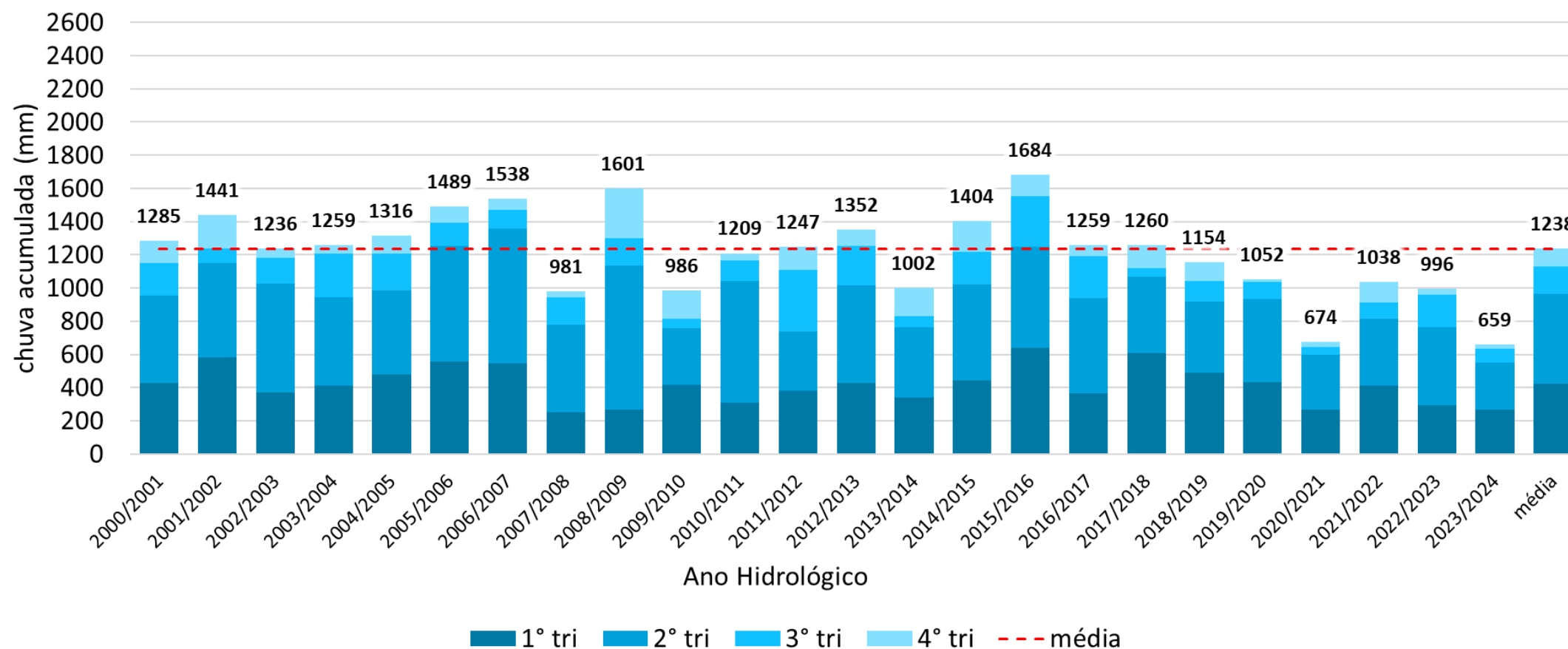
UGRHI 18 - São José dos Dourados | SPI-06



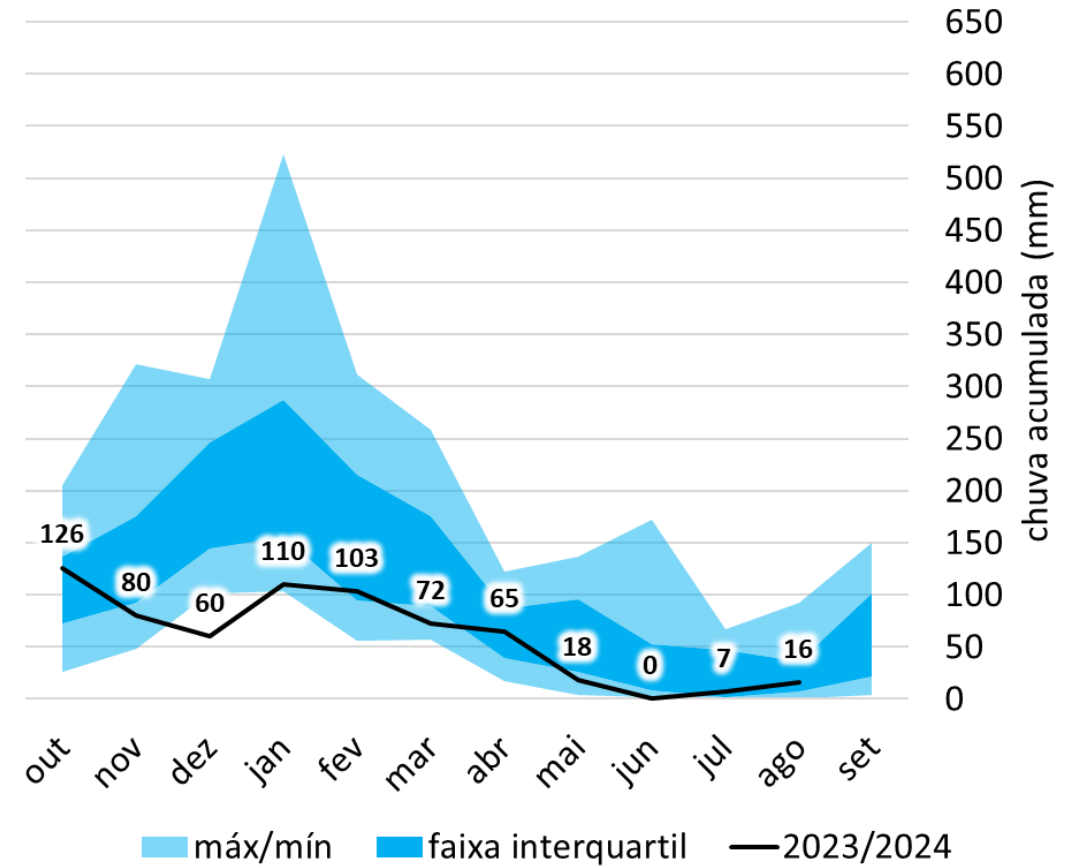
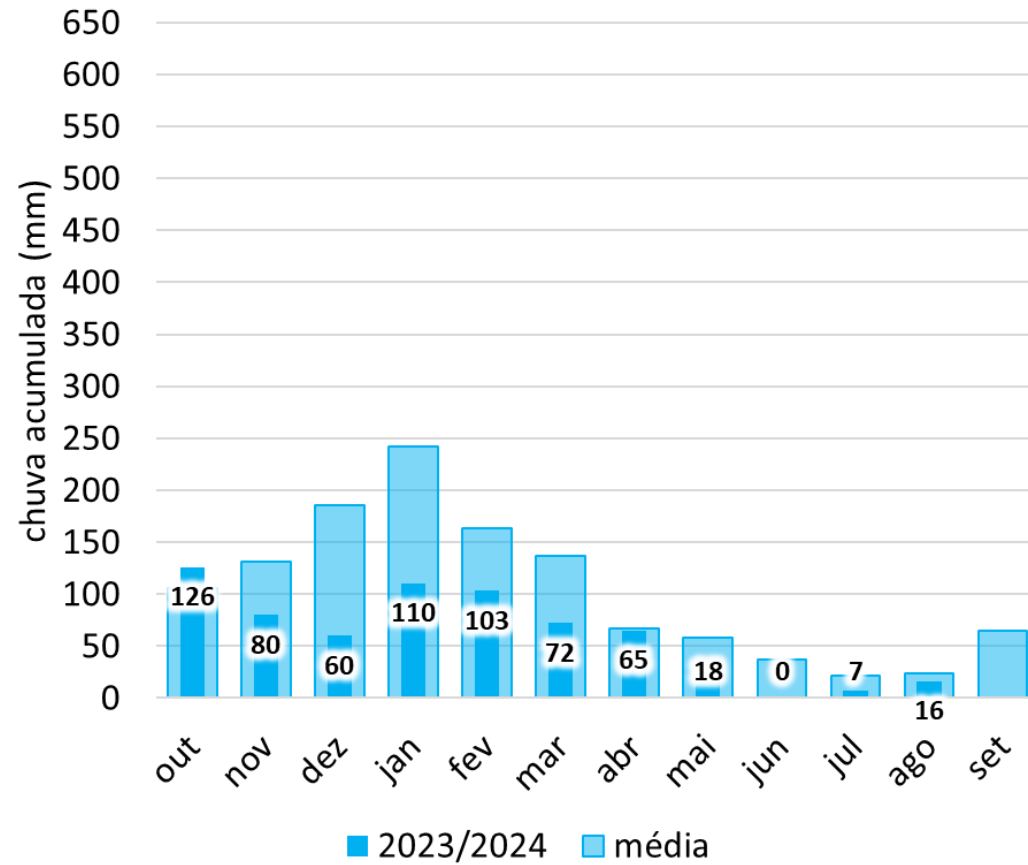
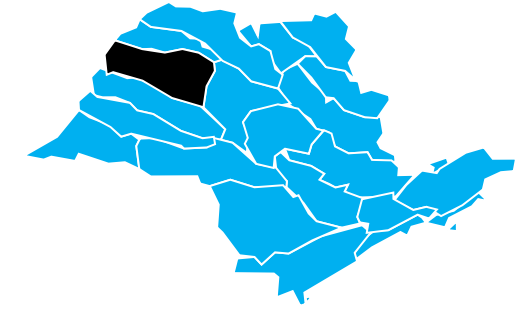
UGRHI 18 - São José dos Dourados | SPI-12



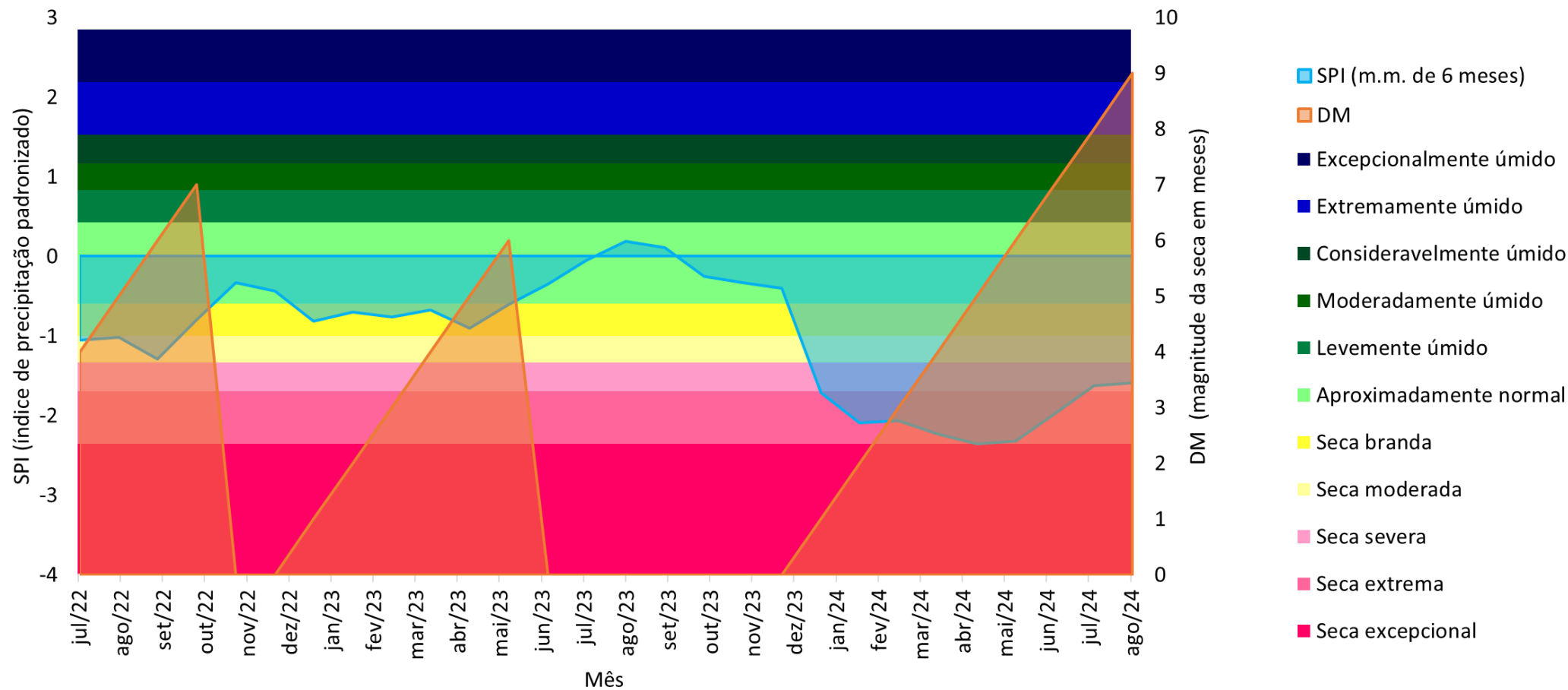
UGRHI 19 - Baixo Tietê



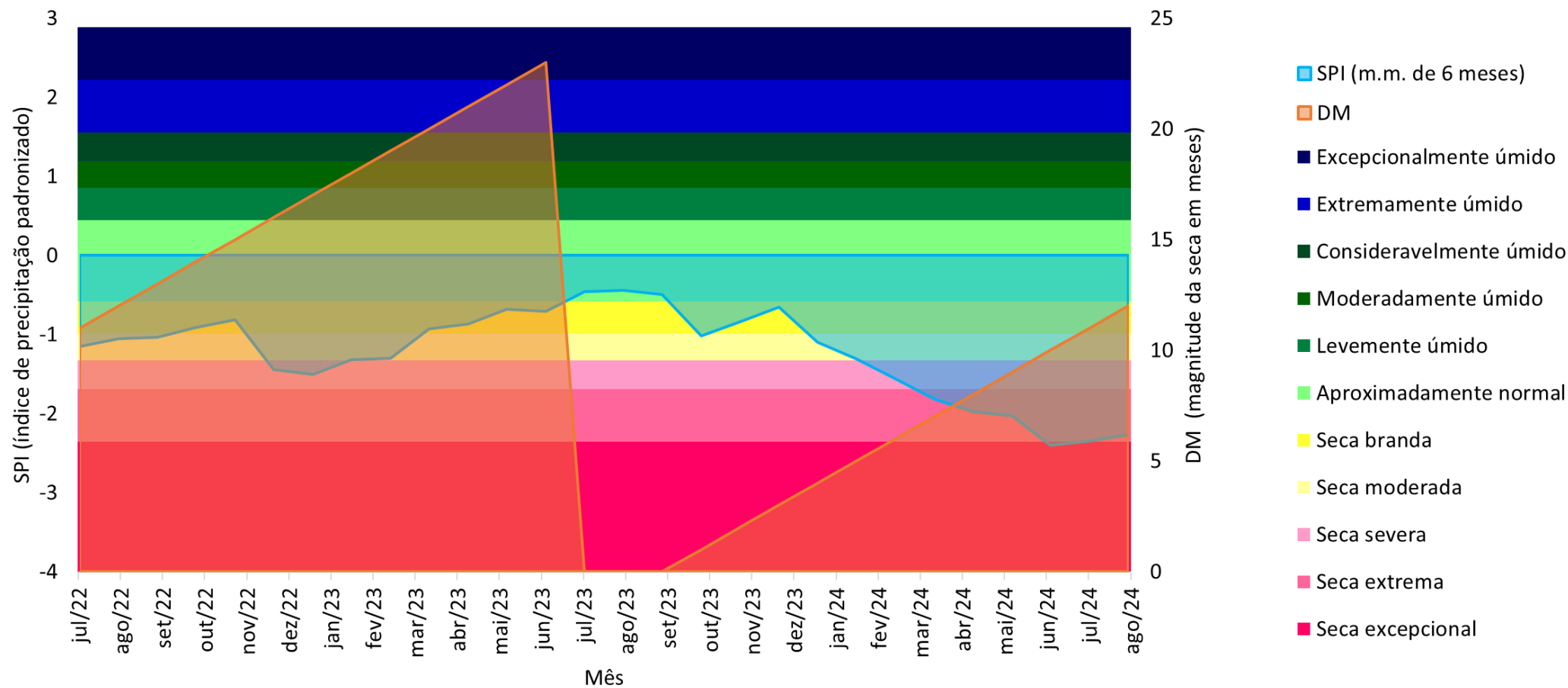
UGRHI 19 - Baixo Tietê



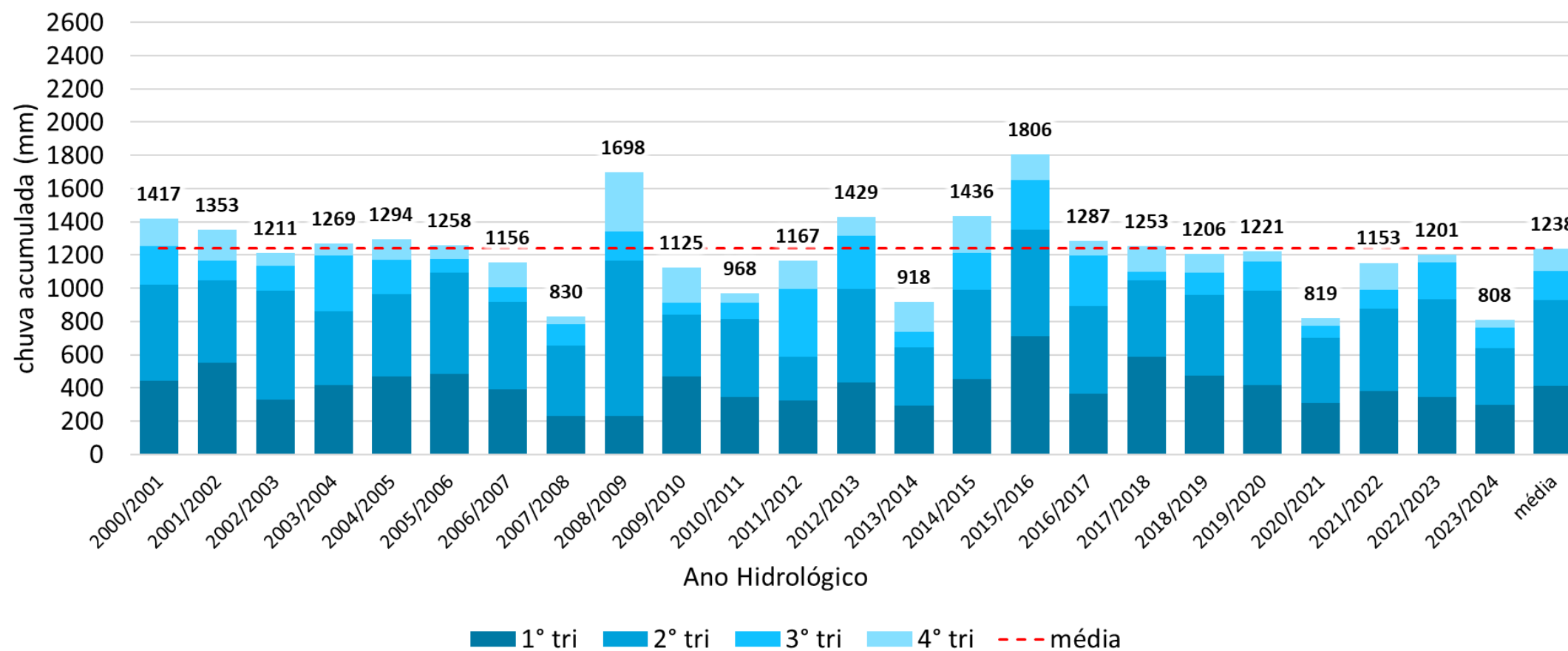
UGRHI 19 - Baixo Tietê | SPI-06



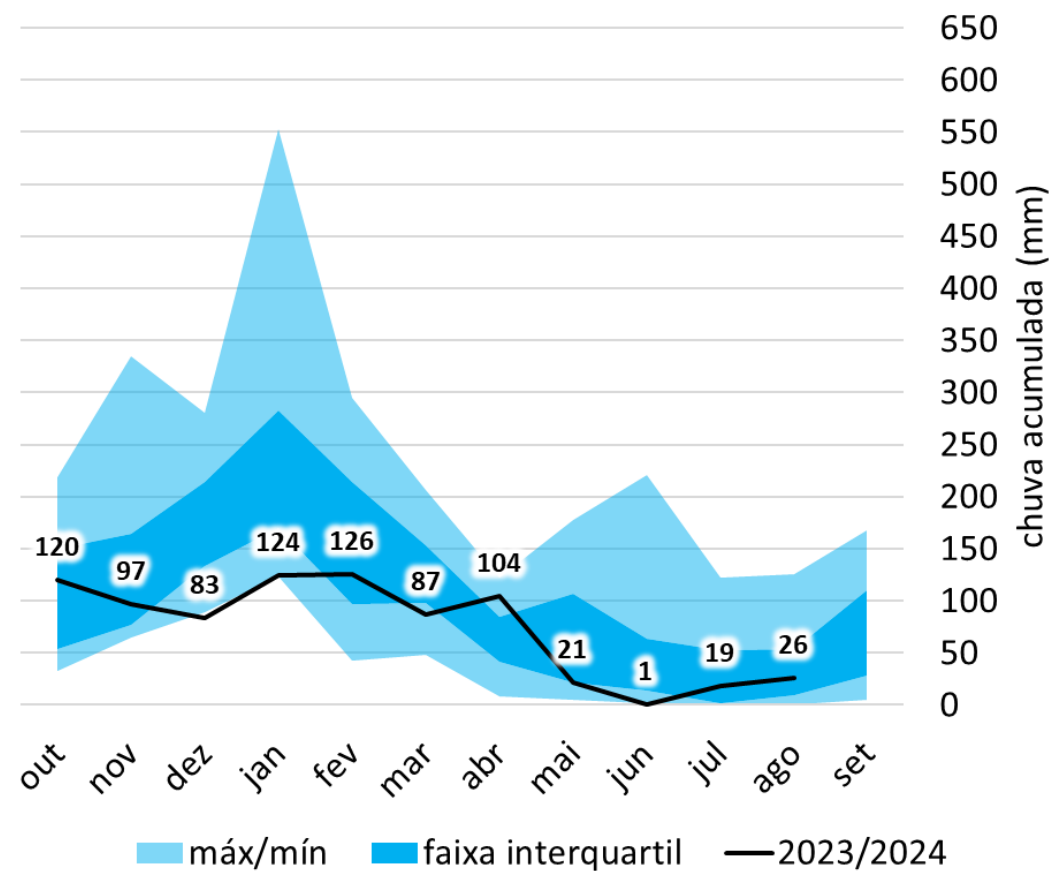
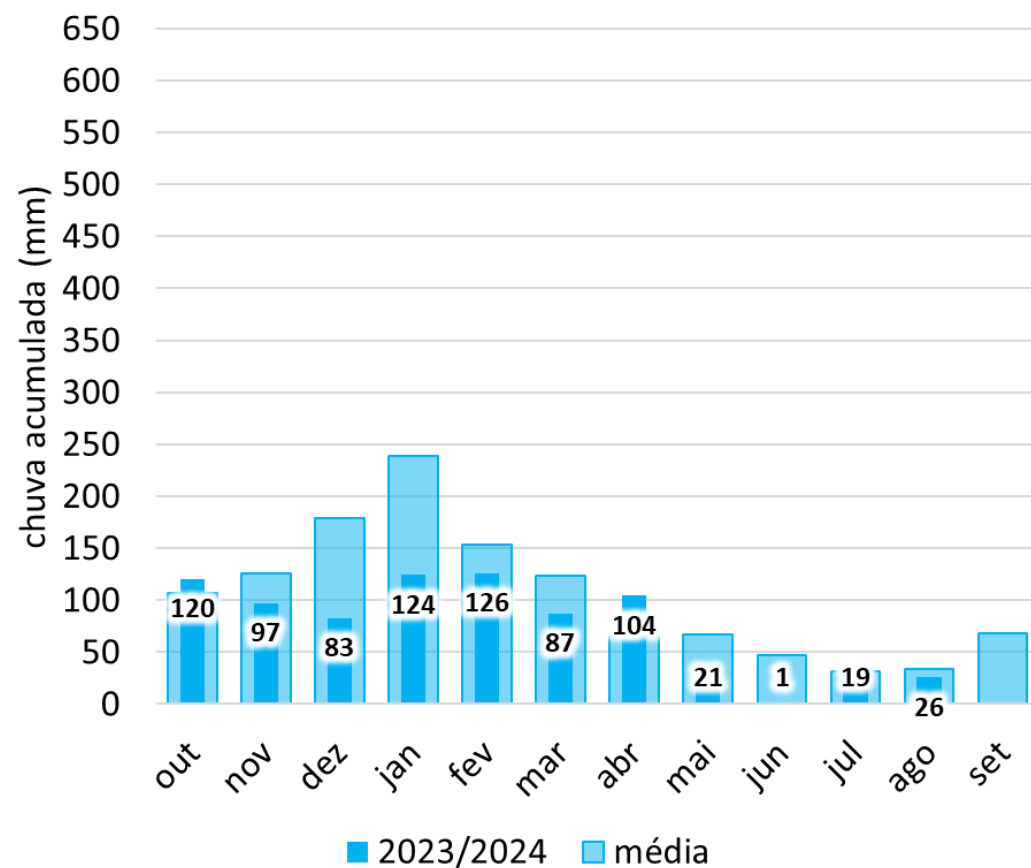
UGRHI 19 - Baixo Tietê | SPI-12



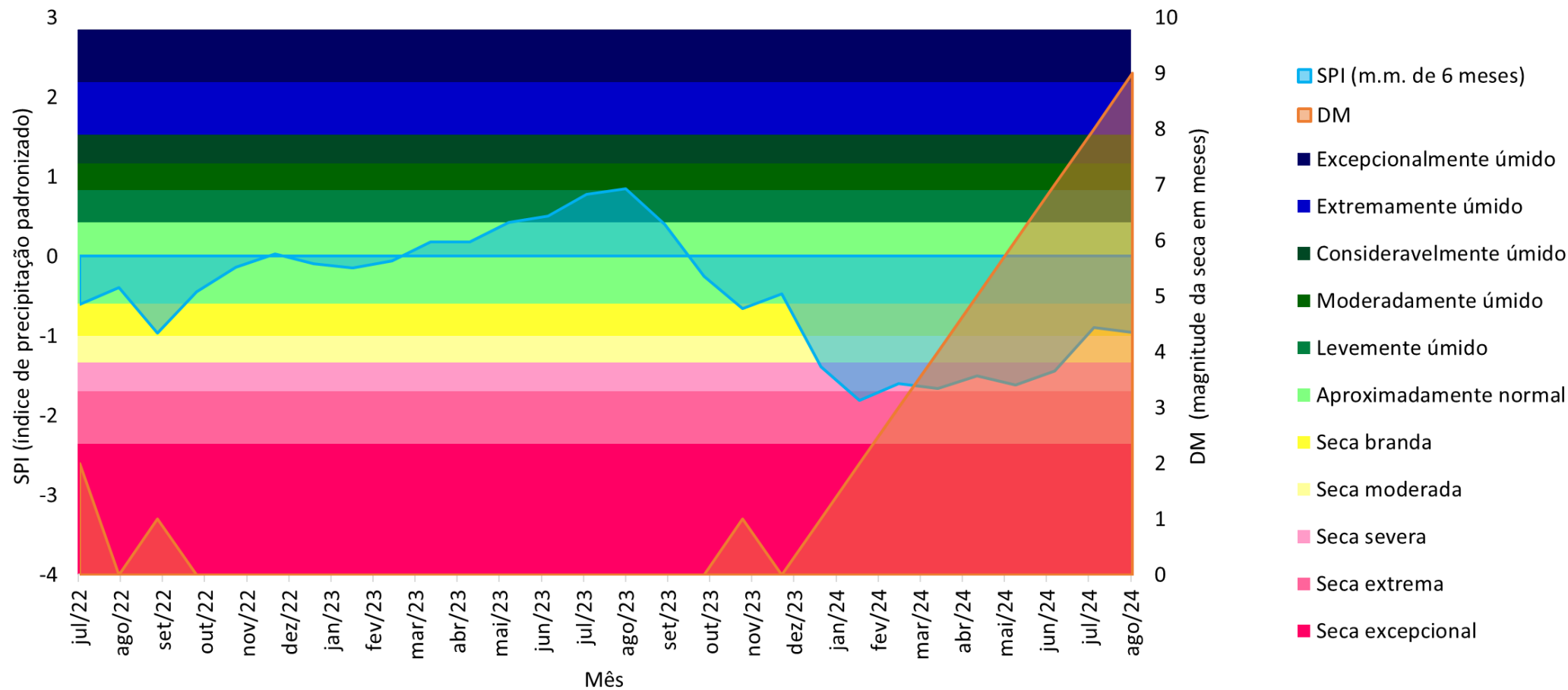
UGRHI 20 - Aguapeí



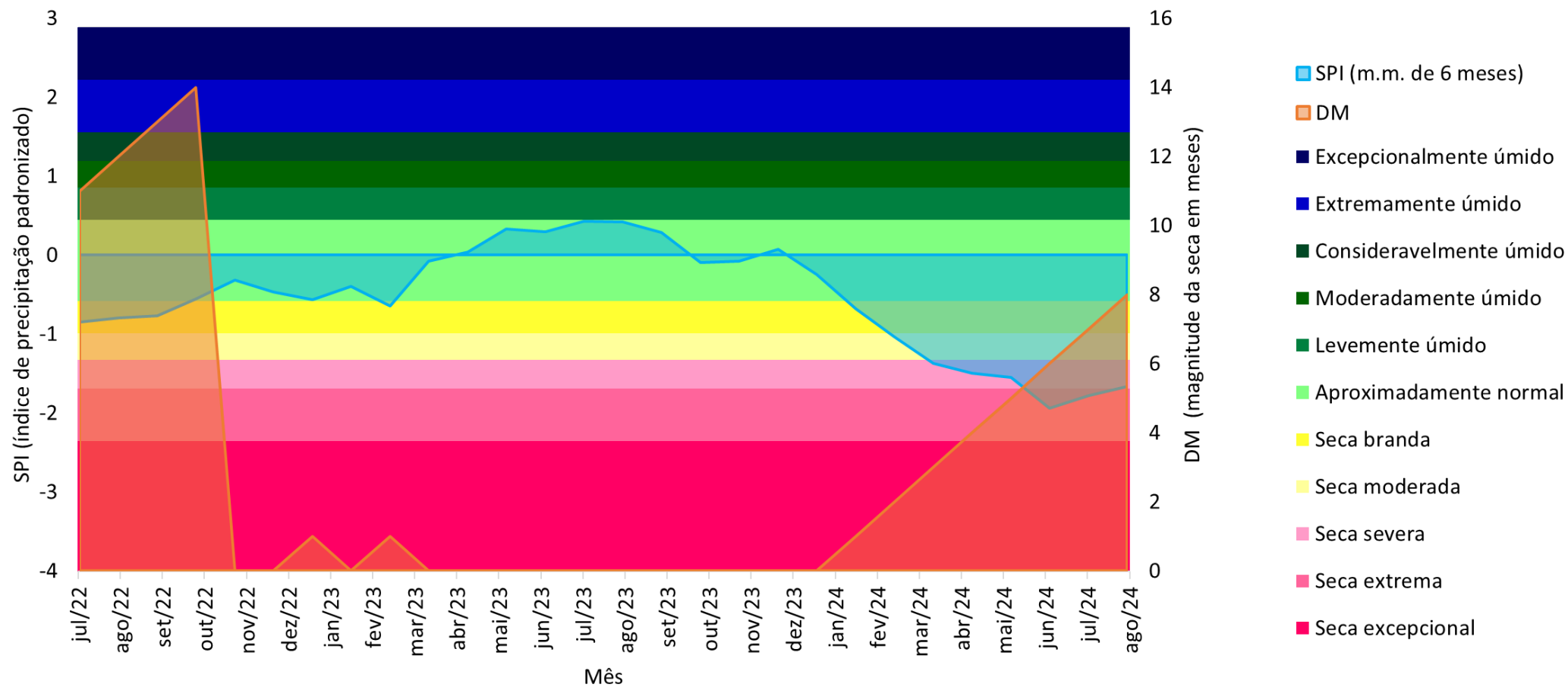
UGRHI 20 - Aguapeí



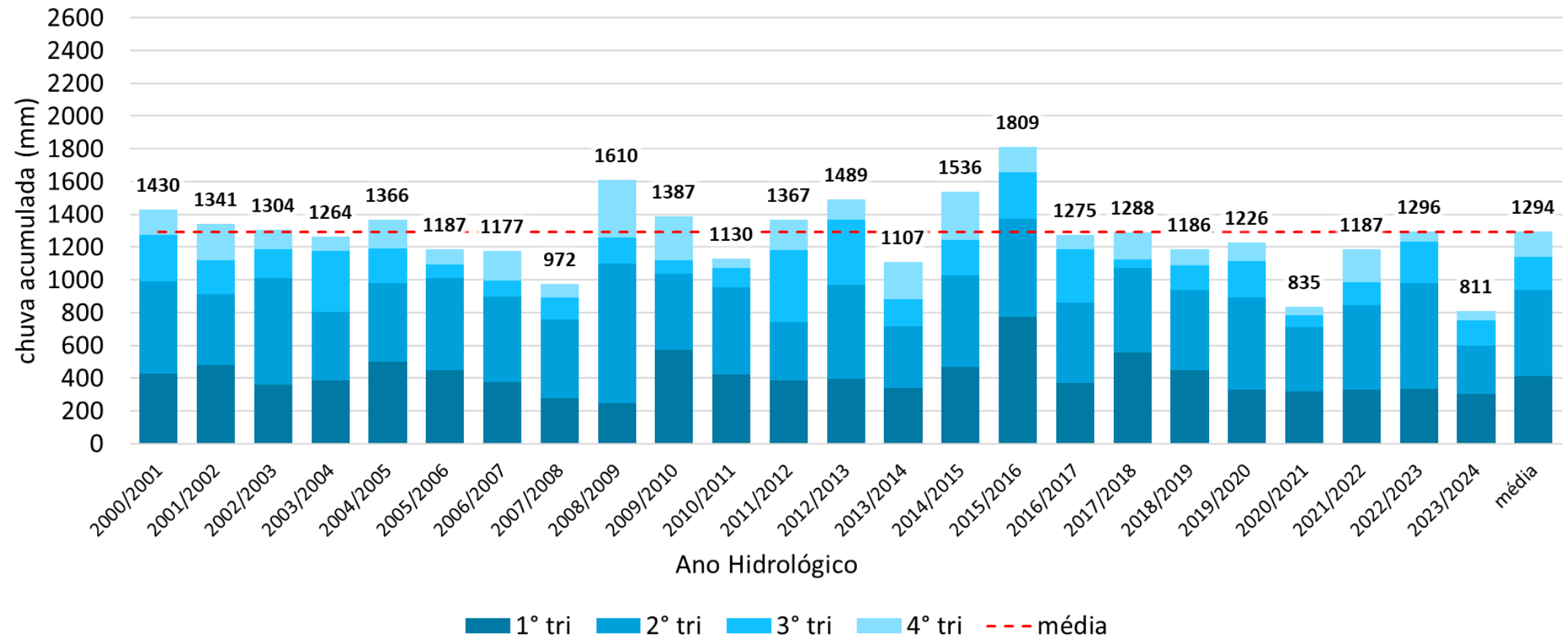
UGRHI 20 - Aguapeí | SPI-06



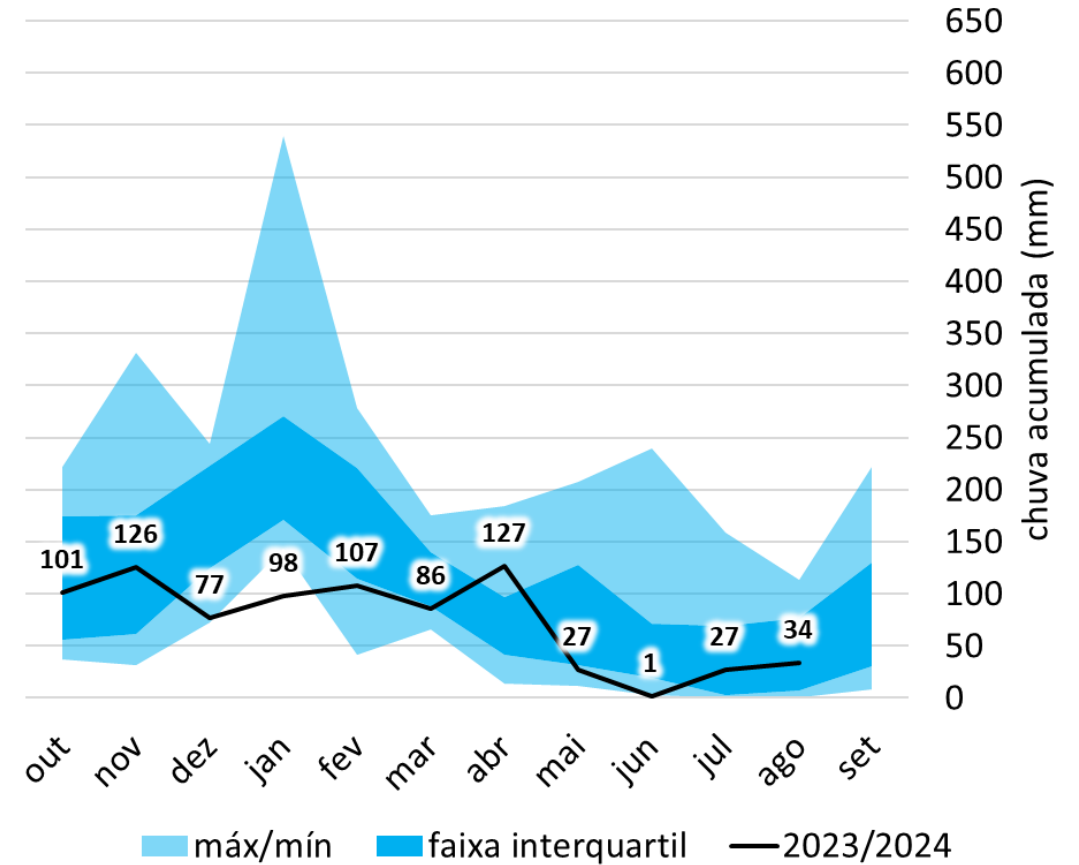
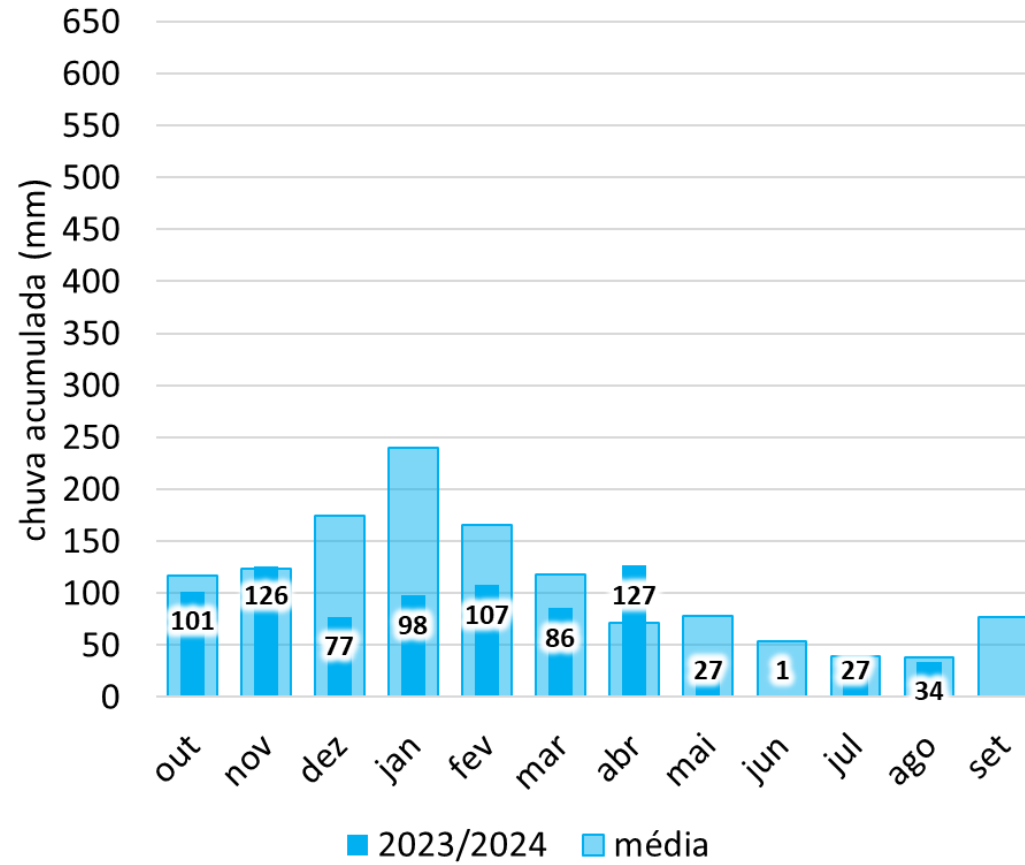
UGRHI 20 - Aguapeí | SPI-12



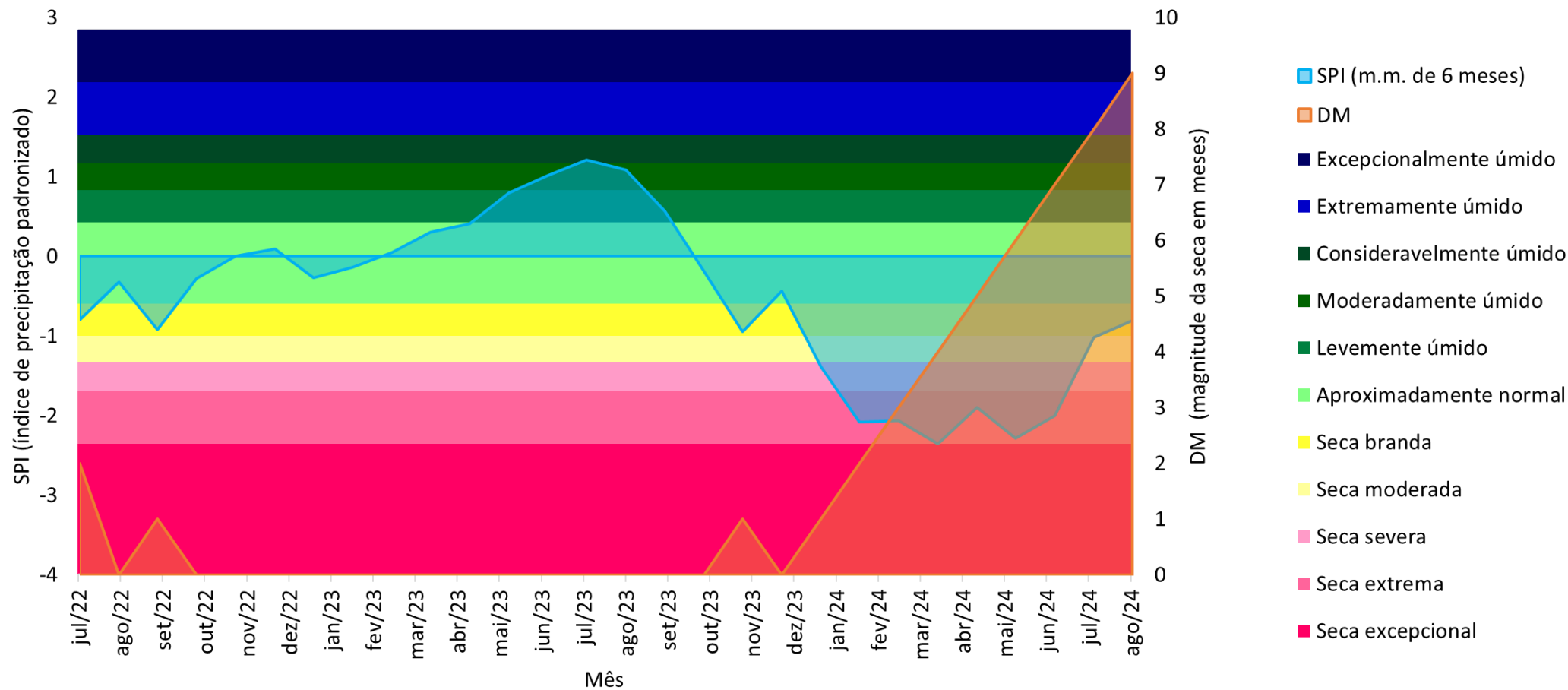
UGRHI 21 - Peixe



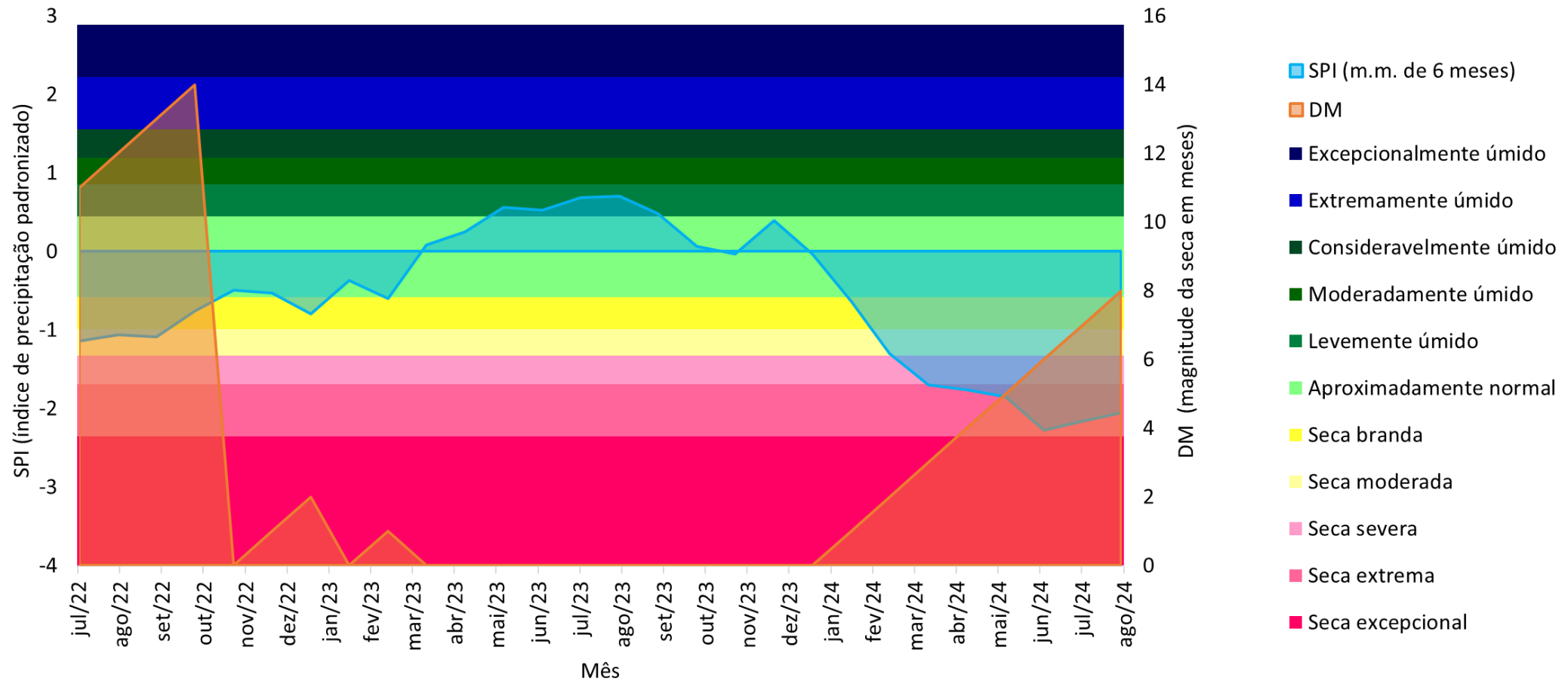
UGRHI 21 - Peixe



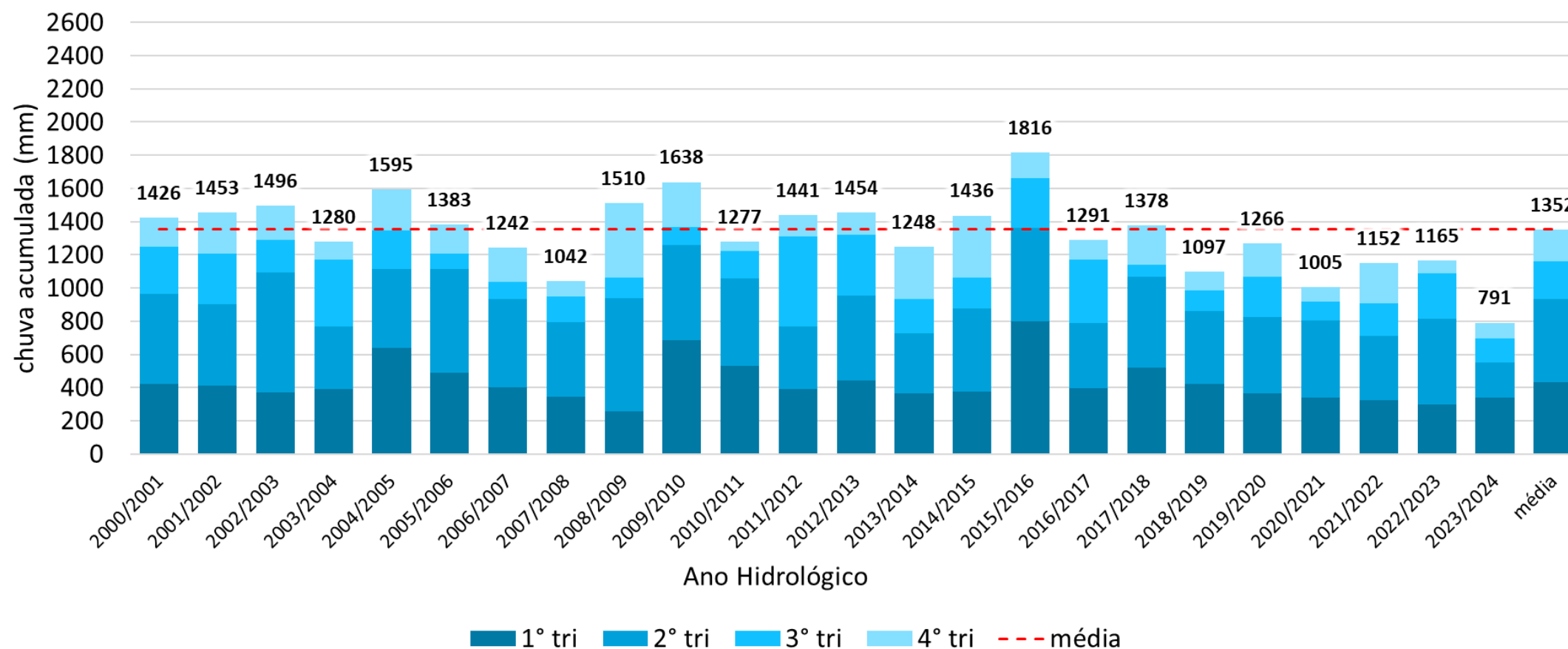
UGRHI 21 – Peixe | SPI-06



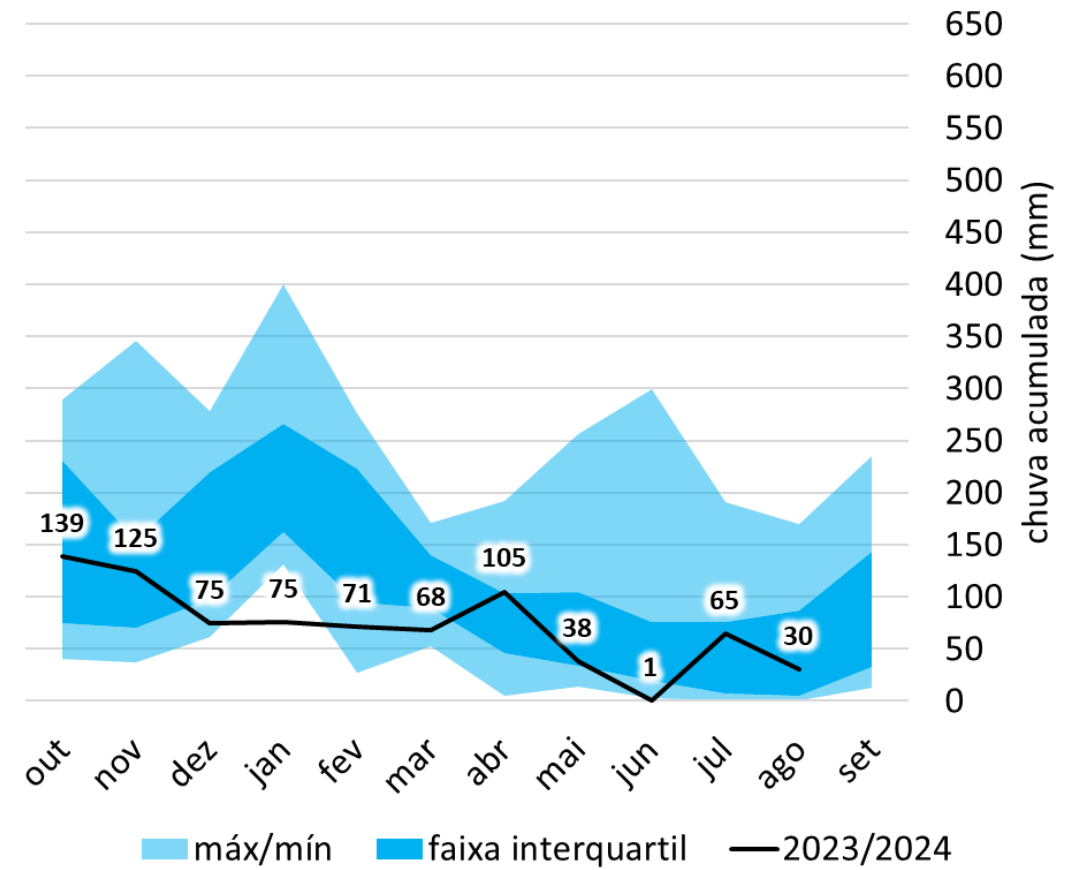
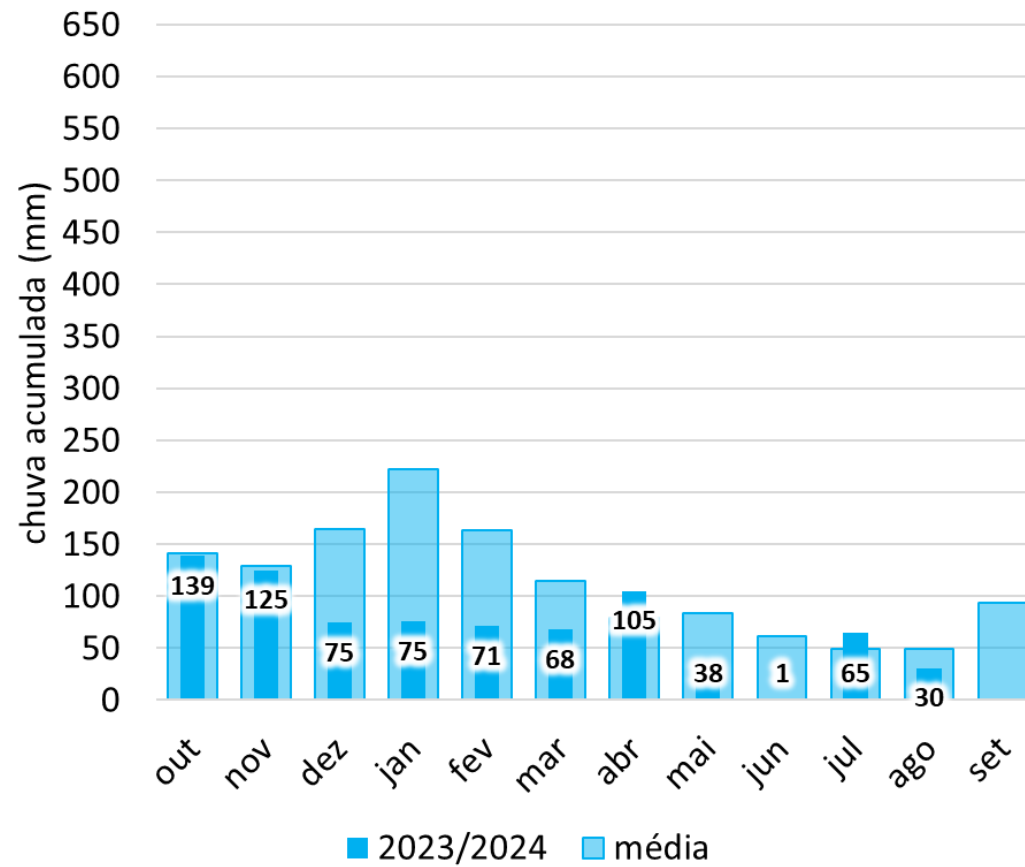
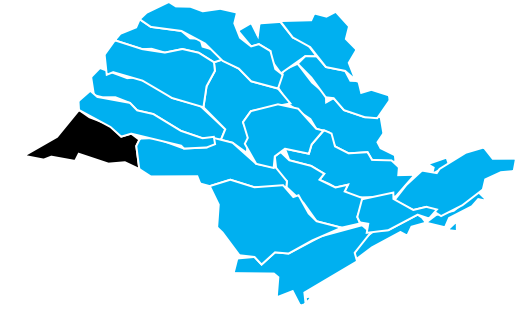
UGRHI 21 – Peixe | SPI-12



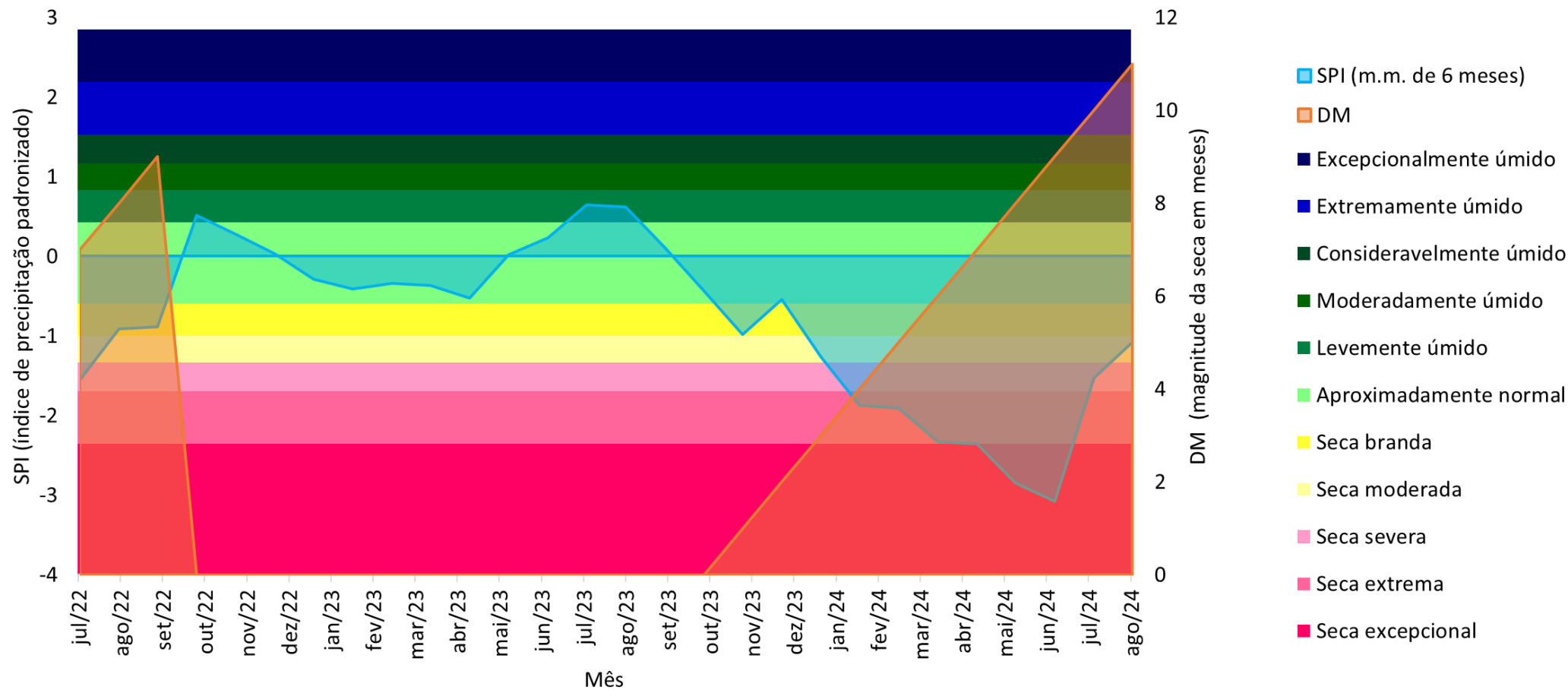
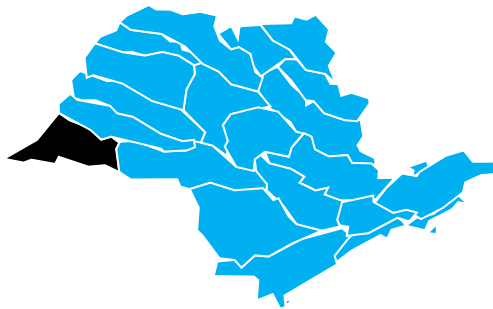
UGRHI 22 - Pontal do Paranapanema



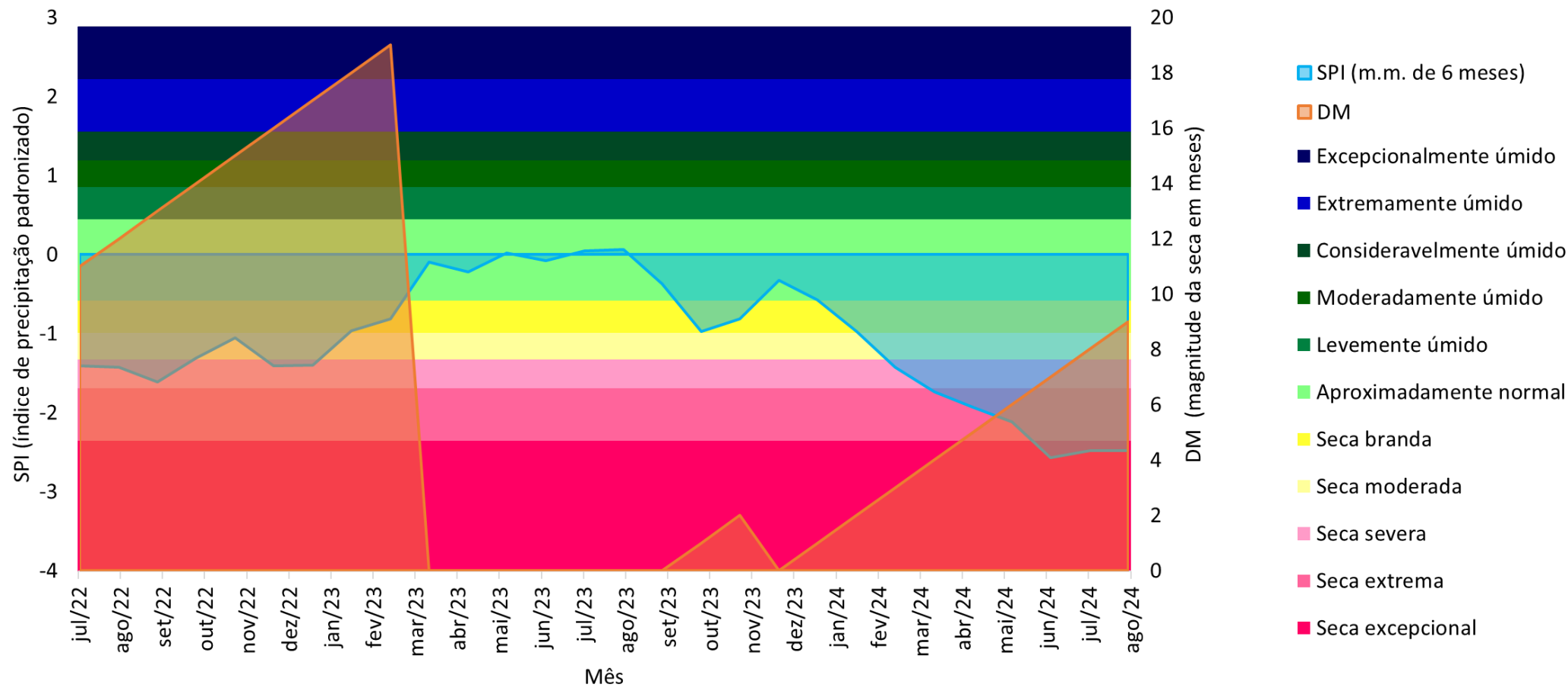
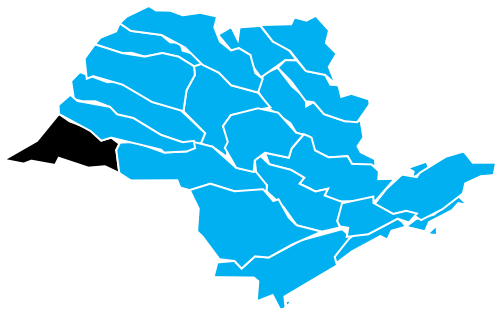
UGRHI 22 - Pontal do Paranapanema

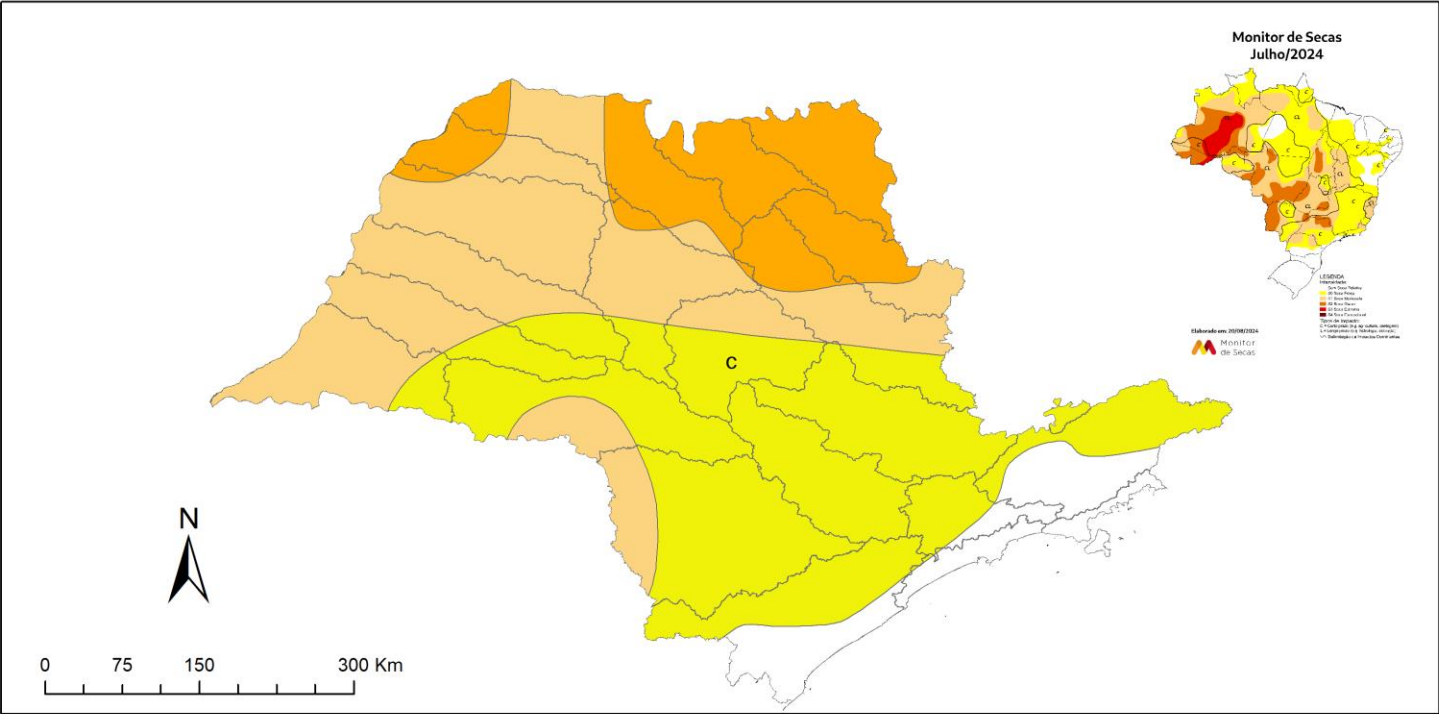


UGRHI 22 - Pontal do Paranapanema | SPI-06



UGRHI 22 - Pontal do Paranapanema | SPI-12





Monitor de Secas

- Acompanhamento regular e periódico (mensal) da atuação da seca
- Classificação da seca por grau de severidade (5 categorias)
- Indicação de impactos (curto e/ou longo prazo)

Legenda

Limites UGRHI

Limite SP

Sem Seca Relativa

Seca Fraca

Seca Moderada

Seca Grave

Seca Extrema

Tipos de impacto:

C = Curto prazo (e.g. agricultura, pastagem)

L = Longo prazo (e.g. hidrologia, ecologia)

Fonte:

Agência Nacional de Águas e Saneamento Básico (Ago/2024)



Monitor de Secas



SÃO PAULO
GOVERNO DO ESTADO
SÃO PAULO SÃO TODOS



siga nossas redes sociais